# PG No. 36 $D_{3\mathrm{d}}(1)$ $\bar{3}m$ (-31m setting) [ trigonal ] (jml basis)

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|\,, \langle \frac{1}{2}, -\frac{1}{2}; s|$
ket: $= |\frac{1}{2}, \frac{1}{2}; s\rangle\,, |\frac{1}{2}, -\frac{1}{2}; s\rangle$

Table 1: (s,s) block.

| No. | multipole | matrix |
|---|---|---|
| 1 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A_{1g})$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 2 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A_{2g})$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 3 | symmetry | $x$ |
| | $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$ |
| 4 | symmetry | $y$ |
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|\,, \langle \frac{1}{2}, -\frac{1}{2}; s|$
ket: $= |\frac{1}{2}, \frac{1}{2}; p\rangle\,, |\frac{1}{2}, -\frac{1}{2}; p\rangle\,, |\frac{3}{2}, \frac{3}{2}; p\rangle\,, |\frac{3}{2}, \frac{1}{2}; p\rangle\,, |\frac{3}{2}, -\frac{1}{2}; p\rangle\,, |\frac{3}{2}, -\frac{3}{2}; p\rangle$

Table 2: (s,p) block.

| No. | multipole | matrix |
|---|---|---|
| 5 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(a)}(A_{2u})$ | $\begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 6 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{6} & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{6}}{12} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 7 | symmetry | $y$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 8 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,0;a)}(A_{2u})$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 9 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & -\frac{1}{4} & 0 & \frac{\sqrt{3}}{12} & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & \frac{1}{4} \end{bmatrix}$ |
| 10 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{i}{4} \end{bmatrix}$ |
| 11 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \end{bmatrix}$ |
| 12 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{4} & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} \end{bmatrix}$ |
| 13 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & \frac{i}{4} \end{bmatrix}$ |
| 14 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 15 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 16 | symmetry | 1 |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_0^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 17 | symmetry | $z$ |
| | $\mathbb{T}_1^{(a)}(A_{2u})$ | $\begin{bmatrix} -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 18 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 19 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 20 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A_{2u})$ | $\begin{bmatrix} -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 21 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{12} & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & -\frac{i}{4} \end{bmatrix}$ |
| 22 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & -\frac{1}{4} & 0 & -\frac{\sqrt{3}}{12} & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & -\frac{1}{4} \end{bmatrix}$ |
| 23 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(1,-1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \end{bmatrix}$ |
| 24 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u, 1)$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & \frac{i}{4} \end{bmatrix}$ |
| 25 | symmetry | $-\sqrt{3}xz$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & \frac{1}{4} \end{bmatrix}$ |
| 26 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 27 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 28 | symmetry | 1 |
| | $\mathbb{M}_{0}^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|, \langle \frac{1}{2}, -\frac{1}{2}; s|$

ket: $= |\frac{3}{2}, \frac{3}{2}; d\rangle, |\frac{3}{2}, \frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{5}{2}; d\rangle, |\frac{5}{2}, \frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

Table 3: (s,d) block.

| No. | multipole | matrix |
|---|---|---|
| 29 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_{2}^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \end{bmatrix}$ |
| 30 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 31 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{10}}{10} & 0 \end{bmatrix}$ |
| 32 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 33 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 34 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \end{bmatrix}$ |
| 35 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{20} & 0 & -\frac{3\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & -\frac{\sqrt{15}i}{15} & 0 \end{bmatrix}$ |
| 36 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} \frac{\sqrt{15}}{20} & 0 & -\frac{3\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & -\frac{\sqrt{15}}{15} & 0 \end{bmatrix}$ |
| 37 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_{2,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 38 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}}{10} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 \\ -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 39 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 40 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_{3}^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 41 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 42 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 43 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 44 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \end{bmatrix}$ |
| 45 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \end{bmatrix}$ |
| 46 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 47 | symmetry | $x$ |
| | $\mathbb{G}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} \frac{\sqrt{3}i}{4} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 48 | symmetry | $y$ |
| | $\mathbb{G}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 49 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \end{bmatrix}$ |
| 50 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{2,1}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{10}}{10} & 0 \end{bmatrix}$ |
| 51 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{T}^{(a)}_{2,2}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 52 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}^{(a)}_{2,1}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 53 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}^{(a)}_{2,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 54 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}^{(1,0;a)}_{2}(A_{1g})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \end{bmatrix}$ |
| 55 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}^{(1,0;a)}_{2,1}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{15}}{20} & 0 & -\frac{3\sqrt{5}}{20} & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & -\frac{\sqrt{15}}{15} & 0 \end{bmatrix}$ |
| 56 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{T}^{(1,0;a)}_{2,2}(E_g,1)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{20} & 0 & \frac{3\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{15}i}{15} & 0 \end{bmatrix}$ |
| 57 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}^{(1,0;a)}_{2,1}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 \\ -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 58 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{2,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 \\ \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 59 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 60 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 61 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 62 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 63 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 64 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \end{bmatrix}$ |
| 65 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \end{bmatrix}$ |
| 66 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 67 | symmetry | $x$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} \frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 68 | symmetry | $y$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} \frac{\sqrt{3}i}{4} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|, \langle \frac{1}{2}, -\frac{1}{2}; s|$

ket: $= |\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 4: (s,f) block.

| No. | multipole | matrix |
|---|---|---|
| 69 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 70 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A_{2u}, 1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 71 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A_{2u}, 2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 72 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u, 1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \end{bmatrix}$ |
| 73 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u, 1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \end{bmatrix}$ |
| 74 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u, 2)$ | $\begin{bmatrix} \frac{\sqrt{14}}{28} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 \end{bmatrix}$ |
| 75 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$ |
| 76 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \end{bmatrix}$ |
| 77 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 78 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\ \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \end{bmatrix}$ |
| 79 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & \frac{\sqrt{210}}{56} & 0 & 0 \end{bmatrix}$ |
| 80 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{21} & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 \end{bmatrix}$ |
| 81 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 82 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 83 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_{1u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 84 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0&0&0&0&0&0&-\frac{\sqrt{2}i}{8}&0&0&0&0&0&\frac{\sqrt{14}i}{8}&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{14}i}{8}&0&0&0&0&0&-\frac{\sqrt{2}i}{8} \end{bmatrix}$ |
| 85 | symmetry | $\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_{2u})$ | $\begin{bmatrix} 0&0&0&0&0&0&\frac{\sqrt{2}}{8}&0&0&0&0&0&\frac{\sqrt{14}}{8}&0 \\ 0&0&0&0&0&0&0&-\frac{\sqrt{14}}{8}&0&0&0&0&0&-\frac{\sqrt{2}}{8} \end{bmatrix}$ |
| 86 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&\frac{\sqrt{6}}{8}&0&\frac{\sqrt{10}}{8}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{10}}{8}&0&-\frac{\sqrt{6}}{8}&0&0 \end{bmatrix}$ |
| 87 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&\frac{\sqrt{6}i}{8}&0&-\frac{\sqrt{10}i}{8}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{10}i}{8}&0&\frac{\sqrt{6}i}{8}&0&0 \end{bmatrix}$ |
| 88 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{1}{2} \\ 0&0&0&0&0&0&-\frac{1}{2}&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 89 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{i}{2} \\ 0&0&0&0&0&0&-\frac{i}{2}&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 90 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&\frac{1}{4}&0&0&0&-\frac{\sqrt{3}}{4}&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{3}}{4}&0&0&0&\frac{1}{4}&0 \end{bmatrix}$ |
| 91 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&-\frac{i}{4}&0&0&0&-\frac{\sqrt{3}i}{4}&0&0 \\ 0&0&0&0&0&0&0&0&\frac{\sqrt{3}i}{4}&0&0&0&\frac{i}{4}&0 \end{bmatrix}$ |
| 92 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} 0&0&-\frac{i}{2}&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&-\frac{i}{2}&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 93 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 94 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 95 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{30}}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 96 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 97 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 98 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 99 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 100 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{T}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 \end{bmatrix}$ |
| 101 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{T}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \end{bmatrix}$ |
| 102 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 \end{bmatrix}$ |
| 103 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \end{bmatrix}$ |
| 104 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 \\ \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} \end{bmatrix}$ |
| 105 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 106 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}^{(1,0;a)}_{3}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \end{bmatrix}$ |
| 107 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{T}^{(1,0;a)}_{3,1}(E_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}i}{21} & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 \end{bmatrix}$ |
| 108 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{T}^{(1,0;a)}_{3,2}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}}{21} & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 \end{bmatrix}$ |
| 109 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 \end{bmatrix}$ |
| 110 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}^{(1,0;a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} \frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 \end{bmatrix}$ |
| 111 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 112 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \end{bmatrix}$ |
| 113 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \end{bmatrix}$ |
| 114 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 \end{bmatrix}$ |
| 115 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & 0 \end{bmatrix}$ |
| 116 | symmetry | $-\frac{\sqrt{35}x^3y}{2} + \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 117 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 118 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & -\frac{i}{4} & 0 \end{bmatrix}$ |
| 119 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & \frac{1}{4} & 0 \end{bmatrix}$ |
| 120 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 121 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 122 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 123 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 124 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p| \, , \langle \frac{1}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, \frac{3}{2}; p| \, , \langle \frac{3}{2}, \frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{3}{2}; p|$
ket: $= |\frac{1}{2}, \frac{1}{2}; p\rangle \, , |\frac{1}{2}, -\frac{1}{2}; p\rangle \, , |\frac{3}{2}, \frac{3}{2}; p\rangle \, , |\frac{3}{2}, \frac{1}{2}; p\rangle \, , |\frac{3}{2}, -\frac{1}{2}; p\rangle \, , |\frac{3}{2}, -\frac{3}{2}; p\rangle$

Table 5: (p,p) block.

| No. | multipole | matrix |
|---|---|---|
| 125 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A_{1g})$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 126 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 127 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{6}i}{12} \\ -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 128 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{6}}{12} \\ -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 129 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$ |
| 130 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$ |
| 131 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_{2}^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 132 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & \frac{i}{4} & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{i}{4} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 133 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 & -\frac{\sqrt{3}}{12} \\ -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{1}{4} & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{1}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 134 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \end{bmatrix}$ |
| 135 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$ |
| 136 | symmetry | 1 |
| | $\mathbb{Q}_0^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 137 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,0;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 138 | symmetry | $x$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} \\ -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 139 | symmetry | $y$ |
| | $\mathbb{G}_{1,2}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 140 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_{2}^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 141 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{1}{4} & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} \\ \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 142 | symmetry | $-\sqrt{3}xz$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & \frac{i}{4} \\ -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 143 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 144 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 145 | symmetry | $z$ |
| | $\mathbb{M}_1^{(a)}(A_{2g})$ | $\begin{bmatrix} \frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{6} & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{6} & 0 & 0 & -\frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \end{bmatrix}$ |
| 146 | symmetry | $x$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(a)}_{1,1}(E_g)$ | $\begin{bmatrix} 0 & \frac{1}{3} & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{2}}{12} & 0 \\ \frac{1}{3} & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{6}}{12} \\ -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{3}}{6} & 0 & \frac{1}{3} & 0 \\ \frac{\sqrt{2}}{12} & 0 & 0 & \frac{1}{3} & 0 & \frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 147 | symmetry | $y$ |
| | $\mathbb{M}^{(a)}_{1,2}(E_g)$ | $\begin{bmatrix} 0 & -\frac{i}{3} & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{2}i}{12} & 0 \\ \frac{i}{3} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{6}i}{12} \\ \frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{12} & \frac{\sqrt{3}i}{6} & 0 & -\frac{i}{3} & 0 \\ \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{i}{3} & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 148 | symmetry | $z$ |
| | $\mathbb{M}^{(1,-1;a)}_{1}(A_{2g})$ | $\begin{bmatrix} -\frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{2\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{18} & 0 & 0 \\ 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 149 | symmetry | $x$ |
| | $\mathbb{M}^{(1,-1;a)}_{1,1}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{18} & \frac{1}{3} & 0 & -\frac{\sqrt{3}}{9} & 0 \\ -\frac{\sqrt{6}}{18} & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & -\frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{9} & \frac{\sqrt{2}}{6} & 0 & \frac{\sqrt{6}}{9} & 0 \\ -\frac{\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{9} & 0 & \frac{\sqrt{2}}{6} \\ 0 & -\frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \end{bmatrix}$ |
| 150 | symmetry | $y$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{18} & \frac{i}{3} & 0 & \frac{\sqrt{3}i}{9} & 0 \\ -\frac{\sqrt{6}i}{18} & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & \frac{i}{3} \\ -\frac{i}{3} & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{9} & \frac{\sqrt{2}i}{6} & 0 & -\frac{\sqrt{6}i}{9} & 0 \\ -\frac{\sqrt{3}i}{9} & 0 & 0 & \frac{\sqrt{6}i}{9} & 0 & -\frac{\sqrt{2}i}{6} \\ 0 & -\frac{i}{3} & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 \end{bmatrix}$ |
| 151 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 152 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$ |
| 153 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_{3}^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 154 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & \frac{\sqrt{30}}{10} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{10} & 0 & -\frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \end{bmatrix}$ |
| 155 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & -\frac{\sqrt{30}i}{10} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 & \frac{\sqrt{10}i}{10} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 156 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 157 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 158 | symmetry | $z$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A_{2g})$ | $\begin{bmatrix} \frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{18} & 0 & 0 & -\frac{\sqrt{30}}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & \frac{\sqrt{30}}{90} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} \end{bmatrix}$ |
| 159 | symmetry | $x$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{9} & \frac{\sqrt{5}}{12} & 0 & -\frac{\sqrt{15}}{36} & 0 \\ \frac{\sqrt{30}}{9} & 0 & 0 & \frac{\sqrt{15}}{36} & 0 & -\frac{\sqrt{5}}{12} \\ \frac{\sqrt{5}}{12} & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{36} & -\frac{\sqrt{10}}{30} & 0 & -\frac{\sqrt{30}}{45} & 0 \\ -\frac{\sqrt{15}}{36} & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 & -\frac{\sqrt{10}}{30} \\ 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 \end{bmatrix}$ |
| 160 | symmetry | $y$ |
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}i}{9} & \frac{\sqrt{5}i}{12} & 0 & \frac{\sqrt{15}i}{36} & 0 \\ \frac{\sqrt{30}i}{9} & 0 & 0 & \frac{\sqrt{15}i}{36} & 0 & \frac{\sqrt{5}i}{12} \\ -\frac{\sqrt{5}i}{12} & 0 & 0 & \frac{\sqrt{10}i}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{36} & -\frac{\sqrt{10}i}{30} & 0 & \frac{\sqrt{30}i}{45} & 0 \\ -\frac{\sqrt{15}i}{36} & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 & \frac{\sqrt{10}i}{30} \\ 0 & -\frac{\sqrt{5}i}{12} & 0 & 0 & -\frac{\sqrt{10}i}{30} & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p| \, , \langle \frac{1}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, \frac{3}{2}; p| \, , \langle \frac{3}{2}, \frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{3}{2}; p|$

ket: $= |\frac{3}{2}, \frac{3}{2}; d\rangle \, , |\frac{3}{2}, \frac{1}{2}; d\rangle \, , |\frac{3}{2}, -\frac{1}{2}; d\rangle \, , |\frac{3}{2}, -\frac{3}{2}; d\rangle \, , |\frac{5}{2}, \frac{5}{2}; d\rangle \, , |\frac{5}{2}, \frac{3}{2}; d\rangle \, , |\frac{5}{2}, \frac{1}{2}; d\rangle \, , |\frac{5}{2}, -\frac{1}{2}; d\rangle \, , |\frac{5}{2}, -\frac{3}{2}; d\rangle \, , |\frac{5}{2}, -\frac{5}{2}; d\rangle$

Table 6: (p,d) block.

| No. | multipole | matrix |
|---|---|---|
| 161 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(A_{2u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 \end{bmatrix}$ |
| 162 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{1}{4} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & \frac{3}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{30} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & -\frac{3}{20} & 0 & \frac{3\sqrt{2}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{30}}{20} \end{bmatrix}$ |
| 163 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & -\frac{3i}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{30} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & -\frac{3i}{20} & 0 & -\frac{3\sqrt{2}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{30}i}{20} \end{bmatrix}$ |
| 164 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 165 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 \end{bmatrix}$ |
| 166 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 167 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{Q}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & \frac{1}{6} & 0 \\ 0 & \frac{\sqrt{2}}{10} & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & -\frac{7\sqrt{2}}{60} & 0 & \frac{1}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{10} & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & -\frac{1}{30} & 0 & \frac{7\sqrt{2}}{60} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{30}}{60} \end{bmatrix}$ |
| 168 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{Q}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & -\frac{i}{6} & 0 \\ 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & -\frac{7\sqrt{2}i}{60} & 0 & -\frac{i}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & -\frac{i}{30} & 0 & -\frac{7\sqrt{2}i}{60} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{30}i}{60} \end{bmatrix}$ |
| 169 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| 170 | symmetry | $\sqrt{15}xyz$ |
| 171 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 172 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| 173 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{Q}_{3,1}^{(a)}(E_u, 2)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{1}{6} & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & 0 & 0 & -\frac{1}{6} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 \end{bmatrix}$$

$\mathbb{Q}_{3,2}^{(a)}(E_u, 2)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{i}{6} & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 \\ \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 & 0 & -\frac{i}{6} \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & 0 \end{bmatrix}$$

$\mathbb{Q}_3^{(1,-1;a)}(A_{1u})$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\mathbb{Q}_3^{(1,-1;a)}(A_{2u}, 1)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{4}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{4}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{15} & 0 \end{bmatrix}$$

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 174 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{18} & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{18} & 0 \\ 0 & \frac{\sqrt{3}}{30} & 0 & 0 & -\frac{\sqrt{5}}{15} & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{30} & 0 & -\frac{1}{10} & 0 & 0 & \frac{7\sqrt{3}}{45} & 0 & -\frac{\sqrt{6}}{45} & 0 & 0 \\ 0 & -\frac{1}{10} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & \frac{\sqrt{6}}{45} & 0 & -\frac{7\sqrt{3}}{45} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & \frac{\sqrt{5}}{15} \end{bmatrix}$ |
| 175 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & -\frac{\sqrt{6}i}{18} & 0 \\ 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{30} & 0 & \frac{i}{10} & 0 & 0 & \frac{7\sqrt{3}i}{45} & 0 & \frac{\sqrt{6}i}{45} & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & \frac{\sqrt{6}i}{45} & 0 & \frac{7\sqrt{3}i}{45} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & -\frac{\sqrt{5}i}{15} \end{bmatrix}$ |
| 176 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{18} & 0 & 0 & 0 & \frac{\sqrt{15}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & 0 & -\frac{\sqrt{3}}{18} \\ 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 & 0 & 0 & \frac{\sqrt{6}}{9} \\ 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 177 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{18} & 0 & 0 & 0 & \frac{\sqrt{15}i}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{18} \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 \\ \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{45} & 0 & 0 & 0 & \frac{\sqrt{6}i}{9} \\ 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 178 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,0;a)}(A_{2u})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 \end{bmatrix}$ |
| 179 | symmetry | $x$ |
| | $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & 0 \\ \frac{1}{5} & 0 & \frac{2\sqrt{3}}{15} & 0 & 0 & -\frac{3}{20} & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 \\ 0 & \frac{2\sqrt{3}}{15} & 0 & \frac{1}{5} & 0 & 0 & -\frac{3\sqrt{2}}{40} & 0 & \frac{3}{20} & 0 \\ 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{20} \end{bmatrix}$ |
| 180 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{5} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 \\ \frac{i}{5} & 0 & -\frac{2\sqrt{3}i}{15} & 0 & 0 & -\frac{3i}{20} & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 \\ 0 & \frac{2\sqrt{3}i}{15} & 0 & -\frac{i}{5} & 0 & 0 & -\frac{3\sqrt{2}i}{40} & 0 & -\frac{3i}{20} & 0 \\ 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{15}i}{20} \end{bmatrix}$ |
| 181 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 182 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 \end{bmatrix}$ |
| 183 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 184 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & \frac{\sqrt{3}}{9} & 0 \\ 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & \frac{\sqrt{10}}{120} & 0 & -\frac{1}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{15} & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & -\frac{7\sqrt{6}}{360} & 0 & \frac{\sqrt{3}}{180} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{5} & 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & -\frac{\sqrt{3}}{180} & 0 & \frac{7\sqrt{6}}{360} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & 0 & 0 & \frac{1}{20} & 0 & -\frac{\sqrt{10}}{120} \end{bmatrix}$ |
| 185 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 & -\frac{\sqrt{3}i}{9} & 0 \\ 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & \frac{\sqrt{10}i}{120} & 0 & \frac{i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{15} & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & -\frac{7\sqrt{6}i}{360} & 0 & -\frac{\sqrt{3}i}{180} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{5} & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & -\frac{\sqrt{3}i}{180} & 0 & -\frac{7\sqrt{6}i}{360} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & \frac{i}{20} & 0 & \frac{\sqrt{10}i}{120} \end{bmatrix}$ |
| 186 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{18} & 0 & 0 & 0 & \frac{\sqrt{30}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & -\frac{\sqrt{3}}{36} & 0 & 0 & 0 & \frac{\sqrt{15}}{180} & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{180} & 0 & 0 & 0 & -\frac{\sqrt{3}}{36} \\ 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 187 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & \frac{\sqrt{3}i}{36} & 0 & 0 & 0 & \frac{\sqrt{15}i}{180} & 0 \\ -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{180} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{36} \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 188 | symmetry | $z$ |
| | $\mathbb{Q}_{1}^{(1,1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{10} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 \end{bmatrix}$ |
| 189 | symmetry | $x$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{10} & 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 \\ 0 & -\frac{1}{5} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{3}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{5}}{20} \end{bmatrix}$ |
| 190 | symmetry | $y$ |
| | $\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{10} & 0 & \frac{i}{5} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 \\ 0 & -\frac{i}{5} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{3}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{5}i}{20} \end{bmatrix}$ |
| 191 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(a)}(A_{1u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 192 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{5}}{15} & 0 & \frac{\sqrt{10}}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{10}}{30} & 0 & \frac{\sqrt{5}}{15} & 0 \\ 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & \frac{\sqrt{5}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{5}}{12} & 0 & \frac{\sqrt{10}}{60} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{6}}{12} \end{bmatrix}$ |
| 193 | symmetry | $-\sqrt{3}xz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{15} & 0 & -\frac{\sqrt{10}i}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & \frac{\sqrt{10}i}{30} & 0 & -\frac{\sqrt{5}i}{15} & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & -\frac{\sqrt{5}i}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{12} & 0 & -\frac{\sqrt{10}i}{60} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{6}i}{12} \end{bmatrix}$ |
| 194 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2,1}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{1}{6} & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 & 0 & 0 & \frac{1}{6} \\ 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & \frac{\sqrt{10}}{15} & 0 \\ \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{15} & 0 & 0 & 0 & \frac{\sqrt{2}}{6} \\ 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 195 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & \frac{i}{6} & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 & \frac{i}{6} \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{10}i}{15} & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{15} & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} \\ 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 196 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_{2}^{(1,-1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{3i}{10} & 0 \end{bmatrix}$ |
| 197 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & -\frac{1}{5} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & -\frac{1}{5} & 0 & -\frac{\sqrt{2}}{5} & 0 \\ 0 & \frac{1}{10} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{3\sqrt{6}}{40} & 0 & 0 & 0 \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{20} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{1}{20} & 0 \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{40} & 0 & -\frac{\sqrt{15}}{20} \end{bmatrix}$ |
| 198 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{6}i}{40} & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & \frac{i}{5} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & -\frac{i}{5} & 0 & \frac{\sqrt{2}i}{5} & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{3\sqrt{6}i}{40} & 0 & 0 & 0 \\ -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{20} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{i}{20} & 0 \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{20} \end{bmatrix}$ |
| 199 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 \\ -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{10} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & \frac{1}{5} & 0 \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 200 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 \\ \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{i}{5} & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 201 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A_{1u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$ |
| 202 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 203 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 204 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{105}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & -\frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{14}}{56} \end{bmatrix}$ |
| 205 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & \frac{\sqrt{210}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56} \end{bmatrix}$ |
| 206 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 207 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 208 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 209 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 210 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_{2}^{(1,0;a)}(A_{1u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 \\ -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 \end{bmatrix}$ |
| 211 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{G}_{2,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & -\frac{2\sqrt{30}}{45} & 0 & -\frac{2\sqrt{15}}{45} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & -\frac{2\sqrt{15}}{45} & 0 & -\frac{2\sqrt{30}}{45} & 0 \\ 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & -\frac{1}{12} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{180} & 0 & -\frac{\sqrt{30}}{72} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & \frac{\sqrt{30}}{72} & 0 & -\frac{\sqrt{15}}{180} & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{40} & 0 & \frac{1}{12} \end{bmatrix}$ |
| 212 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{G}_{2,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & -\frac{2\sqrt{30}i}{45} & 0 & \frac{2\sqrt{15}i}{45} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{2\sqrt{15}i}{45} & 0 & \frac{2\sqrt{30}i}{45} & 0 \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{i}{12} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{180} & 0 & \frac{\sqrt{30}i}{72} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{30}i}{72} & 0 & \frac{\sqrt{15}i}{180} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & -\frac{i}{12} \end{bmatrix}$ |
| 213 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}}{60} & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 \\ \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{45} & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} \\ 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & -\frac{\sqrt{3}}{18} & 0 & 0 & 0 & -\frac{\sqrt{15}}{45} & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{45} & 0 & 0 & 0 & -\frac{\sqrt{3}}{18} \\ 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 214 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & \frac{\sqrt{3}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{45} & 0 \\ -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{45} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{18} \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 215 | symmetry | 1 |
| | $\mathbb{G}_{0}^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 216 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_{2}^{(1,1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{30} & 0 & 0 \\ \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 \end{bmatrix}$ |
| 217 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{42}}{30} & 0 & \frac{\sqrt{14}}{10} & 0 & 0 & -\frac{\sqrt{42}}{90} & 0 & -\frac{\sqrt{21}}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{10} & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & -\frac{\sqrt{21}}{90} & 0 & -\frac{\sqrt{42}}{90} & 0 \\ 0 & \frac{\sqrt{21}}{30} & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{315} & 0 & -\frac{\sqrt{42}}{126} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & \frac{\sqrt{42}}{126} & 0 & -\frac{\sqrt{21}}{315} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & \frac{\sqrt{35}}{105} \end{bmatrix}$ |
| 218 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{42}i}{30} & 0 & -\frac{\sqrt{14}i}{10} & 0 & 0 & -\frac{\sqrt{42}i}{90} & 0 & \frac{\sqrt{21}i}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{10} & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & -\frac{\sqrt{21}i}{90} & 0 & \frac{\sqrt{42}i}{90} & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{315} & 0 & \frac{\sqrt{42}i}{126} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & \frac{\sqrt{42}i}{126} & 0 & \frac{\sqrt{21}i}{315} & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & -\frac{\sqrt{35}i}{105} \end{bmatrix}$ |
| 219 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{G}_{2,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}}{15} & \frac{\sqrt{210}}{180} & 0 & 0 & 0 & -\frac{\sqrt{42}}{180} & 0 \\ \frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{180} & 0 & 0 & 0 & -\frac{\sqrt{210}}{180} \\ 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{30} & -\frac{2\sqrt{105}}{315} & 0 & 0 & 0 & -\frac{4\sqrt{21}}{315} & 0 \\ -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{21}}{315} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{315} \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 220 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{G}_{2,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}i}{15} & -\frac{\sqrt{210}i}{180} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{180} & 0 \\ -\frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{180} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{180} \\ 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & \frac{2\sqrt{105}i}{315} & 0 & 0 & 0 & -\frac{4\sqrt{21}i}{315} & 0 \\ \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}i}{315} & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{315} \\ 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 221 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(A_{2u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 \end{bmatrix}$ |
| 222 | symmetry | $x$ |
| | $\mathbb{T}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{i}{4} & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{6}i}{30} & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & \frac{3i}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{30} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & -\frac{3i}{20} & 0 & \frac{3\sqrt{2}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{30}i}{20} \end{bmatrix}$ |
| 223 | symmetry | $y$ |
| | $\mathbb{T}_{1,2}^{(a)}(E_u)$ | $\begin{bmatrix} \frac{1}{4} & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{12} & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & \frac{3}{20} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{30} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & \frac{3}{20} & 0 & \frac{3\sqrt{2}}{20} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{30}}{20} \end{bmatrix}$ |
| 224 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 225 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 \end{bmatrix}$ |
| 226 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 227 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{T}_{3,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & \frac{i}{6} & 0 \\ 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{10} & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & -\frac{7\sqrt{2}i}{60} & 0 & \frac{i}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & -\frac{i}{30} & 0 & \frac{7\sqrt{2}i}{60} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{30}i}{60} \end{bmatrix}$ |
| 228 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{T}_{3,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & \frac{1}{6} & 0 \\ 0 & \frac{\sqrt{2}}{10} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & \frac{7\sqrt{2}}{60} & 0 & \frac{1}{30} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & \frac{1}{30} & 0 & \frac{7\sqrt{2}}{60} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{30}}{60} \end{bmatrix}$ |
| 229 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{3,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & -\frac{i}{6} & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 & -\frac{i}{6} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 230 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}^{(a)}_{3,2}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{12} \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & -\frac{1}{6} & 0 & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & 0 & 0 & \frac{1}{6} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 231 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}^{(1,-1;a)}_{3}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 232 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}^{(1,-1;a)}_{3}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 \\ \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{4i}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{4i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{15} & 0 \end{bmatrix}$ |
| 233 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 234 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |
| | $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} & 0 & -\frac{\sqrt{3}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{18} & 0 \\ 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{30} & 0 & -\frac{i}{10} & 0 & 0 & \frac{7\sqrt{3}i}{45} & 0 & -\frac{\sqrt{6}i}{45} & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & \frac{\sqrt{6}i}{45} & 0 & -\frac{7\sqrt{3}i}{45} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & \frac{\sqrt{5}i}{15} \end{bmatrix}$ |
| 235 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| | $\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{18} & 0 \\ 0 & \frac{\sqrt{3}}{30} & 0 & 0 & \frac{\sqrt{5}}{15} & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{30} & 0 & -\frac{1}{10} & 0 & 0 & -\frac{7\sqrt{3}}{45} & 0 & -\frac{\sqrt{6}}{45} & 0 & 0 \\ 0 & \frac{1}{10} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & -\frac{\sqrt{6}}{45} & 0 & -\frac{7\sqrt{3}}{45} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & \frac{\sqrt{5}}{15} \end{bmatrix}$ |
| 236 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{18} & 0 & 0 & 0 & \frac{\sqrt{15}i}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{18} \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 & 0 & 0 & \frac{\sqrt{6}i}{9} \\ 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 237 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| 238 | symmetry | $z$ |
| 239 | symmetry | $x$ |
| 240 | symmetry | $y$ |
| 241 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$\mathbb{T}_{3,2}^{(1,-1;a)}(E_u, 2)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{18} & 0 & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & 0 & \frac{\sqrt{3}}{18} \\ 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & \frac{\sqrt{30}}{45} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} \\ 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 \end{bmatrix}$$

$\mathbb{T}_{1}^{(1,0;a)}(A_{2u})$

$$\begin{bmatrix} 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 \end{bmatrix}$$

$\mathbb{T}_{1,1}^{(1,0;a)}(E_u)$

$$\begin{bmatrix} -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{5} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{6}i}{40} & 0 & 0 & 0 \\ -\frac{i}{5} & 0 & -\frac{2\sqrt{3}i}{15} & 0 & 0 & \frac{3i}{20} & 0 & -\frac{3\sqrt{2}i}{40} & 0 & 0 \\ 0 & -\frac{2\sqrt{3}i}{15} & 0 & -\frac{i}{5} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & -\frac{3i}{20} & 0 \\ 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{15}i}{20} \end{bmatrix}$$

$\mathbb{T}_{1,2}^{(1,0;a)}(E_u)$

$$\begin{bmatrix} \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{5} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{6}}{40} & 0 & 0 & 0 \\ \frac{1}{5} & 0 & -\frac{2\sqrt{3}}{15} & 0 & 0 & -\frac{3}{20} & 0 & -\frac{3\sqrt{2}}{40} & 0 & 0 \\ 0 & \frac{2\sqrt{3}}{15} & 0 & -\frac{1}{5} & 0 & 0 & -\frac{3\sqrt{2}}{40} & 0 & -\frac{3}{20} & 0 \\ 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{15}}{20} \end{bmatrix}$$

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 242 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(1,0;a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 \\ \frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 \end{bmatrix}$ |
| 243 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 244 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{9} & 0 & -\frac{\sqrt{3}i}{9} & 0 \\ 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & -\frac{\sqrt{10}i}{120} & 0 & \frac{i}{20} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{15} & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & \frac{7\sqrt{6}i}{360} & 0 & -\frac{\sqrt{3}i}{180} & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{5} & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & \frac{\sqrt{3}i}{180} & 0 & -\frac{7\sqrt{6}i}{360} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & -\frac{i}{20} & 0 & \frac{\sqrt{10}i}{120} \end{bmatrix}$ |
| 245 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & -\frac{\sqrt{3}}{9} & 0 \\ 0 & \frac{\sqrt{6}}{15} & 0 & 0 & \frac{\sqrt{10}}{120} & 0 & \frac{1}{20} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{15} & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & -\frac{7\sqrt{6}}{360} & 0 & -\frac{\sqrt{3}}{180} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{5} & 0 & \frac{\sqrt{6}}{15} & 0 & 0 & -\frac{\sqrt{3}}{180} & 0 & -\frac{7\sqrt{6}}{360} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{15} & 0 & 0 & 0 & 0 & \frac{1}{20} & 0 & \frac{\sqrt{10}}{120} \end{bmatrix}$ |
| 246 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & \frac{\sqrt{3}i}{36} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{180} & 0 \\ -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{180} & 0 & 0 & 0 & \frac{\sqrt{3}i}{36} \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 247 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & \frac{\sqrt{30}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & \frac{\sqrt{3}}{36} & 0 & 0 & 0 & \frac{\sqrt{15}}{180} & 0 \\ -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{180} & 0 & 0 & 0 & -\frac{\sqrt{3}}{36} \\ 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 & 0 \end{bmatrix}$ |
| 248 | symmetry | $z$ |
| | $\mathbb{T}_{1}^{(1,1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 \end{bmatrix}$ |
| 249 | symmetry | $x$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| 250 | symmetry | $y$ |
| 251 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 252 | symmetry | $\sqrt{3}yz$ |
| 253 | symmetry | $-\sqrt{3}xz$ |

$$\mathbb{T}_{1,1}^{(1,1;a)}(E_u)=\begin{bmatrix}
\frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & 0\\
-\frac{\sqrt{3}i}{10} & 0 & -\frac{i}{5} & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{40} & 0 & 0\\
0 & -\frac{i}{5} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{3}i}{20} & 0\\
0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{5}i}{20}
\end{bmatrix}$$

$$\mathbb{T}_{1,2}^{(1,1;a)}(E_u)=\begin{bmatrix}
-\frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & 0\\
\frac{\sqrt{3}}{10} & 0 & -\frac{1}{5} & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{6}}{40} & 0 & 0\\
0 & \frac{1}{5} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{3}}{20} & 0\\
0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{5}}{20}
\end{bmatrix}$$

$$\mathbb{M}_{2}^{(a)}(A_{1u})=\begin{bmatrix}
0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0\\
-\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0
\end{bmatrix}$$

$$\mathbb{M}_{2,1}^{(a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & -\frac{\sqrt{10}i}{30} & 0 & 0\\
0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & -\frac{\sqrt{10}i}{30} & 0 & -\frac{\sqrt{5}i}{15} & 0\\
0 & \frac{\sqrt{10}i}{20} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0\\
-\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & -\frac{\sqrt{5}i}{12} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & \frac{\sqrt{5}i}{12} & 0 & -\frac{\sqrt{10}i}{60} & 0\\
0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{6}i}{12}
\end{bmatrix}$$

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| 254 | symmetry | $\sqrt{3}xy$ |
| 255 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 256 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 257 | symmetry | $\sqrt{3}yz$ |

$\mathbb{M}^{(a)}_{2,2}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{5}}{15} & 0 & -\frac{\sqrt{10}}{30} & 0 & 0 \\
0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & \frac{\sqrt{10}}{30} & 0 & -\frac{\sqrt{5}}{15} & 0 \\
0 & \frac{\sqrt{10}}{20} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 \\
\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & -\frac{\sqrt{5}}{12} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & -\frac{\sqrt{5}}{12} & 0 & -\frac{\sqrt{10}}{60} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{6}}{12}
\end{bmatrix}$$

$\mathbb{M}^{(a)}_{2,1}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & \frac{i}{6} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 \\
-\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 & -\frac{i}{6} \\
0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{20} & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{15} & 0 \\
-\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{15} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} \\
0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}^{(a)}_{2,2}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{5}}{10} & \frac{1}{6} & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 \\
-\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & 0 & 0 & \frac{1}{6} \\
0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}}{20} & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & \frac{\sqrt{10}}{15} & 0 \\
-\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{15} & 0 & 0 & 0 & \frac{\sqrt{2}}{6} \\
0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}^{(1,-1;a)}_{2}(A_{1u})$

$$\begin{bmatrix}
0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 \\
\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 \\
0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\
0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 \\
0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{3}{10} & 0
\end{bmatrix}$$

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{6}i}{40} & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & \frac{i}{5} & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & \frac{i}{5} & 0 & \frac{\sqrt{2}i}{5} & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & -\frac{3\sqrt{6}i}{40} & 0 & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{i}{20} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{i}{20} & 0 \\ 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{20} \end{bmatrix}$ |
| 258 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{40} & 0 & \frac{\sqrt{6}}{40} & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & \frac{1}{5} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{1}{5} & 0 & \frac{\sqrt{2}}{5} & 0 \\ 0 & -\frac{1}{10} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{3\sqrt{6}}{40} & 0 & 0 & 0 \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{20} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{1}{20} & 0 \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{20} \end{bmatrix}$ |
| 259 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 \\ \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} \\ 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{10} & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & -\frac{i}{5} & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 260 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 \\ \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{10} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & \frac{1}{5} & 0 \\ \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \end{bmatrix}$ |
| 261 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \end{bmatrix}$ |
| 262 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 263 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 264 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & \frac{\sqrt{210}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56} \end{bmatrix}$ |
| 265 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & \frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{14}}{56} \end{bmatrix}$ |
| 266 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 267 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 268 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 269 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}^{(1,-1;a)}_{4,2}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \end{bmatrix}$ |
| 270 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}^{(1,0;a)}_{2}(A_{1u})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 \end{bmatrix}$ |
| 271 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{M}^{(1,0;a)}_{2,1}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & -\frac{2\sqrt{30}i}{45} & 0 & -\frac{2\sqrt{15}i}{45} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{40} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{2\sqrt{15}i}{45} & 0 & -\frac{2\sqrt{30}i}{45} & 0 \\ 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & -\frac{i}{12} & 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{180} & 0 & -\frac{\sqrt{30}i}{72} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & \frac{\sqrt{30}i}{72} & 0 & -\frac{\sqrt{15}i}{180} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{i}{12} \end{bmatrix}$ |
| 272 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{M}^{(1,0;a)}_{2,2}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{120} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 & \frac{2\sqrt{30}}{45} & 0 & -\frac{2\sqrt{15}}{45} & 0 & 0 \\ 0 & \frac{\sqrt{10}}{40} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{2\sqrt{15}}{45} & 0 & -\frac{2\sqrt{30}}{45} & 0 \\ 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & \frac{1}{12} & 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{180} & 0 & -\frac{\sqrt{30}}{72} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & -\frac{\sqrt{30}}{72} & 0 & -\frac{\sqrt{15}}{180} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{1}{12} \end{bmatrix}$ |
| 273 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| 274 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 275 | symmetry | $1$ |
| 276 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 277 | symmetry | $\sqrt{3}yz$ |

$$\mathbb{M}_{2,1}^{(1,0;a)}(E_u,2) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 \\ \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{45} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} \\ 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & -\frac{\sqrt{3}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{45} & 0 \\ \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{45} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{18} \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbb{M}_{2,2}^{(1,0;a)}(E_u,2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & \frac{\sqrt{30}}{45} & 0 \\ \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{45} & 0 & 0 & 0 & \frac{\sqrt{6}}{9} \\ 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & -\frac{\sqrt{3}}{18} & 0 & 0 & 0 & \frac{\sqrt{15}}{45} & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{45} & 0 & 0 & 0 & \frac{\sqrt{3}}{18} \\ 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbb{M}_{0}^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbb{M}_{2}^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{30} & 0 & 0 \\ \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 \end{bmatrix}$$

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{42}i}{30} & 0 & -\frac{\sqrt{14}i}{10} & 0 & 0 & \frac{\sqrt{42}i}{90} & 0 & \frac{\sqrt{21}i}{90} & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{10} & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & \frac{\sqrt{21}i}{90} & 0 & \frac{\sqrt{42}i}{90} & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{315} & 0 & \frac{\sqrt{42}i}{126} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & -\frac{\sqrt{42}i}{126} & 0 & \frac{\sqrt{21}i}{315} & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & -\frac{\sqrt{35}i}{105} \end{bmatrix}$ |
| 278 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{42}}{30} & 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & -\frac{\sqrt{42}}{90} & 0 & \frac{\sqrt{21}}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{10} & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & -\frac{\sqrt{21}}{90} & 0 & \frac{\sqrt{42}}{90} & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{14}}{70} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{315} & 0 & \frac{\sqrt{42}}{126} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & \frac{\sqrt{42}}{126} & 0 & \frac{\sqrt{21}}{315} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & -\frac{\sqrt{35}}{105} \end{bmatrix}$ |
| 279 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{M}_{2,1}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}i}{15} & -\frac{\sqrt{210}i}{180} & 0 & 0 & 0 & \frac{\sqrt{42}i}{180} & 0 \\ -\frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{180} & 0 & 0 & 0 & \frac{\sqrt{210}i}{180} \\ 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{30} & \frac{2\sqrt{105}i}{315} & 0 & 0 & 0 & \frac{4\sqrt{21}i}{315} & 0 \\ \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}i}{315} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{315} \\ 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 280 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{M}_{2,2}^{(1,1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}}{15} & -\frac{\sqrt{210}}{180} & 0 & 0 & 0 & -\frac{\sqrt{42}}{180} & 0 \\ -\frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{180} & 0 & 0 & 0 & -\frac{\sqrt{210}}{180} \\ 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{30} & \frac{2\sqrt{105}}{315} & 0 & 0 & 0 & -\frac{4\sqrt{21}}{315} & 0 \\ \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}}{315} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{315} \\ 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{105} & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p|\,, \langle \frac{1}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, \frac{3}{2}; p|\,, \langle \frac{3}{2}, \frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{3}{2}; p|$

ket: $= |\frac{5}{2}, \frac{5}{2}; f\rangle\,, |\frac{5}{2}, \frac{3}{2}; f\rangle\,, |\frac{5}{2}, \frac{1}{2}; f\rangle\,, |\frac{5}{2}, -\frac{1}{2}; f\rangle\,, |\frac{5}{2}, -\frac{3}{2}; f\rangle\,, |\frac{5}{2}, -\frac{5}{2}; f\rangle\,, |\frac{7}{2}, \frac{7}{2}; f\rangle\,, |\frac{7}{2}, \frac{5}{2}; f\rangle\,, |\frac{7}{2}, \frac{3}{2}; f\rangle\,, |\frac{7}{2}, \frac{1}{2}; f\rangle\,, |\frac{7}{2}, -\frac{1}{2}; f\rangle\,, |\frac{7}{2}, -\frac{3}{2}; f\rangle\,, |\frac{7}{2}, -\frac{5}{2}; f\rangle\,, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 7: (p,f) block.

| No. | multipole | matrix |
|---|---|---|
| 281 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \end{bmatrix}$ |
| 282 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{6} & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{6} & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{42} & 0 & \frac{5\sqrt{2}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{2}i}{84} & 0 & \frac{i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & -\frac{\sqrt{10}i}{14} & 0 \end{bmatrix}$ |
| 283 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{6} & 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{6} & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{42} & 0 & \frac{5\sqrt{2}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{2}}{84} & 0 & \frac{1}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & -\frac{\sqrt{10}}{14} & 0 \end{bmatrix}$ |
| 284 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{Q}_{2,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & \frac{2i}{21} & 0 & 0 & 0 & \frac{\sqrt{5}i}{21} & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} \end{bmatrix}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 285 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}^{(a)}_{2,2}(E_g, 2)$ | $\begin{bmatrix} \frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{21} & 0 & 0 & 0 & -\frac{2}{21} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & \frac{2}{21} & 0 & 0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \end{bmatrix}$ |
| 286 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |
| | $\mathbb{Q}^{(a)}_{4}(A_{1g}, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 \end{bmatrix}$ |
| 287 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}^{(a)}_{4}(A_{1g}, 2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 288 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{Q}^{(a)}_{4}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 289 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}i}{28} & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & -\frac{i}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{3i}{28} & 0 \end{bmatrix}$ |
| 290 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{56} & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & -\frac{1}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{3}{28} & 0 \end{bmatrix}$ |
| 291 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 292 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 293 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{3i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{28} & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \end{bmatrix}$ |
| 294 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{3}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \end{bmatrix}$ |
| 295 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 \end{bmatrix}$ |
| 296 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 297 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 298 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ -\frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & -\frac{3\sqrt{3}i}{28} & 0 \end{bmatrix}$ |
| 299 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\ \frac{\sqrt{2}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{2}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & -\frac{3\sqrt{3}}{28} & 0 \end{bmatrix}$ |
| 300 | symmetry | $-\frac{\sqrt{35}x^3y}{2} + \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 301 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 302 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & \frac{i}{8} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & \frac{11\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} \end{bmatrix}$ |
| 303 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & \frac{11\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & \frac{11\sqrt{2}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} \end{bmatrix}$ |
| 304 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_{2}^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 \end{bmatrix}$ |
| 305 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 306 | symmetry | $-\sqrt{3}xz$ |

$$\mathbb{Q}_{2,1}^{(1,0;a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{6}i}{18} & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{10}i}{84} & 0 & -\frac{5i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0\\
0 & \frac{5\sqrt{6}i}{252} & 0 & -\frac{25\sqrt{3}i}{252} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & -\frac{i}{7} & 0 & 0 & 0\\
0 & 0 & \frac{25\sqrt{3}i}{252} & 0 & -\frac{5\sqrt{6}i}{252} & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0\\
0 & 0 & 0 & \frac{5i}{28} & 0 & \frac{5\sqrt{10}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & -\frac{\sqrt{15}i}{21} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 307 | symmetry | $\sqrt{3}xy$ |

$$\mathbb{Q}_{2,2}^{(1,0;a)}(E_g,1)=\begin{bmatrix}
0 & -\frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{6}}{18} & 0 & \frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{5\sqrt{10}}{84} & 0 & -\frac{5}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0\\
0 & -\frac{5\sqrt{6}}{252} & 0 & -\frac{25\sqrt{3}}{252} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & -\frac{1}{7} & 0 & 0 & 0\\
0 & 0 & -\frac{25\sqrt{3}}{252} & 0 & -\frac{5\sqrt{6}}{252} & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & -\frac{\sqrt{15}}{21} & 0 & 0\\
0 & 0 & 0 & -\frac{5}{28} & 0 & \frac{5\sqrt{10}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & -\frac{\sqrt{15}}{21} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 308 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

$$\mathbb{Q}_{2,1}^{(1,0;a)}(E_g,2)=\begin{bmatrix}
-\frac{\sqrt{15}i}{18} & 0 & 0 & 0 & \frac{\sqrt{3}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}i}{18} & 0 & 0 & 0 & \frac{\sqrt{15}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0\\
-\frac{5\sqrt{30}i}{126} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{63} & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0\\
0 & -\frac{5\sqrt{6}i}{63} & 0 & 0 & 0 & -\frac{5\sqrt{30}i}{126} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0\\
0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42}
\end{bmatrix}$$

| | | |
|---|---|---|
| 309 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$$\mathbb{Q}_{2,2}^{(1,0;a)}(E_g,2)=\begin{bmatrix}
-\frac{\sqrt{15}}{18} & 0 & 0 & 0 & -\frac{\sqrt{3}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}}{18} & 0 & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{5}{42} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0\\
-\frac{5\sqrt{30}}{126} & 0 & 0 & 0 & \frac{5\sqrt{6}}{63} & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0\\
0 & -\frac{5\sqrt{6}}{63} & 0 & 0 & 0 & \frac{5\sqrt{30}}{126} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0\\
0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & \frac{\sqrt{105}}{42}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \end{bmatrix}$ |
| 310 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ \frac{3\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \\ 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 311 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ \frac{3\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} \\ 0 & -\frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 312 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{Q}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{5\sqrt{6}i}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 \\ \frac{\sqrt{30}i}{112} & 0 & \frac{5\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{15\sqrt{2}i}{112} & 0 & -\frac{15i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{15i}{56} & 0 & \frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & -\frac{\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & \frac{3\sqrt{5}i}{140} & 0 \end{bmatrix}$ |
| 313 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 314 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 315 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| 316 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| 317 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

$\mathbb{Q}_{4,2}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{5\sqrt{6}}{48} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{48} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0\\
-\frac{\sqrt{30}}{112} & 0 & \frac{5\sqrt{3}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & \frac{1}{14} & 0 & 0 & 0 & 0\\
0 & \frac{15\sqrt{2}}{112} & 0 & -\frac{15}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0\\
0 & 0 & -\frac{15}{56} & 0 & \frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{5}}{35} & 0 & 0\\
0 & 0 & 0 & \frac{5\sqrt{3}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & \frac{3\sqrt{5}}{140} & 0
\end{bmatrix}$$

$\mathbb{Q}_{4,1}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{4,2}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{4,1}^{(1,0;a)}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0\\
0 & 0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0\\
-\frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & -\frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{280} & 0 & 0\\
0 & \frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{10}i}{280} & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0\\
0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,0;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & \frac{\sqrt{10}}{280} & 0 & 0 \\ 0 & \frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & \frac{\sqrt{70}}{280} \end{bmatrix}$ |
| 318 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_{2}^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 \end{bmatrix}$ |
| 319 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{9} & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{5}i}{21} & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{3}i}{63} & 0 & \frac{5\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}i}{63} & 0 & \frac{2\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & -\frac{\sqrt{30}i}{84} & 0 \end{bmatrix}$ |
| 320 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{9} & 0 & \frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}}{21} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{3}}{63} & 0 & \frac{5\sqrt{6}}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}}{63} & 0 & \frac{2\sqrt{3}}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & -\frac{\sqrt{30}}{84} & 0 \end{bmatrix}$ |
| 321 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{18} & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & 0 & 0 \\ \frac{4\sqrt{15}i}{63} & 0 & 0 & 0 & \frac{8\sqrt{3}i}{63} & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 \\ 0 & \frac{8\sqrt{3}i}{63} & 0 & 0 & 0 & \frac{4\sqrt{15}i}{63} & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 \\ 0 & 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} \end{bmatrix}$ |
| 322 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{2}}{21} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & 0 & 0 \\ \frac{4\sqrt{15}}{63} & 0 & 0 & 0 & -\frac{8\sqrt{3}}{63} & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 \\ 0 & \frac{8\sqrt{3}}{63} & 0 & 0 & 0 & -\frac{4\sqrt{15}}{63} & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 \\ 0 & 0 & \frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} \end{bmatrix}$ |
| 323 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 324 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \end{bmatrix}$ |
| 325 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 326 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |
| 327 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 328 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| 329 | symmetry | $\sqrt{15}xyz$ |

$\mathbb{G}_3^{(a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\
\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\
\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\
0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_{3,1}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}i}{21} & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\
\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}i}{24} & 0 & \frac{\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{35}i}{28} & 0
\end{bmatrix}$$

$\mathbb{G}_{3,2}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{21} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\
-\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}}{84} & 0 & \frac{\sqrt{14}}{24} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{35}}{28} & 0
\end{bmatrix}$$

$\mathbb{G}_{3,1}^{(a)}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{42} & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\
-\frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 330 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 331 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| 332 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 333 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{G}_{3,2}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
-\frac{\sqrt{14}}{42} & 0 & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\
0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\
-\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & \frac{5\sqrt{7}}{84} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8}
\end{bmatrix}$$

$\mathbb{G}_3^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 \\
\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{42} & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 \\
\frac{\sqrt{10}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\
0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_3^{(1,-1;a)}(A_{2g}, 1)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{21} & 0 & 0 & 0 \\
0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{10}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{10}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_3^{(1,-1;a)}(A_{2g}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 \\
-\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 \\
-\frac{\sqrt{10}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} \\
0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 334 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$$\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{15}i}{126} & 0 & -\frac{\sqrt{30}i}{126} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{30}i}{126} & 0 & \frac{\sqrt{15}i}{126} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0\\
-\frac{\sqrt{2}i}{42} & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{30}i}{90} & 0 & -\frac{\sqrt{15}i}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{15}i}{315} & 0 & -\frac{\sqrt{30}i}{90} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & \frac{\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & -\frac{5\sqrt{3}i}{42} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 335 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$$\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,1)=\begin{bmatrix}
0 & -\frac{\sqrt{15}}{126} & 0 & -\frac{\sqrt{30}}{126} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{30}}{126} & 0 & \frac{\sqrt{15}}{126} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0\\
\frac{\sqrt{2}}{42} & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{30}}{90} & 0 & -\frac{\sqrt{15}}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{15}}{315} & 0 & -\frac{\sqrt{30}}{90} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & \frac{\sqrt{2}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & -\frac{5\sqrt{3}}{42} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 336 | symmetry | $\sqrt{15}xyz$ |

$$\mathbb{G}_{3,1}^{(1,-1;a)}(E_g,2)=\begin{bmatrix}
\frac{\sqrt{30}i}{252} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{252} & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0\\
0 & -\frac{5\sqrt{6}i}{252} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{252} & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0\\
0 & 0 & 0 & -\frac{\sqrt{2}i}{21} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0\\
\frac{\sqrt{15}i}{63} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{63} & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0\\
0 & -\frac{\sqrt{3}i}{63} & 0 & 0 & 0 & \frac{\sqrt{15}i}{63} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0\\
0 & 0 & -\frac{\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84}
\end{bmatrix}$$

| | | |
|---|---|---|
| 337 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$$\mathbb{G}_{3,2}^{(1,-1;a)}(E_g,2)=\begin{bmatrix}
\frac{\sqrt{30}}{252} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{252} & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0\\
0 & -\frac{5\sqrt{6}}{252} & 0 & 0 & 0 & \frac{\sqrt{30}}{252} & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0\\
0 & 0 & 0 & \frac{\sqrt{2}}{21} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0\\
\frac{\sqrt{15}}{63} & 0 & 0 & 0 & \frac{\sqrt{3}}{63} & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0\\
0 & -\frac{\sqrt{3}}{63} & 0 & 0 & 0 & -\frac{\sqrt{15}}{63} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0\\
0 & 0 & -\frac{\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 338 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$ |
| 339 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 340 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 341 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-\frac{1}{2}&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 342 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{30}i}{60}&0&-\frac{\sqrt{6}i}{12}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{30}i}{20}&0&\frac{\sqrt{2}i}{4}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{2}i}{4}&0&-\frac{\sqrt{30}i}{20}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{6}i}{12}&0&\frac{\sqrt{30}i}{60}&0 \end{bmatrix}$ |
| 343 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&-\frac{\sqrt{30}}{60}&0&-\frac{\sqrt{6}}{12}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&\frac{\sqrt{30}}{20}&0&\frac{\sqrt{2}}{4}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2}}{4}&0&-\frac{\sqrt{30}}{20}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{6}}{12}&0&\frac{\sqrt{30}}{60}&0 \end{bmatrix}$ |
| 344 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{70}i}{20}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{30}i}{20} \\ 0&0&0&0&0&0&\frac{\sqrt{30}i}{20}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&-\frac{\sqrt{70}i}{20}&0&0&0&0&0&0 \end{bmatrix}$ |
| 345 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 346 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{G}_{5,1}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{40} & 0 & 0 & 0 & \frac{\sqrt{70}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} \end{bmatrix}$ |
| 347 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |
| | $\mathbb{G}_{5,2}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{40} & 0 & 0 & 0 & \frac{\sqrt{210}}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{40} & 0 & 0 & 0 & -\frac{\sqrt{70}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & \frac{\sqrt{30}}{120} \end{bmatrix}$ |
| 348 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_{3}^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}}{336} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{25\sqrt{14}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 349 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \end{bmatrix}$ |
| 350 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{G}_3^{(1,0;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & \frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}i}{336} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{25\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 351 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{63} & 0 & \frac{\sqrt{42}i}{63} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{336} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{63} & 0 & -\frac{\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 \\ \frac{5\sqrt{70}i}{336} & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}i}{144} & 0 & \frac{5\sqrt{21}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{21}i}{504} & 0 & \frac{5\sqrt{42}i}{144} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & -\frac{5\sqrt{70}i}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{105}i}{84} & 0 \end{bmatrix}$ |
| 352 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{21}}{63} & 0 & \frac{\sqrt{42}}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{336} & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{63} & 0 & -\frac{\sqrt{21}}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 \\ -\frac{5\sqrt{70}}{336} & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{42}}{144} & 0 & \frac{5\sqrt{21}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}}{504} & 0 & \frac{5\sqrt{42}}{144} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & -\frac{5\sqrt{70}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{105}}{84} & 0 \end{bmatrix}$ |
| 353 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 354 | symmetry | $\sqrt{15}xyz$ |
| 355 | symmetry | $z$ |
| 356 | symmetry | $x$ |
| 357 | symmetry | $y$ |

$\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{42}i}{126} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{126} & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 \\
0 & \frac{\sqrt{210}i}{126} & 0 & 0 & 0 & \frac{\sqrt{42}i}{126} & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 \\
0 & 0 & 0 & \frac{5\sqrt{70}i}{168} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\
-\frac{25\sqrt{21}i}{504} & 0 & 0 & 0 & \frac{5\sqrt{105}i}{504} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\
0 & \frac{5\sqrt{105}i}{504} & 0 & 0 & 0 & -\frac{25\sqrt{21}i}{504} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\
0 & 0 & \frac{5\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24}
\end{bmatrix}$$

$\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{42}}{126} & 0 & 0 & 0 & \frac{\sqrt{210}}{126} & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 \\
0 & \frac{\sqrt{210}}{126} & 0 & 0 & 0 & -\frac{\sqrt{42}}{126} & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{70}}{168} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\
-\frac{25\sqrt{21}}{504} & 0 & 0 & 0 & -\frac{5\sqrt{105}}{504} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\
0 & \frac{5\sqrt{105}}{504} & 0 & 0 & 0 & \frac{25\sqrt{21}}{504} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\
0 & 0 & \frac{5\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24}
\end{bmatrix}$$

$\mathbb{G}_{1}^{(1,1;a)}(A_{2g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_{1,1}^{(1,1;a)}(E_g)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 358 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3}{7} & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 \\ -\frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}}{112} & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ \frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 359 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 \end{bmatrix}$ |
| 360 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{G}_{3}^{(1,1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3i}{7} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{112} & 0 \\ \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}i}{112} & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 \\ -\frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 361 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 362 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 363 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| 364 | symmetry | $\sqrt{15}xyz$ |
| 365 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{G}_{3,1}^{(1,1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{112} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{6}i}{7} & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{112} & 0 & 0\\
-\frac{3\sqrt{10}i}{112} & 0 & \frac{9i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{3}i}{56} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{9i}{56} & 0 & \frac{3\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{15}i}{84} & 0
\end{bmatrix}$$

$\mathbb{G}_{3,2}^{(1,1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{3}}{7} & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{2}}{112} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{6}}{7} & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0\\
\frac{3\sqrt{10}}{112} & 0 & \frac{9}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{9}{56} & 0 & \frac{3\sqrt{10}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{15}}{84} & 0
\end{bmatrix}$$

$\mathbb{G}_{3,1}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{14} & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0\\
0 & \frac{\sqrt{30}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{3i}{56} & 0\\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0\\
\frac{5\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0\\
0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & \frac{5\sqrt{3}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0\\
0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168}
\end{bmatrix}$$

$\mathbb{G}_{3,2}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{30}}{14} & 0 & 0 & \frac{3}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0\\
0 & \frac{\sqrt{30}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{3}{56} & 0\\
0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0\\
\frac{5\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0\\
0 & -\frac{\sqrt{15}}{56} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0\\
0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{42}}{168}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \end{bmatrix}$ |
| 366 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}_{2,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{6} & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{6} & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{42} & 0 & -\frac{5\sqrt{2}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{2}}{84} & 0 & -\frac{1}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & \frac{\sqrt{10}}{14} & 0 \end{bmatrix}$ |
| 367 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{T}_{2,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{6} & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{6} & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{42} & 0 & \frac{5\sqrt{2}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{2}i}{84} & 0 & \frac{i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & -\frac{\sqrt{10}i}{14} & 0 \end{bmatrix}$ |
| 368 | symmetry | $\sqrt{3}xy$ |
| | $\mathbb{T}_{2,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{21} & 0 & 0 & 0 & -\frac{2}{21} & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & -\frac{2}{21} & 0 & 0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \end{bmatrix}$ |
| 369 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{2,2}(E_g,2)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & -\frac{2i}{21} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & \frac{2i}{21} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{21} & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} \end{bmatrix}$ |
| 370 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{T}^{(a)}_{4}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 \end{bmatrix}$ |
| 371 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}^{(a)}_{4}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 372 | symmetry | $\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{T}^{(a)}_{4}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 373 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{4,1}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 \\ \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{56} & 0 & -\frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & \frac{1}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{3}{28} & 0 \end{bmatrix}$ |
| 374 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| | $\mathbb{T}^{(a)}_{4,2}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{56} & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & -\frac{i}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{3i}{28} & 0 \end{bmatrix}$ |
| 375 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{T}^{(a)}_{4,1}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 376 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{T}^{(a)}_{4,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 377 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 378 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$$\mathbb{T}^{(a)}_{4,1}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{3}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 \\
0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56}
\end{bmatrix}$$

| No. | multipole | matrix |
|---|---|---|
| 379 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

$$\mathbb{T}^{(a)}_{4,2}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 \\
0 & -\frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 \\
0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56}
\end{bmatrix}$$

| No. | multipole | matrix |
|---|---|---|
| 380 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

$$\mathbb{T}^{(1,-1;a)}_{4}(A_{1g},1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\
0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0
\end{bmatrix}$$

| No. | multipole | matrix |
|---|---|---|
| 381 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$$\mathbb{T}^{(1,-1;a)}_{4}(A_{1g},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\
-\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\
0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,-1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 382 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{T}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\ \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{112} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{30}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & \frac{3\sqrt{3}}{28} & 0 \end{bmatrix}$ |
| 383 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |
| | $\mathbb{T}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\ \frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & -\frac{3\sqrt{3}i}{28} & 0 \end{bmatrix}$ |
| 384 | symmetry | $-\frac{\sqrt{35}x^3y}{2} + \frac{\sqrt{35}xy^3}{2}$ |
| | $\mathbb{T}_{4,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 385 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,-1;a)}_{4,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 386 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{T}^{(1,-1;a)}_{4,1}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & -\frac{1}{8} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & -\frac{11\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & \frac{11\sqrt{2}}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} \end{bmatrix}$ |
| 387 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |
| | $\mathbb{T}^{(1,-1;a)}_{4,2}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & -\frac{i}{8} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & \frac{11\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{2}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} \end{bmatrix}$ |
| 388 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}^{(1,0;a)}_{2}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 \end{bmatrix}$ |
| 389 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 390 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{T}^{(1,0;a)}_{2,1}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{3}}{9} & 0 & \frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{18} & 0 & \frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{10}}{84} & 0 & -\frac{5}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{6}}{252} & 0 & -\frac{25\sqrt{3}}{252} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & -\frac{1}{7} & 0 & 0 & 0 \\
0 & 0 & \frac{25\sqrt{3}}{252} & 0 & -\frac{5\sqrt{6}}{252} & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 \\
0 & 0 & 0 & \frac{5}{28} & 0 & \frac{5\sqrt{10}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & -\frac{\sqrt{15}}{21} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 391 | symmetry | $\sqrt{3}xy$ |

$\mathbb{T}^{(1,0;a)}_{2,2}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{3}i}{9} & 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{18} & 0 & -\frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{10}i}{84} & 0 & \frac{5i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{6}i}{252} & 0 & \frac{25\sqrt{3}i}{252} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & \frac{i}{7} & 0 & 0 & 0 \\
0 & 0 & \frac{25\sqrt{3}i}{252} & 0 & \frac{5\sqrt{6}i}{252} & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 \\
0 & 0 & 0 & \frac{5i}{28} & 0 & -\frac{5\sqrt{10}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & \frac{\sqrt{15}i}{21} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 392 | symmetry | $\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$ |

$\mathbb{T}^{(1,0;a)}_{2,1}(E_g,2)$

$$\begin{bmatrix}
-\frac{\sqrt{15}}{18} & 0 & 0 & 0 & \frac{\sqrt{3}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{18} & 0 & 0 & 0 & \frac{\sqrt{15}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \\
-\frac{5\sqrt{30}}{126} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{63} & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 \\
0 & -\frac{5\sqrt{6}}{63} & 0 & 0 & 0 & -\frac{5\sqrt{30}}{126} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 \\
0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & -\frac{\sqrt{105}}{42}
\end{bmatrix}$$

| | | |
|---|---|---|
| 393 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$\mathbb{T}^{(1,0;a)}_{2,2}(E_g,2)$

$$\begin{bmatrix}
\frac{\sqrt{15}i}{18} & 0 & 0 & 0 & \frac{\sqrt{3}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{18} & 0 & 0 & 0 & \frac{\sqrt{15}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 \\
\frac{5\sqrt{30}i}{126} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{63} & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 \\
0 & \frac{5\sqrt{6}i}{63} & 0 & 0 & 0 & -\frac{5\sqrt{30}i}{126} & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 \\
0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \end{bmatrix}$ |
| 394 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{112} & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ -\frac{3\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} \\ 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 395 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 \\ \frac{3\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} \\ 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 396 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{5\sqrt{6}}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{48} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 \\ \frac{\sqrt{30}}{112} & 0 & \frac{5\sqrt{3}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{15\sqrt{2}}{112} & 0 & -\frac{15}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{15}{56} & 0 & \frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{3\sqrt{5}}{140} & 0 \end{bmatrix}$ |
| 397 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 398 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 399 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |
| 400 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| 401 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

$\mathbb{T}^{(1,0;a)}_{4,2}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{5\sqrt{6}i}{48} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0\\
\frac{\sqrt{30}i}{112} & 0 & -\frac{5\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0\\
0 & -\frac{15\sqrt{2}i}{112} & 0 & \frac{15i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0\\
0 & 0 & \frac{15i}{56} & 0 & -\frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{5}i}{35} & 0 & 0\\
0 & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{3\sqrt{5}i}{140} & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{4,1}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{4,2}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{4,1}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0\\
0 & 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0\\
-\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & \frac{15}{56} & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & -\frac{\sqrt{10}}{280} & 0 & 0\\
0 & \frac{15}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{10}}{280} & 0 & 0 & 0 & \frac{11\sqrt{30}}{840} & 0\\
0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{280}
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 \\ \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{280} & 0 & 0 \\ 0 & -\frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & -\frac{\sqrt{10}i}{280} & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} \end{bmatrix}$ |
| 402 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}^{(1,1;a)}_{2}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 \end{bmatrix}$ |
| 403 | symmetry | $\sqrt{3}yz$ |
| | $\mathbb{T}^{(1,1;a)}_{2,1}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{9} & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}}{21} & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{3}}{63} & 0 & -\frac{5\sqrt{6}}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}}{63} & 0 & -\frac{2\sqrt{3}}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & \frac{\sqrt{30}}{84} & 0 \end{bmatrix}$ |
| 404 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{T}^{(1,1;a)}_{2,2}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{9} & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}i}{21} & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{3}i}{63} & 0 & \frac{5\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{63} & 0 & \frac{2\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & -\frac{\sqrt{30}i}{84} & 0 \end{bmatrix}$ |
| 405 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 406 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 407 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 408 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| 409 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{T}_{2,1}^{(1,1;a)}(E_g, 2)$

$$\begin{bmatrix}
\frac{\sqrt{30}}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{18} & 0 & 0 & 0 & -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{2}}{21} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & 0 & 0 \\
-\frac{4\sqrt{15}}{63} & 0 & 0 & 0 & -\frac{8\sqrt{3}}{63} & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 \\
0 & -\frac{8\sqrt{3}}{63} & 0 & 0 & 0 & -\frac{4\sqrt{15}}{63} & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 \\
0 & 0 & -\frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{168}
\end{bmatrix}$$

$\mathbb{T}_{2,2}^{(1,1;a)}(E_g, 2)$

$$\begin{bmatrix}
-\frac{\sqrt{30}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{2}i}{21} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{168} & 0 & 0 & 0 \\
\frac{4\sqrt{15}i}{63} & 0 & 0 & 0 & -\frac{8\sqrt{3}i}{63} & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 \\
0 & \frac{8\sqrt{3}i}{63} & 0 & 0 & 0 & -\frac{4\sqrt{15}i}{63} & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 \\
0 & 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{168} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168}
\end{bmatrix}$$

$\mathbb{M}_3^{(a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 \\
\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\
\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\
0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_3^{(a)}(A_{2g}, 1)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 \\ \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 410 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| | $\mathbb{M}_{3,1}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{21} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{24} & 0 & \frac{\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & \frac{\sqrt{14}}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{35}}{28} & 0 \end{bmatrix}$ |
| 411 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{M}_{3,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{7}i}{21} & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\ \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{24} & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{35}i}{28} & 0 \end{bmatrix}$ |
| 412 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3,1}^{(a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ -\frac{5\sqrt{7}}{84} & 0 & 0 & 0 & \frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{7}}{84} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \end{bmatrix}$ |
| 413 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 414 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 415 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| 416 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 417 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{M}_{3,2}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{42} & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\
\frac{5\sqrt{7}i}{84} & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{84} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8}
\end{bmatrix}$$

$\mathbb{M}_{3}^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{42} & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 \\
-\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 \\
-\frac{\sqrt{10}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \\
0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{3}^{(1,-1;a)}(A_{2g}, 1)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 \\
0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{10}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{10}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0
\end{bmatrix}$$

$\mathbb{M}_{3}^{(1,-1;a)}(A_{2g}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 \\
-\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 \\
-\frac{\sqrt{10}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\
0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{126} & 0 & -\frac{\sqrt{30}}{126} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{126} & 0 & \frac{\sqrt{15}}{126} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 \\ -\frac{\sqrt{2}}{42} & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{90} & 0 & -\frac{\sqrt{15}}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{315} & 0 & -\frac{\sqrt{30}}{90} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & \frac{\sqrt{2}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & -\frac{5\sqrt{3}}{42} & 0 \end{bmatrix}$ |
| 418 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}i}{126} & 0 & \frac{\sqrt{30}i}{126} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{126} & 0 & -\frac{\sqrt{15}i}{126} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 \\ -\frac{\sqrt{2}i}{42} & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{90} & 0 & \frac{\sqrt{15}i}{315} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{315} & 0 & \frac{\sqrt{30}i}{90} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & -\frac{\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & \frac{5\sqrt{3}i}{42} & 0 \end{bmatrix}$ |
| 419 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} \frac{\sqrt{30}}{252} & 0 & 0 & 0 & \frac{5\sqrt{6}}{252} & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 \\ 0 & -\frac{5\sqrt{6}}{252} & 0 & 0 & 0 & -\frac{\sqrt{30}}{252} & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{21} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{63} & 0 & 0 & 0 & -\frac{\sqrt{3}}{63} & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{63} & 0 & 0 & 0 & \frac{\sqrt{15}}{63} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & \frac{\sqrt{210}}{84} \end{bmatrix}$ |
| 420 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{30}i}{252} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{252} & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{252} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{252} & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{21} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{63} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{63} & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{63} & 0 & 0 & 0 & \frac{\sqrt{15}i}{63} & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} \end{bmatrix}$ |
| 421 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 422 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \end{bmatrix}$ |
| 423 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 424 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 425 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{i}{2} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&\frac{i}{2}&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 426 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{30}}{60}&0&-\frac{\sqrt{6}}{12}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{30}}{20}&0&\frac{\sqrt{2}}{4}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{2}}{4}&0&-\frac{\sqrt{30}}{20}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{6}}{12}&0&\frac{\sqrt{30}}{60}&0&0 \end{bmatrix}$ |
| 427 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{30}i}{60}&0&\frac{\sqrt{6}i}{12}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{30}i}{20}&0&-\frac{\sqrt{2}i}{4}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{2}i}{4}&0&\frac{\sqrt{30}i}{20}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{6}i}{12}&0&-\frac{\sqrt{30}i}{60}&0&0 \end{bmatrix}$ |
| 428 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{70}}{20}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{30}}{20} \\ 0&0&0&0&0&0&\frac{\sqrt{30}}{20}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&-\frac{\sqrt{70}}{20}&0&0&0&0&0&0 \end{bmatrix}$ |
| 429 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 430 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{40} & 0 & 0 & 0 & -\frac{\sqrt{210}}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{40} & 0 & 0 & 0 & \frac{\sqrt{70}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} \end{bmatrix}$ |
| 431 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{40} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{40} & 0 & 0 & 0 & \frac{\sqrt{70}i}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} \end{bmatrix}$ |
| 432 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_{3}^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & -\frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}i}{336} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{25\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 433 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 434 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 435 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| 436 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |
| 437 | symmetry | $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{M}_3^{(1,0;a)}(A_{2g},1)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 \\
0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_3^{(1,0;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 \\
-\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{48} \\
0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{14}}{336} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\
-\frac{25\sqrt{14}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\
0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{3,1}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{21}}{63} & 0 & -\frac{\sqrt{42}}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{336} & 0 & \frac{5\sqrt{14}}{112} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{63} & 0 & \frac{\sqrt{21}}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{210}}{336} & 0 & 0 \\
-\frac{5\sqrt{70}}{336} & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{42}}{144} & 0 & -\frac{5\sqrt{21}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{5\sqrt{21}}{504} & 0 & -\frac{5\sqrt{42}}{144} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & \frac{5\sqrt{70}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{105}}{84} & 0
\end{bmatrix}$$

$\mathbb{M}_{3,2}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{21}i}{63} & 0 & \frac{\sqrt{42}i}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{63} & 0 & -\frac{\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 \\
-\frac{5\sqrt{70}i}{336} & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{42}i}{144} & 0 & \frac{5\sqrt{21}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{5\sqrt{21}i}{504} & 0 & \frac{5\sqrt{42}i}{144} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & -\frac{5\sqrt{70}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{105}i}{84} & 0
\end{bmatrix}$$

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} \frac{\sqrt{42}}{126} & 0 & 0 & 0 & \frac{\sqrt{210}}{126} & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{126} & 0 & 0 & 0 & -\frac{\sqrt{42}}{126} & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{70}}{168} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ \frac{25\sqrt{21}}{504} & 0 & 0 & 0 & -\frac{5\sqrt{105}}{504} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ 0 & -\frac{5\sqrt{105}}{504} & 0 & 0 & 0 & \frac{25\sqrt{21}}{504} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & -\frac{5\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \end{bmatrix}$ |
| 438 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{42}i}{126} & 0 & 0 & 0 & \frac{\sqrt{210}i}{126} & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{126} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{126} & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{70}i}{168} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ -\frac{25\sqrt{21}i}{504} & 0 & 0 & 0 & -\frac{5\sqrt{105}i}{504} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 \\ 0 & \frac{5\sqrt{105}i}{504} & 0 & 0 & 0 & \frac{25\sqrt{21}i}{504} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & \frac{5\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \end{bmatrix}$ |
| 439 | symmetry | $z$ |
| | $\mathbb{M}_{1}^{(1,1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 440 | symmetry | $x$ |
| | $\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 441 | symmetry | $y$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{1,2}^{(1,1;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{20} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 442 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{3i}{7} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 \\ \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{2}i}{112} & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 \\ -\frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 443 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & \frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 \end{bmatrix}$ |
| 444 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_{3}^{(1,1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3}{7} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 \\ \frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}}{112} & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ -\frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & -\frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 445 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{7} & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{2}}{112} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{7} & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 \\ -\frac{3\sqrt{10}}{112} & 0 & \frac{9}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9}{56} & 0 & \frac{3\sqrt{10}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{15}}{84} & 0 \end{bmatrix}$ |
| 446 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}i}{7} & 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{2}i}{112} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{7} & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 \\ -\frac{3\sqrt{10}i}{112} & 0 & -\frac{9i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9i}{56} & 0 & -\frac{3\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{15}i}{84} & 0 \end{bmatrix}$ |
| 447 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} & 0 & 0 & \frac{3}{56} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{14} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{3}{56} & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 \\ \frac{5\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & 0 & \frac{5\sqrt{3}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} \end{bmatrix}$ |
| 448 | symmetry | $\sqrt{15}xyz$ |
| | $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)$ | $\begin{bmatrix} \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{14} & 0 & 0 & -\frac{3i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{3i}{56} & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & \frac{5\sqrt{3}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} \end{bmatrix}$ |

bra: $=\langle\frac{3}{2},\frac{3}{2};d|\,,\langle\frac{3}{2},\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{5}{2};d|\,,\langle\frac{5}{2},\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},-\frac{5}{2};d|$
ket: $=|\frac{3}{2},\frac{3}{2};d\rangle\,,|\frac{3}{2},\frac{1}{2};d\rangle\,,|\frac{3}{2},-\frac{1}{2};d\rangle\,,|\frac{3}{2},-\frac{3}{2};d\rangle\,,|\frac{5}{2},\frac{5}{2};d\rangle\,,|\frac{5}{2},\frac{3}{2};d\rangle\,,|\frac{5}{2},\frac{1}{2};d\rangle\,,|\frac{5}{2},-\frac{1}{2};d\rangle\,,|\frac{5}{2},-\frac{3}{2};d\rangle\,,|\frac{5}{2},-\frac{5}{2};d\rangle$

Table 8: (d,d) block.

| No. | multipole | matrix |
|---|---|---|
| 449 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A_{1g})$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 450 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(a)}(A_{1g})$ | $\begin{bmatrix} -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \end{bmatrix}$ |
| 451 | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|

452 symmetry $-\sqrt{3}xz$

$$\mathbb{Q}_{2,1}^{(a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{7}i}{10} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0\\
-\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{7}i}{70} & 0\\
0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & -\frac{\sqrt{105}i}{70}\\
-\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{7}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0\\
-\frac{3\sqrt{42}i}{140} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0\\
0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & -\frac{\sqrt{105}i}{35}\\
0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0
\end{bmatrix}$$

453 symmetry $\sqrt{3}xy$

$$\mathbb{Q}_{2,2}^{(a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{7}}{10} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0\\
\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{14}}{28} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{7}}{70} & 0\\
0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & -\frac{\sqrt{105}}{70}\\
-\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{7}}{70} & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & \frac{\sqrt{42}}{35} & 0 & 0 & 0\\
\frac{3\sqrt{42}}{140} & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{14}}{28} & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0\\
0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & -\frac{\sqrt{105}}{35}\\
0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 454 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 455 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

$\mathbb{Q}_{2,1}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{10} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 \\
-\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} \\
0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & -\frac{\sqrt{210}i}{70} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 \\
-\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} \\
0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{2,2}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}}{10} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 \\
-\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \\
0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{70} & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 \\
-\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\
0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 456 | $\mathbb{Q}_4^{(a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} \end{bmatrix}$ |
| 457 | $\mathbb{Q}_4^{(a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 458 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |
| 459 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |

$\mathbb{Q}_4^{(a)}(A_{2g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\
0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{4,1}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & -\frac{\sqrt{42}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{70}i}{140} \\
\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{14} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\
0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{70}i}{70} \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 460 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 461 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |

$\mathbb{Q}_{4,2}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{21}}{14} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & -\frac{\sqrt{42}}{28} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{70}}{140}\\
\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0\\
-\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{21}}{14} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0\\
0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{70}}{70}\\
0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0
\end{bmatrix}$$

$\mathbb{Q}_{4,1}^{(a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 462 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{Q}_{4,1}^{(a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} \\ 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 \end{bmatrix}$ |
| 463 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 464 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 465 | symmetry | $\sqrt{3}yz$ |

$\mathbb{Q}_{4,2}^{(a)}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{14} & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\
-\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} \\
0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_2^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
-\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15}
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{30}i}{100} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} & 0 & -\frac{3\sqrt{2}i}{20} \\ -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{100} & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & -\frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\ -\frac{9\sqrt{5}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{5} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 \end{bmatrix}$ |
| 466 | symmetry | $-\sqrt{3}xz$ |
| | $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{50} & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & \frac{9\sqrt{5}}{100} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{100} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{30}}{100} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{100} & 0 & -\frac{3\sqrt{2}}{20} \\ -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{100} & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ \frac{9\sqrt{5}}{100} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & \frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{\sqrt{2}}{5} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 \end{bmatrix}$ |
| 467 | symmetry | $\sqrt{3}xy$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 468 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 469 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

$$\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{50} & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} \\ 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}i}{50} & \frac{i}{5} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 \\ -\frac{3\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 & -\frac{i}{5} \\ 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 \end{bmatrix}$$

$$\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} & 0 \\ -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} \\ 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & \frac{1}{5} & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 \\ -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & \frac{1}{5} \\ 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 470 | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)$ | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| 471 | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)$ | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 \end{bmatrix}$ |
| 472 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{70}i}{280} \\ \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & -\frac{\sqrt{70}i}{35} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 \end{bmatrix}$ |
| 473 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 474 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 475 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |

$$\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)=\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{42}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{280} \\ \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & -\frac{\sqrt{70}}{35} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 \end{bmatrix}$$

$$\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)=\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 476 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |
| | $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 \\ -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 \end{bmatrix}$ |
| 477 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 478 | symmetry | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 \end{bmatrix}$ |
| 479 | symmetry | $\mathbb{Q}_{0}^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} \end{bmatrix}$ |
| 480 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 480 | symmetry | $\sqrt{3}yz$ |
| 481 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{Q}_2^{(1,1;a)}(A_{1g})$

$$\begin{bmatrix}
\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \\
-\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70}
\end{bmatrix}$$

$\mathbb{Q}_{2,1}^{(1,1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 \\
\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{25} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{2\sqrt{105}i}{525} & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{175} & 0 & -\frac{2\sqrt{7}i}{35} \\
-\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{105}i}{525} & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{70}i}{350} & 0 & 0 & 0 \\
-\frac{3\sqrt{70}i}{175} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 \\
0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{350} & 0 & -\frac{3\sqrt{7}i}{70} \\
0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 482 | symmetry | $\sqrt{3}xy$ |
| 483 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$\mathbb{Q}_{2,2}^{(1,1;a)}(E_g, 1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{105}}{25} & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 & 0 \\
-\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{525} & 0 & \frac{\sqrt{210}}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & \frac{2\sqrt{105}}{525} & 0 \\
0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 & -\frac{2\sqrt{7}}{35} \\
-\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{105}}{525} & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & \frac{3\sqrt{70}}{350} & 0 & 0 & 0 \\
\frac{3\sqrt{70}}{175} & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & \frac{3\sqrt{70}}{350} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}}{105} & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{350} & 0 \\
0 & 0 & \frac{2\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{350} & 0 & -\frac{3\sqrt{7}}{70} \\
0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0
\end{bmatrix}$$

$\mathbb{Q}_{2,1}^{(1,1;a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{25} & \frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 \\
\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & \frac{4\sqrt{21}i}{105} \\
0 & \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{70}i}{175} & -\frac{3\sqrt{14}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 \\
-\frac{2\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} \\
0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{140} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 484 | symmetry | $z$ |
| 485 | symmetry | $x$ |

$\mathbb{Q}_{2,2}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{25} & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & -\frac{8\sqrt{105}}{525} & 0 \\
\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & -\frac{4\sqrt{21}}{105} \\
0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 \\
0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{70}}{175} & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 \\
-\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} \\
0 & -\frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_1^{(1,0;a)}(A_{2g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 486 | symmetry | $\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 487 | symmetry | $y$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{2}}{4} \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{20} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | | | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 488 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(1,0;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 489 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 490 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |

$$\mathbb{G}_3^{(1,0;a)}(A_{2g},2)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8}\\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)=\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{3\sqrt{5}i}{20} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{30}i}{120} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & -\frac{7\sqrt{30}i}{120} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{20} & 0 & \frac{\sqrt{2}i}{8}\\
\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{7\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{15}i}{60} & 0 & \frac{3\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{7\sqrt{30}i}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 491 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 492 | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{3\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{30}}{120} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & -\frac{7\sqrt{30}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{8} \\ \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{5}}{20} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{60} & 0 & \frac{3\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{7\sqrt{30}}{120} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 493 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 494 | symmetry | $\sqrt{15}xyz$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 494 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 495 | symmetry | $\sqrt{3}yz$ |

$\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_2^{(1,0;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 496 | symmetry | $-\sqrt{3}xz$ |
| 497 | symmetry | $\sqrt{3}xy$ |

$\mathbb{T}_{2,1}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{84} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & -\frac{\sqrt{70}}{28} \\
\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{7}}{28} & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{21}}{84} & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_{2,2}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & -\frac{\sqrt{42}i}{84} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & \frac{\sqrt{70}i}{28} \\
-\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{7}i}{28} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{21}i}{84} & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 498 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 499 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 500 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| 501 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$\mathbb{T}_4^{(1,0;a)}(A_{1g},1)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\mathbb{T}_4^{(1,0;a)}(A_{1g},2)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 502 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{105}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & -\frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{14}}{56} \\ -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 503 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 504 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 505 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |

$\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & \frac{\sqrt{210}i}{56} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56}\\
\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{105}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 506 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| | $\mathbb{T}^{(1,0;a)}_{4,1}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 507 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 508 | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 509 | $\mathbb{M}^{(a)}_1(A_{2g})$ | $\begin{bmatrix} \frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$ |

| No. | multipole | matrix |
|---|---|---|
| 509 | symmetry | $z$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 510 | symmetry | $\mathbb{M}^{(a)}_{1,1}(E_g)$ $=$ $\begin{bmatrix} 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{1}{10} & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{50} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & \frac{\sqrt{30}}{100} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & \frac{\sqrt{15}}{50} & 0 \\ 0 & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & \frac{1}{10} \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & \frac{1}{5} & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{100} & 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{100} & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{1}{5} \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 \end{bmatrix}$ |
| | | $y$ |
| | | $\mathbb{M}^{(a)}_{1,2}(E_g)$ $=$ $\begin{bmatrix} 0 & -\frac{3\sqrt{15}i}{50} & 0 & 0 & -\frac{i}{10} & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & 0 \\ \frac{3\sqrt{15}i}{50} & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 & -\frac{\sqrt{15}i}{50} & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{25} & 0 & -\frac{3\sqrt{15}i}{50} & 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & -\frac{\sqrt{15}i}{50} & 0 \\ 0 & 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & -\frac{i}{10} \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{50} & 0 & 0 & \frac{i}{5} & 0 & -\frac{2\sqrt{10}i}{25} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{100} & 0 & \frac{\sqrt{30}i}{100} & 0 & 0 & \frac{2\sqrt{10}i}{25} & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & -\frac{2\sqrt{10}i}{25} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{25} & 0 & -\frac{i}{5} \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 \end{bmatrix}$ |
| 511 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 512 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| 513 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{M}_3^{(a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_3^{(a)}(A_{2g},1)$

$$\begin{bmatrix}
-\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10}
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 514 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |
| 515 | symmetry | $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$ |

$$\mathbb{M}_3^{(a)}(A_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbb{M}_{3,1}^{(a)}(E_g,1) = \begin{bmatrix} 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{3\sqrt{15}}{50} & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{25} & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & -\frac{7\sqrt{10}}{100} & 0 & \frac{\sqrt{5}}{50} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & \frac{7\sqrt{10}}{100} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{50} & 0 & -\frac{\sqrt{6}}{20} \\ \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{10}}{100} & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & 0 \\ -\frac{3\sqrt{15}}{50} & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 & \frac{\sqrt{15}}{50} & 0 & \frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & \frac{\sqrt{5}}{50} & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & \frac{\sqrt{15}}{50} & 0 \\ 0 & 0 & \frac{7\sqrt{10}}{100} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{50} & 0 & -\frac{\sqrt{6}}{10} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 516 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |
| 517 | symmetry | $\sqrt{15}xyz$ |

$\mathbb{M}_{3,2}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{10}i}{25} & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & 0\\
-\frac{\sqrt{10}i}{25} & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & -\frac{7\sqrt{10}i}{100} & 0 & -\frac{\sqrt{5}i}{50} & 0 & 0\\
0 & \frac{\sqrt{30}i}{25} & 0 & \frac{\sqrt{10}i}{25} & 0 & 0 & -\frac{\sqrt{5}i}{50} & 0 & -\frac{7\sqrt{10}i}{100} & 0\\
0 & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{50} & 0 & \frac{\sqrt{6}i}{20}\\
-\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0\\
0 & \frac{7\sqrt{10}i}{100} & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & 0\\
-\frac{3\sqrt{15}i}{50} & 0 & \frac{\sqrt{5}i}{50} & 0 & 0 & \frac{\sqrt{15}i}{50} & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0\\
0 & \frac{\sqrt{5}i}{50} & 0 & -\frac{3\sqrt{15}i}{50} & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & -\frac{\sqrt{15}i}{50} & 0\\
0 & 0 & \frac{7\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{50} & 0 & \frac{\sqrt{6}i}{10}\\
0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0
\end{bmatrix}$$

$\mathbb{M}_{3,1}^{(a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0\\
0 & 0 & 0 & -\frac{1}{5} & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & \frac{1}{10} & 0\\
\frac{1}{5} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}}{10}\\
0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0\\
0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0\\
0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{6}}{10} & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0\\
\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}}{20}\\
0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 518 | symmetry | $z$ |
| 519 | symmetry | $x$ |

$\mathbb{M}_{3,2}^{(a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 \\
0 & 0 & 0 & -\frac{i}{5} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{i}{10} & 0 \\
-\frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\
0 & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\
0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{10} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 \\
-\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\
0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_1^{(1,-1;a)}(A_{2g})$

$$\begin{bmatrix}
-\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10}
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 520 | symmetry | $y$ |
| 521 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 & 0 \\
-\frac{\sqrt{30}}{50} & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{25} & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{15}}{25} & 0 & -\frac{\sqrt{30}}{25} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & -\frac{\sqrt{2}}{5} \\
\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{25} & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\
-\frac{\sqrt{5}}{25} & 0 & \frac{\sqrt{15}}{25} & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 \\
0 & -\frac{\sqrt{15}}{25} & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & \frac{2\sqrt{5}}{25} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{\sqrt{2}}{10} \\
0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0
\end{bmatrix}$$

$\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$

$$\begin{bmatrix}
0 & \frac{\sqrt{30}i}{50} & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & 0 \\
-\frac{\sqrt{30}i}{50} & 0 & \frac{\sqrt{10}i}{25} & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & \frac{\sqrt{15}i}{25} & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{25} & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & \frac{\sqrt{15}i}{25} & 0 & \frac{\sqrt{30}i}{25} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{5} \\
-\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & -\frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{25} & 0 & -\frac{\sqrt{15}i}{25} & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{25} & 0 & -\frac{\sqrt{5}i}{25} & 0 & 0 & \frac{3\sqrt{10}i}{50} & 0 & -\frac{2\sqrt{5}i}{25} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & -\frac{\sqrt{2}i}{10} \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 522 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| 523 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{M}_3^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}i}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_3^{(1,-1;a)}(A_{2g},1)$

$$\begin{bmatrix}
\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35}
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 524 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |
| 525 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$$\mathbb{M}_3^{(1,-1;a)}(A_{2g},2)=\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35}\\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{35} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35}\\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{35} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{210}}{350} & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & \frac{6\sqrt{35}}{175} & 0 & 0 & 0\\
\frac{\sqrt{210}}{350} & 0 & -\frac{3\sqrt{70}}{350} & 0 & 0 & \frac{\sqrt{210}}{75} & 0 & -\frac{2\sqrt{105}}{525} & 0 & 0\\
0 & -\frac{3\sqrt{70}}{350} & 0 & \frac{\sqrt{210}}{350} & 0 & 0 & \frac{2\sqrt{105}}{525} & 0 & -\frac{\sqrt{210}}{75} & 0\\
0 & 0 & \frac{\sqrt{210}}{350} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & \frac{\sqrt{14}}{35}\\
-\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{210}}{75} & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 & 0\\
\frac{6\sqrt{35}}{175} & 0 & \frac{2\sqrt{105}}{525} & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & \frac{6\sqrt{70}}{175} & 0 & 0\\
0 & -\frac{2\sqrt{105}}{525} & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 & \frac{6\sqrt{70}}{175} & 0 & \frac{3\sqrt{35}}{175} & 0\\
0 & 0 & -\frac{\sqrt{210}}{75} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & -\frac{3\sqrt{14}}{35}\\
0 & 0 & 0 & \frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|

526 symmetry

$$-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$$

$\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}}{70} & \frac{2\sqrt{105}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 \\
-\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & \frac{2\sqrt{105}}{105} \\
0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\
0 & \frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 \\
-\frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} \\
0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0
\end{bmatrix}$$

527 symmetry

$$\sqrt{15}xyz$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 528 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |
| 529 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

$$\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)=\begin{bmatrix}
0 & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{21}i}{70} & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0\\
\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105}\\
0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0\\
0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0\\
0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0\\
0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0\\
\frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70}\\
0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0\\
0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}_{5}^{(1,-1;a)}(A_{1g})=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{6} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3}\\
0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{6} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} \end{bmatrix}$ |
| 530 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 \end{bmatrix}$ |
| 531 | symmetry | $\frac{3\sqrt{14}x^5}{16}-\frac{15\sqrt{14}x^3y^2}{8}+\frac{15\sqrt{14}xy^4}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 532 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 533 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & \frac{\sqrt{105}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{21} & 0 & -\frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \end{bmatrix}$ |
| 534 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & -\frac{\sqrt{105}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{21} & 0 & \frac{\sqrt{210}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \end{bmatrix}$ |
| 535 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 536 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |
| | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 537 | symmetry | $\frac{\sqrt{105}x^4z}{4}-\frac{\sqrt{105}x^2z^3}{2}-\frac{\sqrt{105}y^4z}{4}+\frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 538 | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,4)$ | symmetry | $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |
| 539 | $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,4)$ | symmetry | $z$ |

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{6}}{12}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{30}}{12}&0&0\\
0&0&0&0&-\frac{\sqrt{6}}{12}&0&0&0&-\frac{\sqrt{30}}{12}&0\\
0&0&0&0&0&\frac{\sqrt{30}}{12}&0&0&0&\frac{\sqrt{6}}{12}\\
0&0&0&0&0&0&-\frac{\sqrt{30}}{12}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{6}}{12}&0&0
\end{bmatrix}$$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{6}i}{12}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{30}i}{12}&0&0\\
0&0&0&0&\frac{\sqrt{6}i}{12}&0&0&0&-\frac{\sqrt{30}i}{12}&0\\
0&0&0&0&0&-\frac{\sqrt{30}i}{12}&0&0&0&\frac{\sqrt{6}i}{12}\\
0&0&0&0&0&0&\frac{\sqrt{30}i}{12}&0&0&0\\
0&0&0&0&0&0&0&-\frac{\sqrt{6}i}{12}&0&0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 540 | symmetry | $x$ |
| 541 | symmetry | $y$ |

$\mathbb{M}_1^{(1,1;a)}(A_{2g})$

$$\begin{bmatrix}
\frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35}
\end{bmatrix}$$

$\mathbb{M}_{1,1}^{(1,1;a)}(E_g)$

$$\begin{bmatrix}
0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & -\frac{\sqrt{35}}{100} & 0 & 0 & 0 \\
\frac{\sqrt{210}}{50} & 0 & \frac{\sqrt{70}}{25} & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & -\frac{\sqrt{105}}{100} & 0 & 0 \\
0 & \frac{\sqrt{70}}{25} & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{105}}{100} & 0 & -\frac{\sqrt{210}}{100} & 0 \\
0 & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{100} & 0 & -\frac{\sqrt{14}}{20} \\
\frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}}{100} & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & -\frac{4\sqrt{35}}{175} & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{100} & 0 & \frac{\sqrt{105}}{100} & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{100} & 0 & \frac{\sqrt{35}}{100} & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & -\frac{4\sqrt{35}}{175} & 0 \\
0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{\sqrt{14}}{35} \\
0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0
\end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 542 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 543 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

$$\mathbb{M}_{1,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & \frac{\sqrt{35}i}{100} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{50} & 0 & -\frac{\sqrt{70}i}{25} & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & \frac{\sqrt{105}i}{100} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{25} & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & \frac{\sqrt{105}i}{100} & 0 & \frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{100} & 0 & \frac{\sqrt{14}i}{20} \\ -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & -\frac{\sqrt{14}i}{35} & 0 & \frac{4\sqrt{35}i}{175} & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{100} & 0 & -\frac{\sqrt{105}i}{100} & 0 & 0 & -\frac{4\sqrt{35}i}{175} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{100} & 0 & -\frac{\sqrt{35}i}{100} & 0 & 0 & -\frac{3\sqrt{70}i}{175} & 0 & \frac{4\sqrt{35}i}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{175} & 0 & \frac{\sqrt{14}i}{35} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{35} & 0 \end{bmatrix}$$

$$\mathbb{M}_{3}^{(1,1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & \frac{6\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}i}{280} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 544 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 545 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$$\mathbb{M}_3^{(1,1;a)}(A_{2g},1) = \begin{bmatrix} -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 \\ 0 & \frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 \\ 0 & 0 & -\frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{525} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} \end{bmatrix}$$

$$\mathbb{M}_3^{(1,1;a)}(A_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{280} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 546 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 547 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$$\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)=\begin{bmatrix}
0 & -\frac{6\sqrt{70}}{175} & 0 & 0 & -\frac{3\sqrt{42}}{280} & 0 & \frac{9\sqrt{105}}{700} & 0 & 0 & 0\\
-\frac{6\sqrt{70}}{175} & 0 & \frac{6\sqrt{210}}{175} & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & -\frac{3\sqrt{35}}{700} & 0 & 0\\
0 & \frac{6\sqrt{210}}{175} & 0 & -\frac{6\sqrt{70}}{175} & 0 & 0 & \frac{3\sqrt{35}}{700} & 0 & -\frac{3\sqrt{70}}{200} & 0\\
0 & 0 & -\frac{6\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}}{700} & 0 & \frac{3\sqrt{42}}{280}\\
-\frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{105} & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{70}}{200} & 0 & 0 & \frac{\sqrt{42}}{105} & 0 & -\frac{\sqrt{105}}{525} & 0 & 0 & 0\\
\frac{9\sqrt{105}}{700} & 0 & \frac{3\sqrt{35}}{700} & 0 & 0 & -\frac{\sqrt{105}}{525} & 0 & -\frac{2\sqrt{210}}{525} & 0 & 0\\
0 & -\frac{3\sqrt{35}}{700} & 0 & -\frac{9\sqrt{105}}{700} & 0 & 0 & -\frac{2\sqrt{210}}{525} & 0 & -\frac{\sqrt{105}}{525} & 0\\
0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{525} & 0 & \frac{\sqrt{42}}{105}\\
0 & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{105} & 0
\end{bmatrix}$$

$$\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)=\begin{bmatrix}
0 & \frac{6\sqrt{70}i}{175} & 0 & 0 & -\frac{3\sqrt{42}i}{280} & 0 & -\frac{9\sqrt{105}i}{700} & 0 & 0 & 0\\
-\frac{6\sqrt{70}i}{175} & 0 & -\frac{6\sqrt{210}i}{175} & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & \frac{3\sqrt{35}i}{700} & 0 & 0\\
0 & \frac{6\sqrt{210}i}{175} & 0 & \frac{6\sqrt{70}i}{175} & 0 & 0 & \frac{3\sqrt{35}i}{700} & 0 & \frac{3\sqrt{70}i}{200} & 0\\
0 & 0 & -\frac{6\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}i}{700} & 0 & -\frac{3\sqrt{42}i}{280}\\
\frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{105} & 0 & 0 & 0 & 0\\
0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 & \frac{\sqrt{42}i}{105} & 0 & \frac{\sqrt{105}i}{525} & 0 & 0 & 0\\
\frac{9\sqrt{105}i}{700} & 0 & -\frac{3\sqrt{35}i}{700} & 0 & 0 & -\frac{\sqrt{105}i}{525} & 0 & \frac{2\sqrt{210}i}{525} & 0 & 0\\
0 & -\frac{3\sqrt{35}i}{700} & 0 & \frac{9\sqrt{105}i}{700} & 0 & 0 & -\frac{2\sqrt{210}i}{525} & 0 & \frac{\sqrt{105}i}{525} & 0\\
0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{525} & 0 & -\frac{\sqrt{42}i}{105}\\
0 & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{105} & 0
\end{bmatrix}$$

*continued …*

Table 8

| No. | multipole | matrix |
|---|---|---|
| 548 | symmetry | $\sqrt{15}xyz$ |

$$\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)=\begin{bmatrix}
0 & 0 & \frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0\\
0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{7}}{140} & 0\\
\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{140} & 0 & 0 & 0 & \frac{3\sqrt{35}}{140}\\
0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{210} & 0 & 0 & 0\\
0 & 0 & -\frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & -\frac{\sqrt{210}}{210} & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0\\
-\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 & 0 & \frac{\sqrt{210}}{210}\\
0 & -\frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 & 0\\
0 & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{210} & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)=\begin{bmatrix}
0 & 0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0\\
0 & 0 & 0 & -\frac{6\sqrt{7}i}{35} & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0\\
-\frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{140}\\
0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0 & 0\\
0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & \frac{\sqrt{210}i}{210} & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0\\
\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 & 0 & 0 & \frac{\sqrt{210}i}{210}\\
0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & 0\\
0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0
\end{bmatrix}$$

bra: $=\langle\frac{3}{2},\frac{3}{2};d|\,,\langle\frac{3}{2},\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{5}{2};d|\,,\langle\frac{5}{2},\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},-\frac{5}{2};d|$

ket: $=|\frac{5}{2},\frac{5}{2};f\rangle\,,|\frac{5}{2},\frac{3}{2};f\rangle\,,|\frac{5}{2},\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{3}{2};f\rangle\,,|\frac{5}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{7}{2};f\rangle\,,|\frac{7}{2},\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{3}{2};f\rangle\,,|\frac{7}{2},\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{3}{2};f\rangle\,,|\frac{7}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},-\frac{7}{2};f\rangle$

Table 9: (d,f) block.

| No. | multipole | matrix |
|---|---|---|
| 549 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 550 | $\mathbb{Q}_1^{(a)}(A_{2u})$ | $\begin{bmatrix} 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 \end{bmatrix}$ |
| 551 | $\mathbb{Q}_{1,1}^{(a)}(E_u)$ | $\begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$ |
| 552 | symmetry | $y$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

552 symmetry

$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$

$\mathbb{Q}_{1,2}^{(a)}(E_u)$

$$\begin{bmatrix}
-\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{30}i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{42}i}{28}
\end{bmatrix}$$

$\mathbb{Q}_3^{(a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\
\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\
0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 \\
\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\
0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

553 symmetry

$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 554 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 555 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{Q}_3^{(a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0
\end{bmatrix}$$

$\mathbb{Q}_3^{(a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\
\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\
0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\
0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

556 symmetry $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$

$\mathbb{Q}_{3,1}^{(a)}(E_u, 1)$

$$\begin{bmatrix}
\frac{3\sqrt{70}}{280} & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{21}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}}{140} & 0 & \frac{\sqrt{42}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{7}}{140} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{105}}{84} & 0 \\
0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}}{70} & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}}{70} & 0 & -\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{35} & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{105}}{84} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{3}}{12}
\end{bmatrix}$$

557 symmetry $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$

$\mathbb{Q}_{3,2}^{(a)}(E_u, 1)$

$$\begin{bmatrix}
\frac{3\sqrt{70}i}{280} & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{40} & 0 & -\frac{\sqrt{21}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{140} & 0 & -\frac{\sqrt{42}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{105}i}{84} & 0 \\
0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}i}{70} & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}i}{35} & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{105}i}{84} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{3}i}{12}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 558 | symmetry | $\sqrt{15}xyz$ |
| 559 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$\mathbb{Q}_{3,1}^{(a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 \\
0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 \\
0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} \\
0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 \\
-\frac{\sqrt{14}}{28} & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\
0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \\
0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\
0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{3,2}^{(a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\
0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\
\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\
0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\
0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 \\ -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} \\ 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| $\mathbb{Q}_5^{(a)}(A_{1u})$ | | |
| 560 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |
| $\mathbb{Q}_5^{(a)}(A_{2u}, 1)$ | | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \end{bmatrix}$ |
| 561 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 562 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| 563 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

$\mathbb{Q}_5^{(a)}(A_{2u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\
-\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} \\
0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{5,1}^{(a)}(E_u, 1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 564 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| | $\mathbb{Q}_{5,1}^{(a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & -\frac{\sqrt{3}}{30} & 0 \\ 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{3}}{420} & 0 & -\frac{13\sqrt{15}}{420} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}}{140} & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{420} & 0 & \frac{11\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & \frac{13\sqrt{15}}{420} & 0 & -\frac{23\sqrt{3}}{420} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{105}}{420} \end{bmatrix}$ |
| 565 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 566 | $\mathbb{Q}_{5,2}^{(a)}(E_u,2)$ | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| 567 | $\mathbb{Q}_{5,1}^{(a)}(E_u,3)$ | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & \frac{\sqrt{3}i}{30} & 0\\
0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{3}i}{420} & 0 & \frac{13\sqrt{15}i}{420} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{140} & 0 & -\frac{\sqrt{30}i}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & -\frac{11\sqrt{2}i}{140} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & \frac{13\sqrt{15}i}{420} & 0 & \frac{23\sqrt{3}i}{420} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{105}i}{420}
\end{bmatrix}$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 568 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{Q}^{(a)}_{5,2}(E_u,3)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{Q}^{(a)}_{5,1}(E_u,4)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{60} \\
0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 \\
\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{2\sqrt{42}}{105} & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 & 0 & -\frac{2\sqrt{42}}{105} & 0 \\
0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{30} \\
0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 569 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(a)}(E_u,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{60} & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} \\ 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & 0 & \frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & -\frac{2\sqrt{42}i}{105} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 570 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \\ \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 571 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 572 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 573 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{Q}_3^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0
\end{bmatrix}$$

$\mathbb{Q}_3^{(1,-1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 \\
\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\
0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\
\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

574 symmetry $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$

$\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{5}}{70} & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{30} & 0 & \frac{\sqrt{6}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}}{210} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & \frac{\sqrt{30}}{28} & 0 \\
0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{5}}{70} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{140} & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \\
0 & 0 & -\frac{3}{35} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & -\frac{\sqrt{30}}{42} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & \frac{\sqrt{42}}{42}
\end{bmatrix}$$

575 symmetry $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$

$\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{70} & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{30} & 0 & -\frac{\sqrt{6}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{210} & 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{30}i}{28} & 0 \\
0 & -\frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{5}i}{70} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{2}i}{140} & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{35} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{140} & 0 & -\frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & \frac{\sqrt{30}i}{42} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & -\frac{\sqrt{42}i}{42}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 576 | symmetry | $\sqrt{15}xyz$ |
| 577 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 \\
-\frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{30}}{210} & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 \\
0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 & 0 & \frac{3}{28} & 0 \\
0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\
0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\
-\frac{3}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & \frac{3}{28} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 \\
0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\
0 & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\
0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
\frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 \\
0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

578 symmetry $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$

$\mathbb{Q}_5^{(1,-1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{50} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 \\
-\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} \\
0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0
\end{bmatrix}$$

579 symmetry $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$

$\mathbb{Q}_5^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 \\
-\frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0 \\ -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} \\ 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 580 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| | $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 581 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 582 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| | $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{3\sqrt{10}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{10}}{100} & 0 \\ 0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}}{350} & 0 & \frac{13\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}}{175} & 0 & -\frac{1}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & \frac{1}{35} & 0 & -\frac{11\sqrt{15}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{70} & 0 & \frac{23\sqrt{10}}{350} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{14}}{70} \end{bmatrix}$ |
| 583 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 584 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| 585 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

$\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{100} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{10}i}{100} & 0\\
0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{15}i}{210} & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}i}{350} & 0 & -\frac{13\sqrt{2}i}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{6}i}{42} & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}i}{175} & 0 & \frac{i}{35} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & \frac{i}{35} & 0 & \frac{11\sqrt{15}i}{175} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{70} & 0 & -\frac{23\sqrt{10}i}{350} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & \frac{\sqrt{14}i}{70}
\end{bmatrix}$$

$\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{35} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{175} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 586 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,4) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{200} & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{40} & 0 & 0 & 0 & \frac{\sqrt{10}}{200} \\
0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\
\frac{\sqrt{210}}{420} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & \frac{8\sqrt{35}}{175} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 & 0 & \frac{8\sqrt{35}}{175} & 0 \\
0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 & -\frac{\sqrt{10}}{25} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 587 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 588 | symmetry | $z$ |
| 589 | symmetry | $x$ |

$$\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,4)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{200} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{40} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{40} & 0 & 0 & 0 & \frac{\sqrt{10}i}{200}\\
0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{10}i}{25} & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 & 0\\
-\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0\\
0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & \frac{8\sqrt{35}i}{175} & 0\\
0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{25}\\
0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{Q}_{1}^{(1,0;a)}(A_{2u})=\begin{bmatrix}
0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 590 | symmetry | $y$ |
| 591 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |

$\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)$

$$\begin{bmatrix}
\frac{\sqrt{10}}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{10}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{6}{35} & 0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{2}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{6}{35} & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{21}}{28}
\end{bmatrix}$$

$\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)$

$$\begin{bmatrix}
\frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 592 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| 593 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{Q}_3^{(1,0;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\
-\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\
0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 \\
-\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} \\
0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_3^{(1,0;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\
-\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| 594 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |

$$\mathbb{Q}_3^{(1,0;a)}(A_{2u},2)=\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0\\
-\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4}\\
0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0\\
-\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{48}\\
0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 595 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$$\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{210}}{560} & 0 & \frac{3\sqrt{21}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{7}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}}{280} & 0 & -\frac{\sqrt{14}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{21}}{280} & 0 & \frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & \frac{\sqrt{35}}{28} & 0\\
0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{210}}{70} & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{168} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{336} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{42}}{35} & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{336} & 0 & \frac{\sqrt{210}}{336} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{84} & 0 & \frac{\sqrt{35}}{168} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & -\frac{1}{24}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

596 symmetry $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$

$\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{210}i}{560} & 0 & -\frac{3\sqrt{21}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{7}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}i}{280} & 0 & \frac{\sqrt{14}i}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{21}i}{280} & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{35}i}{28} & 0\\
0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{210}i}{70} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{7}i}{84} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{21}i}{70} & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{42}i}{35} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{336} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{84} & 0 & -\frac{\sqrt{35}i}{168} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{i}{24}
\end{bmatrix}$$

$\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0\\
\frac{\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0\\
0 & -\frac{\sqrt{35}}{280} & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0\\
0 & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8}\\
0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0\\
\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0\\
0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0\\
0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24}\\
0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0
\end{bmatrix}$$

597 symmetry $\sqrt{15}xyz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 \end{bmatrix}$ |
| 598 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & -\frac{\sqrt{30}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 \\ \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{300} \\ 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 599 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 \end{bmatrix}$ |
| 600 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & -\frac{\sqrt{30}}{300} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{700} & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{300} \\ 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 601 | symmetry | $\frac{3\sqrt{14}x^5}{16}-\frac{15\sqrt{14}x^3y^2}{8}+\frac{15\sqrt{14}xy^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 602 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |
| | $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 603 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

604 symmetry $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$

$\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & \frac{3\sqrt{10}}{50} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & -\frac{\sqrt{10}}{50} & 0\\
0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{10}}{4200} & 0 & -\frac{13\sqrt{2}}{840} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{6}}{14} & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}}{2100} & 0 & \frac{1}{420} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{3}}{7} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & -\frac{1}{420} & 0 & \frac{11\sqrt{15}}{2100} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}}{840} & 0 & -\frac{23\sqrt{10}}{4200} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & \frac{\sqrt{14}}{840}
\end{bmatrix}$$

605 symmetry $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$

$\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{50} & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{10}i}{50} & 0\\
0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{840} & 0 & -\frac{\sqrt{6}i}{168} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{10}i}{4200} & 0 & \frac{13\sqrt{2}i}{840} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{2100} & 0 & -\frac{i}{420} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & -\frac{i}{420} & 0 & -\frac{11\sqrt{15}i}{2100} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}i}{840} & 0 & \frac{23\sqrt{10}i}{4200} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & -\frac{\sqrt{14}i}{840}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 606 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |
| 607 | symmetry | $\frac{\sqrt{105}x^4z}{4}-\frac{\sqrt{105}x^2z^3}{2}-\frac{\sqrt{105}y^4z}{4}+\frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2100} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{150}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{50} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2100} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{150}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 608 | symmetry | $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |
| 609 | symmetry | $z$ |

$\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & \frac{3\sqrt{70}}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 & \frac{\sqrt{10}}{100} \\
0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{10}}{300} & 0 & 0 & 0 & \frac{\sqrt{14}}{210} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{2\sqrt{35}}{525} & 0 & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 \\
0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 & 0 & -\frac{2\sqrt{35}}{525} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{210} & 0 & 0 & 0 & \frac{\sqrt{10}}{300} \\
0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{420} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} \\
0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{10}i}{300} & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 \\
0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{210} & 0 & 0 & 0 & \frac{\sqrt{10}i}{300} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{420} & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 610 | $\mathbb{Q}_1^{(1,1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 \end{bmatrix}$ |
| 611 | $\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 612 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 613 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

$\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{5}i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{6i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56}
\end{bmatrix}$$

$\mathbb{Q}_3^{(1,1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{6}i}{112} & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \\
-\frac{15\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\
0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 \\
\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\
0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 614 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 615 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{Q}_3^{(1,1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\
\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0
\end{bmatrix}$$

$\mathbb{Q}_3^{(1,1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}}{112} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 \\
-\frac{15\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\
0 & -\frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5}{42} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 \\
\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\
0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 616 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 617 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$$\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)=\begin{bmatrix}
-\frac{3\sqrt{30}}{112} & 0 & \frac{9\sqrt{3}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{2}}{16} & 0 & -\frac{3}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0\\
0 & 0 & \frac{3}{56} & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{9\sqrt{3}}{56} & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0\\
0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{30}}{42} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{56} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}}{42} & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{2}}{112} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{6}}{21} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{112} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & \frac{\sqrt{5}}{56} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{7}}{56}
\end{bmatrix}$$

$$\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)=\begin{bmatrix}
-\frac{3\sqrt{30}i}{112} & 0 & -\frac{9\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{i}{14} & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{2}i}{16} & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0\\
0 & 0 & \frac{3i}{56} & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{9\sqrt{3}i}{56} & 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{5}i}{28} & 0\\
0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{i}{28} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{2}i}{112} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{112} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{5}i}{56} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{7}i}{56}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 618 | symmetry | $\sqrt{15}xyz$ |
| 619 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{15}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & \frac{15}{56} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\
0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{1}{56} & 0 \\
0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\
0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
-\frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 \\
0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 620 | symmetry | $\sqrt{3}yz$ |
| 621 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{G}_2^{(a)}(A_{1u})$

$$\begin{bmatrix}
0 & \frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\
-\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0
\end{bmatrix}$$

$\mathbb{G}_{2,1}^{(a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{6}}{14} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{70} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & \frac{1}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & \frac{1}{14} & 0 \\
0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & \frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & -\frac{1}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{35}}{28}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 622 | symmetry | $\sqrt{3}xy$ |
| 623 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$\mathbb{G}_{2,2}^{(a)}(E_u, 1)$

$$\begin{bmatrix}
\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\
0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28}
\end{bmatrix}$$

$\mathbb{G}_{2,1}^{(a)}(E_u, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\
0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 \\
0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\
0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\
0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 \\
0 & 0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 624 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| 625 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

$\mathbb{G}_{2,2}^{(a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\
0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(a)}(A_{1u},1)$

$$\begin{bmatrix}
0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 \\
\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 626 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| 627 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

$\mathbb{G}_4^{(a)}(A_{1u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & -\frac{3i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 \\
\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{20} \\
0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 \\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(a)}(A_{2u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & \frac{3}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\
-\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3}{20} \\
0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{3\sqrt{6}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 \\
-\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

628 symmetry $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$

$$\mathbb{G}^{(a)}_{4,1}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{35} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{3}}{35} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & -\frac{9\sqrt{3}}{140} & 0\\
0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{2}}{14} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{13\sqrt{3}}{140} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{280} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{2}}{280} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & -\frac{13\sqrt{3}}{140} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{105}}{140}
\end{bmatrix}$$

629 symmetry $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$

$$\mathbb{G}^{(a)}_{4,2}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{35} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & -\frac{3\sqrt{3}i}{35} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & \frac{9\sqrt{3}i}{140} & 0\\
0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{2}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{13\sqrt{3}i}{140} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{280} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{2}i}{280} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & \frac{13\sqrt{3}i}{140} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{105}i}{140}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 630 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |
| 631 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |

$$\mathbb{G}_{4,1}^{(a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{4,2}^{(a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| 632 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$$\mathbb{G}^{(a)}_{4,1}(E_u,3) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & \frac{33\sqrt{2}}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 \\
0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & -\frac{3\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 \\
0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}}{280} \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 & 0 & 0 & -\frac{17}{140} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{17}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 633 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

$$\mathbb{G}^{(a)}_{4,2}(E_u,3) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & -\frac{17i}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{17i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}_2^{(1,-1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 \\
\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0
\end{bmatrix}$$

634 symmetry $\sqrt{3}yz$

$\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{3\sqrt{10}}{140} & 0 & -\frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{140} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{9}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 \\
0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}}{35} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & \frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & \frac{2}{35} & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & -\frac{\sqrt{15}}{140} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & -\frac{\sqrt{21}}{28}
\end{bmatrix}$$

635 symmetry $-\sqrt{3}xz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & -\frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\
0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\
0 & 0 & 0 & \frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28}
\end{bmatrix}$$

636 symmetry $\sqrt{3}xy$

$\mathbb{G}_{2,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 \\
0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 \\
0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\
0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 \\
0 & -\frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 \\
0 & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\
0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0
\end{bmatrix}$$

637 symmetry $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 638 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| 639 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

$\mathbb{G}_{2,2}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 \\
0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 \\
0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} \\
0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 \\
0 & \frac{3i}{35} & 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & \frac{\sqrt{30}i}{35} & 0 \\
0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} \\
0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(1,-1;a)}(A_{1u},1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 \\
-\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 640 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| 641 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

$\mathbb{G}_4^{(1,-1;a)}(A_{1u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\
-\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\
0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 \\
-\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(1,-1;a)}(A_{2u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\
\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\
0 & -\frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{24} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 642 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| 643 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

$\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{15}}{168} & 0 & 0 & 0 & 0 & \frac{3}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & -\frac{1}{7} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{168} & 0 & -\frac{\sqrt{6}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & \frac{3}{28} & 0 \\
0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & -\frac{5\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}}{42} & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & \frac{13}{56} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{6}}{336} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{13}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}}{84} & 0 & \frac{\sqrt{35}}{56}
\end{bmatrix}$$

$\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{15}i}{168} & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{5}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & \frac{i}{7} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{168} & 0 & \frac{\sqrt{6}i}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{3i}{28} & 0 \\
0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 & \frac{5\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & \frac{13i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{6}i}{336} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{13i}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}i}{84} & 0 & -\frac{\sqrt{35}i}{56}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 644 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |
| | $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 645 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 646 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |
| 647 | symmetry | $-\frac{5x^6}{16} - \frac{15x^4y^2}{16} + \frac{45x^4z^2}{8} - \frac{15x^2y^4}{16} + \frac{45x^2y^2z^2}{4} - \frac{15x^2z^4}{2} - \frac{5y^6}{16} + \frac{45y^4z^2}{8} - \frac{15y^2z^4}{2} + z^6$ |

$\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 \\
0 & \frac{\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\
0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{9}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}}{168} & 0 & 0 \\
0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & -\frac{17\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{9}{56} & 0 \\
0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} \\
0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\
0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{9i}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & \frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & -\frac{9i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\
0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A_{1u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 \end{bmatrix}$ |
| 648 | $\mathbb{G}_6^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 649 | | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 650 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |
| 651 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

$\mathbb{G}_6^{(1,-1;a)}(A_{1u},3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_6^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 652 | $\mathbb{G}_6^{(1,-1;a)}(A_{2u},2)$ | $\frac{15\sqrt{154}x^4yz}{16}-\frac{15\sqrt{154}x^2y^3z}{8}+\frac{3\sqrt{154}y^5z}{16}$ $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{77}}{44}&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{231}}{44}&0&0\\ 0&0&0&0&0&0&\frac{\sqrt{22}}{44}&0&0&0&0&\frac{\sqrt{154}}{44}&0\\ 0&0&0&0&0&0&0&-\frac{\sqrt{154}}{44}&0&0&0&0&-\frac{\sqrt{22}}{44}\\ 0&0&0&0&0&0&0&0&\frac{\sqrt{231}}{44}&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{77}}{44}&0&0&0 \end{bmatrix}$ |
| 653 | $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)$ | $\frac{3\sqrt{154}x^5z}{16}-\frac{15\sqrt{154}x^3y^2z}{8}+\frac{15\sqrt{154}xy^4z}{16}$ $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{21}}{12}&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{15}}{12}\\ 0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&\frac{\sqrt{15}}{12}&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&-\frac{\sqrt{21}}{12}&0&0&0&0&0 \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 654 | symmetry | $\frac{5\sqrt{21}x^4yz}{8}+\frac{5\sqrt{21}x^2y^3z}{4}-\frac{5\sqrt{21}x^2yz^3}{2}+\frac{5\sqrt{21}y^5z}{8}-\frac{5\sqrt{21}y^3z^3}{2}+\sqrt{21}yz^5$ |
| | $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{264} & 0 & \frac{\sqrt{462}}{264} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{264} & 0 & -\frac{5\sqrt{154}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{132} & 0 & \frac{5\sqrt{77}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{132} & 0 & -\frac{\sqrt{1155}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{264} & 0 & \frac{\sqrt{770}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{264} & 0 & -\frac{\sqrt{22}}{264} \end{bmatrix}$ |
| 655 | symmetry | $-\frac{5\sqrt{21}x^5z}{8}-\frac{5\sqrt{21}x^3y^2z}{4}+\frac{5\sqrt{21}x^3z^3}{2}-\frac{5\sqrt{21}xy^4z}{8}+\frac{5\sqrt{21}xy^2z^3}{2}-\sqrt{21}xz^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 656 | symmetry | $\frac{3\sqrt{7}x^5y}{4} - \frac{15\sqrt{7}x^3yz^2}{2} - \frac{3\sqrt{7}xy^5}{4} + \frac{15\sqrt{7}xy^3z^2}{2}$ |
| 657 | symmetry | $-\frac{3\sqrt{7}x^6}{16} + \frac{15\sqrt{7}x^4y^2}{16} + \frac{15\sqrt{7}x^4z^2}{8} + \frac{15\sqrt{7}x^2y^4}{16} - \frac{45\sqrt{7}x^2y^2z^2}{4} - \frac{3\sqrt{7}y^6}{16} + \frac{15\sqrt{7}y^4z^2}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 658 | symmetry | $\frac{\sqrt{210}x^5y}{16}+\frac{\sqrt{210}x^3y^3}{8}-\sqrt{210}x^3yz^2+\frac{\sqrt{210}xy^5}{16}-\sqrt{210}xy^3z^2+\sqrt{210}xyz^4$ |
| 659 | symmetry | $\frac{\sqrt{210}x^6}{32}+\frac{\sqrt{210}x^4y^2}{32}-\frac{\sqrt{210}x^4z^2}{2}-\frac{\sqrt{210}x^2y^4}{32}+\frac{\sqrt{210}x^2z^4}{2}-\frac{\sqrt{210}y^6}{32}+\frac{\sqrt{210}y^4z^2}{2}-\frac{\sqrt{210}y^2z^4}{2}$ |

$\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{154}i}{44}&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2310}i}{132}&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{165}i}{66}\\
0&0&0&0&0&0&-\frac{\sqrt{165}i}{66}&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{2310}i}{132}&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{154}i}{44}&0&0&0&0&0
\end{bmatrix}$$

$\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,4)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{77}}{66}&0&0&0&0\\
0&0&0&0&0&0&\frac{\sqrt{11}}{66}&0&0&0&-\frac{\sqrt{385}}{66}&0&0&0\\
0&0&0&0&0&0&0&-\frac{\sqrt{154}}{66}&0&0&0&\frac{\sqrt{462}}{66}&0&0\\
0&0&0&0&0&0&0&0&\frac{\sqrt{462}}{66}&0&0&0&-\frac{\sqrt{154}}{66}&0\\
0&0&0&0&0&0&0&0&0&-\frac{\sqrt{385}}{66}&0&0&0&\frac{\sqrt{11}}{66}\\
0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{77}}{66}&0&0&0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 660 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| 661 | symmetry | $\sqrt{3}yz$ |

$\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,4)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{77}i}{66}&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{11}i}{66}&0&0&0&-\frac{\sqrt{385}i}{66}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{154}i}{66}&0&0&0&\frac{\sqrt{462}i}{66}&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{462}i}{66}&0&0&0&-\frac{\sqrt{154}i}{66}&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{385}i}{66}&0&0&0&\frac{\sqrt{11}i}{66}\\
0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{77}i}{66}&0&0&0
\end{bmatrix}$$

$\mathbb{G}_{2}^{(1,0;a)}(A_{1u})$

$$\begin{bmatrix}
0&\frac{\sqrt{15}i}{35}&0&0&0&0&0&0&-\frac{\sqrt{6}i}{14}&0&0&0&0&0\\
0&0&\frac{\sqrt{10}i}{70}&0&0&0&0&0&0&-\frac{3\sqrt{30}i}{70}&0&0&0&0\\
0&0&0&-\frac{\sqrt{10}i}{70}&0&0&0&0&0&0&-\frac{3\sqrt{30}i}{70}&0&0&0\\
0&0&0&0&-\frac{\sqrt{15}i}{35}&0&0&0&0&0&0&-\frac{\sqrt{6}i}{14}&0&0\\
-\frac{\sqrt{15}i}{14}&0&0&0&0&0&0&-\frac{\sqrt{10}i}{28}&0&0&0&0&0&0\\
0&\frac{\sqrt{15}i}{70}&0&0&0&0&0&0&-\frac{\sqrt{6}i}{28}&0&0&0&0&0\\
0&0&\frac{2\sqrt{15}i}{35}&0&0&0&0&0&0&-\frac{\sqrt{5}i}{70}&0&0&0&0\\
0&0&0&\frac{2\sqrt{15}i}{35}&0&0&0&0&0&0&\frac{\sqrt{5}i}{70}&0&0&0\\
0&0&0&0&\frac{\sqrt{15}i}{70}&0&0&0&0&0&0&\frac{\sqrt{6}i}{28}&0&0\\
0&0&0&0&0&-\frac{\sqrt{15}i}{14}&0&0&0&0&0&0&\frac{\sqrt{10}i}{28}&0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

662 symmetry $-\sqrt{3}xz$

$\mathbb{G}_{2,1}^{(1,0;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\
0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168}
\end{bmatrix}$$

663 symmetry $\sqrt{3}xy$

$\mathbb{G}_{2,2}^{(1,0;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{i}{14} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{210} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 \\
0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & \frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & \frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & -\frac{\sqrt{6}i}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{168}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 664 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 665 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

$$\mathbb{G}_{2,1}^{(1,0;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{2,2}^{(1,0;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\
0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\
0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 666 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| 667 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$\mathbb{G}_4^{(1,0;a)}(A_{1u},1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{2}i}{140} & 0 & 0 \\
\frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(1,0;a)}(A_{1u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 \\
\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}i}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} \\
0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & -\frac{3\sqrt{30}i}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 \\
\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 668 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |
| 669 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |

$$\mathbb{G}_{\mathrm{T}_4}^{(1,0;a)}(A_{2u})=\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{560} & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0\\
-\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{100}\\
0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{200} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & \frac{3\sqrt{30}}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2800} & 0\\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{400}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{4,1}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{10}}{560} & 0 & -\frac{1}{56} & 0 & 0 & 0 & 0 & \frac{27\sqrt{15}}{700} & 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{6}}{112} & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}}{175} & 0 & -\frac{9}{140} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}}{56} & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & -\frac{9}{140} & 0 & -\frac{9\sqrt{15}}{175} & 0 & 0\\
0 & 0 & 0 & \frac{1}{56} & 0 & \frac{\sqrt{10}}{560} & 0 & 0 & 0 & 0 & \frac{9\sqrt{3}}{70} & 0 & \frac{27\sqrt{15}}{700} & 0\\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{280} & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{10}}{70} & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{15}}{1400} & 0 & -\frac{\sqrt{3}}{140} & 0 & 0 & 0 & 0\\
0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{2800} & 0 & -\frac{\sqrt{6}}{80} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{10}}{2800} & 0 & 0\\
0 & 0 & 0 & -\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{140} & 0 & \frac{13\sqrt{15}}{1400} & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{21}}{280}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 670 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 671 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |

$$\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{10}i}{560} & 0 & \frac{i}{56} & 0 & 0 & 0 & 0 & \frac{27\sqrt{15}i}{700} & 0 & -\frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{6}i}{112} & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{175} & 0 & \frac{9i}{140} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & -\frac{9i}{140} & 0 & \frac{9\sqrt{15}i}{175} & 0 & 0\\
0 & 0 & 0 & \frac{i}{56} & 0 & -\frac{\sqrt{10}i}{560} & 0 & 0 & 0 & 0 & \frac{9\sqrt{3}i}{70} & 0 & -\frac{27\sqrt{15}i}{700} & 0\\
0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{280} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{10}i}{70} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{15}i}{1400} & 0 & \frac{\sqrt{3}i}{140} & 0 & 0 & 0 & 0\\
0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{2800} & 0 & \frac{\sqrt{6}i}{80} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & \frac{\sqrt{10}i}{2800} & 0 & 0\\
0 & 0 & 0 & -\frac{3i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{140} & 0 & -\frac{13\sqrt{15}i}{1400} & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & \frac{\sqrt{21}i}{280}
\end{bmatrix}$$

$$\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{350} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{100}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{350} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 672 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| 673 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

$\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}i}{700} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_{4,1}^{(1,0;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & -\frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{280} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{30}}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{15}}{280} & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & \frac{99\sqrt{10}}{1400} & 0 \\
0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{210}}{1400} \\
0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & -\frac{\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & \frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & \frac{17\sqrt{5}}{1400} & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & \frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{15}}{1400} & 0 \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}}{1400} \\
0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}^{(1,0;a)}_{4,2}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 \\
0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{\sqrt{15}i}{280} & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{1400} \\
0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & \frac{17\sqrt{5}i}{1400} & 0 & 0 \\
0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{15}i}{1400} & 0 \\
0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{1400} \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0
\end{bmatrix}$$

674 symmetry 1

$\mathbb{G}^{(1,1;a)}_{0}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

675 symmetry $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}_{2}^{(1,1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & \frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\
\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{105} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0
\end{bmatrix}$$

676 symmetry $\sqrt{3}yz$

$\mathbb{G}_{2,1}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{2\sqrt{15}}{35} & 0 & \frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}}{7} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 \\
0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{15}}{70} & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & -\frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & -\frac{1}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & -\frac{\sqrt{15}}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & \frac{\sqrt{10}}{210} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & \frac{\sqrt{14}}{42}
\end{bmatrix}$$

677 symmetry $-\sqrt{3}xz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}_{2,2}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\
0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42}
\end{bmatrix}$$

678 symmetry $\sqrt{3}xy$

$\mathbb{G}_{2,1}^{(1,1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\
\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & \frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\
0 & \frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\
0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\
0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\
0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\
0 & 0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\
0 & 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0
\end{bmatrix}$$

679 symmetry $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 680 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$$\mathbb{G}_{2,2}^{(1,1;a)}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0\\
-\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0\\
0 & -\frac{8i}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0\\
0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280}\\
0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}i}{105} & 0 & 0 & 0\\
\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{105} & 0 & 0\\
0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{105} & 0\\
0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{105}\\
0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{4}^{(1,1;a)}(A_{1u},1)=\begin{bmatrix}
0 & -\frac{\sqrt{330}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{70} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{2\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{2\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{70} & 0 & 0\\
-\frac{\sqrt{330}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{33}i}{1155} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{385} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{385} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{33}i}{1155} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 681 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 682 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| 683 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

$\mathbb{G}_4^{(1,1;a)}(A_{1u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{70} & \frac{\sqrt{330}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1050} & 0 \\
\frac{\sqrt{385}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{300} \\
0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & -\frac{3\sqrt{55}i}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 \\
-\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{550} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{G}_4^{(1,1;a)}(A_{2u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{70} & -\frac{\sqrt{330}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 \\
-\frac{\sqrt{385}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{300} \\
0 & \frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & \frac{3\sqrt{55}}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 \\
\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{550} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{770} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 684 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| 685 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

$\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{165}}{210} & 0 & -\frac{\sqrt{66}}{42} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{700} & 0 & \frac{\sqrt{22}}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{11}}{14} & 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{175} & 0 & -\frac{\sqrt{66}}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{22}}{14} & 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{420} & 0 & -\frac{\sqrt{110}}{175} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}}{42} & 0 & \frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{70} & 0 & \frac{3\sqrt{110}}{700} & 0\\
0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{770} & 0 & \frac{\sqrt{66}}{231} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{165}}{210} & 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{110}}{3850} & 0 & -\frac{\sqrt{22}}{385} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{11550} & 0 & -\frac{\sqrt{11}}{110} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{110} & 0 & -\frac{\sqrt{165}}{11550} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}}{84} & 0 & \frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{385} & 0 & \frac{13\sqrt{110}}{3850} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{231} & 0 & -\frac{\sqrt{154}}{770}
\end{bmatrix}$$

$\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{165}i}{210} & 0 & \frac{\sqrt{66}i}{42} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{700} & 0 & -\frac{\sqrt{22}i}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{11}i}{14} & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{175} & 0 & \frac{\sqrt{66}i}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{420} & 0 & \frac{\sqrt{110}i}{175} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}i}{42} & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{70} & 0 & -\frac{3\sqrt{110}i}{700} & 0\\
0 & \frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{770} & 0 & -\frac{\sqrt{66}i}{231} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{165}i}{210} & 0 & -\frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{110}i}{3850} & 0 & \frac{\sqrt{22}i}{385} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{11550} & 0 & \frac{\sqrt{11}i}{110} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{110} & 0 & \frac{\sqrt{165}i}{11550} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{385} & 0 & -\frac{13\sqrt{110}i}{3850} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{231} & 0 & \frac{\sqrt{154}i}{770}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 686 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |
| 687 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |

$$\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{150} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}}{1155} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{1925} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{275} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{150} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}i}{1155} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}i}{1925} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{275} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{G}_{4,1}^{(1,1;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & \frac{\sqrt{385}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{110}}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & -\frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 \\
0 & -\frac{\sqrt{22}}{14} & 0 & 0 & 0 & \frac{\sqrt{110}}{70} & 0 & 0 & \frac{\sqrt{55}}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 \\
0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}}{700} \\
0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & -\frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}}{385} & 0 & 0 & 0 \\
-\frac{\sqrt{165}}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & \frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}}{11550} & 0 & 0 \\
0 & \frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & \frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}}{3850} & 0 \\
0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1925} \\
0 & 0 & 0 & -\frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 & 0 & 0
\end{bmatrix}$$

688 symmetry

$$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$$

$\mathbb{G}_{4,2}^{(1,1;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}i}{14} & 0 & 0 & 0 & \frac{\sqrt{110}i}{70} & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{700} \\
0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & \frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & -\frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0
\end{bmatrix}$$

689 symmetry

$z$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 690 | symmetry | $x$ |
| 691 | symmetry | $y$ |

$\mathbb{T}_1^{(a)}(A_{2u})$

$$\begin{bmatrix}
0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0
\end{bmatrix}$$

$\mathbb{T}_{1,1}^{(a)}(E_u)$

$$\begin{bmatrix}
-\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{30}i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{42}i}{28}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(a)}_{1,2}(E_u)$ | $\begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$ |
| 692 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}^{(a)}_{3}(A_{1u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\ 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 693 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 694 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 695 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{T}_3^{(a)}(A_{2u}, 1)$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\
\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0
\end{bmatrix}$$

$\mathbb{T}_3^{(a)}(A_{2u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\
\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\
0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 \\
\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\
0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

696 symmetry $-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$

$$\mathbb{T}^{(a)}_{3,1}(E_u,1)=\begin{bmatrix}
\frac{3\sqrt{70}i}{280} & 0 & -\frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{42}i}{40} & 0 & \frac{\sqrt{21}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{21}i}{140} & 0 & \frac{\sqrt{42}i}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{105}i}{84} & 0\\
0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{70}i}{70} & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{7}i}{70} & 0 & -\frac{\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{14}i}{35} & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{105}i}{84} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{3}i}{12}
\end{bmatrix}$$

697 symmetry $-\frac{\sqrt{15}x^2z}{2} + \frac{\sqrt{15}y^2z}{2}$

$$\mathbb{T}^{(a)}_{3,2}(E_u,1)=\begin{bmatrix}
-\frac{3\sqrt{70}}{280} & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{21}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{21}}{140} & 0 & \frac{\sqrt{42}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{105}}{84} & 0\\
0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{70}}{70} & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{7}}{70} & 0 & -\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{14}}{35} & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{70}}{56} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{105}}{84} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{3}}{12}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 698 | symmetry | $\sqrt{15}xyz$ |
| 699 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$\mathbb{T}^{(a)}_{3,1}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0\\
-\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0\\
0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0\\
0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24}\\
0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0\\
-\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0\\
0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0\\
0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12}\\
0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(a)}_{3,2}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0\\
-\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0\\
0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0\\
0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24}\\
0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0\\
-\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0\\
0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0\\
0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{12}\\
0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 700 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| 701 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

$\mathbb{T}_5^{(a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\
\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} \\
0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_5^{(a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 \\
-\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 702 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| 703 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

$\mathbb{T}_5^{(a)}(A_{2u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 \\
-\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} \\
0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_{5,1}^{(a)}(E_u, 1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 704 | $\mathbb{T}^{(a)}_{5,2}(E_u,1)$ | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| 705 | $\mathbb{T}^{(a)}_{5,1}(E_u,2)$ | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & -\frac{\sqrt{3}i}{30} & 0\\
0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{3}i}{420} & 0 & -\frac{13\sqrt{15}i}{420} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{140} & 0 & \frac{\sqrt{30}i}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & \frac{11\sqrt{2}i}{140} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & \frac{13\sqrt{15}i}{420} & 0 & -\frac{23\sqrt{3}i}{420} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{105}i}{420}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 706 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| 707 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

$\mathbb{T}^{(a)}_{5,2}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{3}}{10} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & -\frac{\sqrt{3}}{30} & 0\\
0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{3}}{420} & 0 & -\frac{13\sqrt{15}}{420} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{140} & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{10}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 & \frac{11\sqrt{2}}{140} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{15}}{420} & 0 & -\frac{23\sqrt{3}}{420} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{105}}{420}
\end{bmatrix}$$

$\mathbb{T}^{(a)}_{5,1}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 708 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |
| | $\mathbb{T}^{(a)}_{5,2}(E_u,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 709 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |
| | $\mathbb{T}^{(a)}_{5,1}(E_u,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} \\ 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & \frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 & -\frac{2\sqrt{42}i}{105} & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 710 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 711 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

$\mathbb{T}^{(a)}_{5,2}(E_u, 4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{20} & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} \\
0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 \\
\frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{2\sqrt{42}}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 \\
0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} \\
0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,-1;a)}_{3}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 \\
-\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\
0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\
-\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 712 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 713 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{T}_3^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\
\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0
\end{bmatrix}$$

$\mathbb{T}_3^{(1,-1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 \\
\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\
0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 \\
\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 714 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 715 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{70} & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{3}i}{30} & 0 & \frac{\sqrt{6}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{6}i}{210} & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{\sqrt{30}i}{28} & 0\\
0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{5}i}{70} & 0 & -\frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0\\
0 & -\frac{3\sqrt{2}i}{140} & 0 & -\frac{3i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0\\
0 & 0 & -\frac{3i}{35} & 0 & -\frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & -\frac{\sqrt{30}i}{42} & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & \frac{\sqrt{42}i}{42}
\end{bmatrix}$$

$\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{5}}{70} & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{3}}{30} & 0 & \frac{\sqrt{6}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{6}}{210} & 0 & \frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & \frac{\sqrt{30}}{28} & 0\\
0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{5}}{70} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{2}}{140} & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0\\
0 & 0 & \frac{3}{35} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0\\
0 & 0 & 0 & \frac{3\sqrt{2}}{140} & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & -\frac{\sqrt{30}}{42} & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & \frac{\sqrt{42}}{42}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 716 | symmetry | $\sqrt{15}xyz$ |
| 717 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\
-\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3i}{28} & 0 \\
0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\
0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
-\frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 \\
0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 \\
-\frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 \\
0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 \\
0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\
0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\
-\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 \\
0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\
0 & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 718 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

$\mathbb{T}_5^{(1,-1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0\\
\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{25}\\
0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0\\
-\frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 719 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 720 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |
| 721 | symmetry | $-\frac{15\sqrt{14}x^4y}{16} + \frac{15\sqrt{14}x^2y^3}{8} - \frac{3\sqrt{14}y^5}{16}$ |

$\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{175} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{175} & 0 \\
-\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{25} \\
0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 722 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ | $\mathbb{T}^{(1,-1;a)}_{5,2}(E_u,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 723 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ | $\mathbb{T}^{(1,-1;a)}_{5,1}(E_u,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{3\sqrt{10}i}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{10}i}{100} & 0 \\ 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{210} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}i}{350} & 0 & \frac{13\sqrt{2}i}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}i}{175} & 0 & -\frac{i}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & \frac{i}{35} & 0 & -\frac{11\sqrt{15}i}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{70} & 0 & \frac{23\sqrt{10}i}{350} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{14}i}{70} \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 724 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |
| 725 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

$\mathbb{T}^{(1,-1;a)}_{5,2}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & \frac{3\sqrt{10}}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{10}}{100} & 0 \\
0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & \frac{23\sqrt{10}}{350} & 0 & \frac{13\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}}{175} & 0 & -\frac{1}{35} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{21} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{1}{35} & 0 & -\frac{11\sqrt{15}}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}}{70} & 0 & \frac{23\sqrt{10}}{350} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{14}}{70}
\end{bmatrix}$$

$\mathbb{T}^{(1,-1;a)}_{5,1}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 726 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{T}^{(1,-1;a)}_{5,2}(E_u,3)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 727 | symmetry | $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$ |

$$\mathbb{T}^{(1,-1;a)}_{5,1}(E_u,4)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 & \frac{\sqrt{10}i}{200} \\
0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & \frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & \frac{8\sqrt{35}i}{175} & 0 \\
0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{25} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 728 | $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{200} & 0 & 0 & 0 & \frac{\sqrt{14}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}}{200} \\ 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{210}}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{8\sqrt{35}}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 & \frac{\sqrt{10}}{25} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 729 | $\mathbb{T}_{1}^{(1,0;a)}(A_{2u})$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 \end{bmatrix}$ |

| No. | multipole |
|---|---|
| 729 | $x$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 730 | symmetry | $y$ |
| 731 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$\mathbb{T}^{(1,0;a)}_{1,1}(E_u)$

$$\begin{bmatrix}
-\frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28}
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{1,2}(E_u)$

$$\begin{bmatrix}
\frac{\sqrt{10}}{20} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{10}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{6}{35} & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{2}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{6}{35} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{21}}{28}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 732 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| 733 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{T}_3^{(1,0;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\
-\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\
0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 \\
-\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} \\
0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}_3^{(1,0;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 734 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |

$$\mathbb{T}_3^{(1,0;a)}(A_{2u},2)=\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0\\
\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4}\\
0 & \frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0\\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48}\\
0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & 0
\end{bmatrix}$$

| No. | multipole | matrix |
|---|---|---|
| 735 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$$\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
\frac{\sqrt{210}i}{560} & 0 & -\frac{3\sqrt{21}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{7}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{7}i}{280} & 0 & \frac{\sqrt{14}i}{80} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{3\sqrt{21}i}{280} & 0 & -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{\sqrt{35}i}{28} & 0\\
0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{24} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{210}i}{70} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{168} & 0 & \frac{\sqrt{7}i}{84} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{21}i}{70} & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{336} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{336} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & -\frac{\sqrt{35}i}{168} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{i}{24}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

736 symmetry $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$

$$\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{21}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{7}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}}{280} & 0 & \frac{\sqrt{14}}{80} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{21}}{280} & 0 & -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{35}}{28} & 0\\
0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{210}}{70} & 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{168} & 0 & \frac{\sqrt{7}}{84} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{14}}{336} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{42}}{35} & 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{336} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{84} & 0 & -\frac{\sqrt{35}}{168} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & \frac{1}{24}
\end{bmatrix}$$

736 symmetry $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$

$$\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0\\
-\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0\\
0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0\\
0 & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8}\\
0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0\\
-\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0\\
0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & \frac{\sqrt{7}i}{168} & 0\\
0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & \frac{\sqrt{2}i}{24}\\
0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

737 symmetry $\sqrt{15}xyz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

738 symmetry $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$

$\mathbb{T}_{3,2}^{(1,0;a)}(E_u, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\
0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 \\
0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 \\
0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \\
0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0
\end{bmatrix}$$

739 symmetry $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$

$\mathbb{T}_5^{(1,0;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & -\frac{\sqrt{30}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 \\
\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{300} \\
0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 740 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ | $\mathbb{T}_5^{(1,0;a)}(A_{2u},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 \end{bmatrix}$ |
| 741 | symmetry | $\frac{3\sqrt{14}x^5}{16}-\frac{15\sqrt{14}x^3y^2}{8}+\frac{15\sqrt{14}xy^4}{16}$ | $\mathbb{T}_5^{(1,0;a)}(A_{2u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{30}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 \\ -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{300} \\ 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 742 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |
| 743 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |

$\mathbb{T}^{(1,0;a)}_{5,1}(E_u,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{5,2}(E_u,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 744 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ | $\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{50} & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & \frac{\sqrt{10}i}{50} & 0\\
0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{840} & 0 & -\frac{\sqrt{6}i}{168} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}i}{4200} & 0 & \frac{13\sqrt{2}i}{840} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}i}{2100} & 0 & -\frac{i}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & \frac{i}{420} & 0 & -\frac{11\sqrt{15}i}{2100} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{840} & 0 & \frac{23\sqrt{10}i}{4200} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{168} & 0 & -\frac{\sqrt{14}i}{840}
\end{bmatrix}$$

| | | | |
|---|---|---|---|
| 745 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ | $\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & -\frac{3\sqrt{10}}{50} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & \frac{\sqrt{10}}{50} & 0\\
0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{840} & 0 & -\frac{\sqrt{6}}{168} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{23\sqrt{10}}{4200} & 0 & \frac{13\sqrt{2}}{840} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}}{2100} & 0 & -\frac{1}{420} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{3}}{7} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & -\frac{1}{420} & 0 & -\frac{11\sqrt{15}}{2100} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}}{840} & 0 & \frac{23\sqrt{10}}{4200} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & -\frac{\sqrt{14}}{840}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 746 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |
| 747 | symmetry | $\frac{\sqrt{105}x^4z}{4}-\frac{\sqrt{105}x^2z^3}{2}-\frac{\sqrt{105}y^4z}{4}+\frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{T}^{(1,0;a)}_{5,1}(E_u,3)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{50} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2100} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{150}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{T}^{(1,0;a)}_{5,2}(E_u,3)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{50} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

748 symmetry $-\frac{\sqrt{105}x^3yz}{2} - \frac{\sqrt{105}xy^3z}{2} + \sqrt{105}xyz^3$

$\mathbb{T}^{(1,0;a)}_{5,1}(E_u, 4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} \\
0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{10}i}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & \frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}i}{525} & 0 \\
0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{300} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{420} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,0;a)}_{5,2}(E_u, 4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & \frac{3\sqrt{70}}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{100} & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & \frac{\sqrt{10}}{100} \\
0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{10}}{300} & 0 & 0 & 0 & \frac{\sqrt{14}}{210} & 0 & 0 & 0 \\
\frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 & 0 & -\frac{2\sqrt{35}}{525} & 0 \\
0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 & \frac{\sqrt{10}}{300} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0
\end{bmatrix}$$

749 symmetry $z$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{T}_1^{(1,1;a)}(A_{2u})$

$$\begin{bmatrix}
0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0
\end{bmatrix}$$

750 symmetry $x$

$\mathbb{T}_{1,1}^{(1,1;a)}(E_u)$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{5}i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{6i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{42}i}{56}
\end{bmatrix}$$

751 symmetry $y$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 752 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| 753 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

$\mathbb{T}^{(1,1;a)}_{1,2}(E_u)$

$$\begin{bmatrix}
-\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56}
\end{bmatrix}$$

$\mathbb{T}^{(1,1;a)}_{3}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}}{112} & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 \\
\frac{15\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\
0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5}{42} & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 \\
-\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\
0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 754 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| 755 | symmetry | $-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$ |

$\mathbb{T}_3^{(1,1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\
\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0
\end{bmatrix}$$

$\mathbb{T}_3^{(1,1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}i}{112} & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 \\
-\frac{15\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\
0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{56} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5i}{42} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{112} & 0 \\
\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} \\
0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 756 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |
| 757 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{T}^{(1,1;a)}_{3,1}(E_u,1)$

$$\begin{bmatrix}
-\frac{3\sqrt{30}i}{112} & 0 & \frac{9\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{3i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{3i}{56} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{3}i}{56} & 0 & \frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 \\
0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{30}i}{42} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{56} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{2}i}{112} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & \frac{\sqrt{5}i}{56} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{7}i}{56}
\end{bmatrix}$$

$\mathbb{T}^{(1,1;a)}_{3,2}(E_u,1)$

$$\begin{bmatrix}
\frac{3\sqrt{30}}{112} & 0 & \frac{9\sqrt{3}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{16} & 0 & -\frac{3}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{3}{56} & 0 & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{3}}{56} & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0 \\
0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{30}}{42} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}}{42} & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{2}}{112} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{21} & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{112} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{5}}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{7}}{56}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 758 | symmetry | $\sqrt{15}xyz$ |
| 759 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{T}^{(1,1;a)}_{3,1}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 \\
0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
-\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 \\
0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{T}^{(1,1;a)}_{3,2}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\
0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 \\
0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\
0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 760 | symmetry | $\sqrt{3}yz$ |
| 761 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{M}_2^{(a)}(A_{1u})$

$$\begin{bmatrix}
0 & \frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\
-\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0
\end{bmatrix}$$

$\mathbb{M}_{2,1}^{(a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\
0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 762 | symmetry | $\sqrt{3}xy$ |
| 763 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$\mathbb{M}^{(a)}_{2,2}(E_u, 1)$

$$\begin{bmatrix}
\frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{70} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & -\frac{1}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & -\frac{1}{14} & 0 \\
0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & \frac{1}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{35}}{28}
\end{bmatrix}$$

$\mathbb{M}^{(a)}_{2,1}(E_u, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\
0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 764 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$$\mathbb{M}^{(a)}_{2,2}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0\\
-\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0\\
0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0\\
0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28}\\
0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0\\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0\\
0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0\\
0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14}\\
0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}^{(a)}_{4}(A_{1u},1)=\begin{bmatrix}
0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0\\
\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{35} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 765 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 766 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| 767 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

$\mathbb{M}_4^{(a)}(A_{1u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{3}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\
\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3}{20} \\
0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{3\sqrt{6}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 \\
\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_4^{(a)}(A_{2u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{3i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 \\
\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{20} \\
0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{280} & 0 \\
\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{M}_{4,1}^{(a)}(E_u,1)$ | $\begin{bmatrix} \frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{35} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & -\frac{3\sqrt{3}i}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & \frac{9\sqrt{3}i}{140} & 0 \\ 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{3}i}{140} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{280} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{2}i}{280} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & \frac{13\sqrt{3}i}{140} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{105}i}{140} \end{bmatrix}$ |
| 768 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| | $\mathbb{M}_{4,2}^{(a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{35} & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & -\frac{3\sqrt{3}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & \frac{9\sqrt{3}}{140} & 0 \\ 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{13\sqrt{3}}{140} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{280} & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{2}}{280} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & \frac{13\sqrt{3}}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{140} \end{bmatrix}$ |
| 769 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 770 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |
| 771 | symmetry | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ |

$$\mathbb{M}_{4,1}^{(a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}_{4,2}^{(a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

772 symmetry $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$

$\mathbb{M}^{(a)}_{4,1}(E_u, 3)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & \frac{33\sqrt{2}i}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{280} \\
0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & \frac{17i}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{17i}{140} & 0 & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0
\end{bmatrix}$$

773 symmetry $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$

$\mathbb{M}^{(a)}_{4,2}(E_u, 3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}}{280} \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 & 0 & 0 & -\frac{17}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{17}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{M}_2^{(1,-1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\
\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0
\end{bmatrix}$$

774 symmetry $\sqrt{3}yz$

$\mathbb{M}_{2,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\
0 & 0 & 0 & -\frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\
0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28}
\end{bmatrix}$$

775 symmetry $-\sqrt{3}xz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 776 | symmetry | $\sqrt{3}xy$ |
| 777 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{3\sqrt{10}}{140} & 0 & \frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{140} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{6}{35} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{9}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 \\
0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}}{35} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & -\frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & \frac{2}{35} & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & \frac{\sqrt{15}}{140} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & \frac{\sqrt{21}}{28}
\end{bmatrix}$$

$\mathbb{M}_{2,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & \frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 \\
0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 \\
0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} \\
0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 \\
0 & \frac{3i}{35} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \\
0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\
0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 778 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$$\mathbb{M}_{2,2}^{(1,-1;a)}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0\\
\frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0\\
0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0\\
0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35}\\
0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0\\
\frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0\\
0 & \frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0\\
0 & 0 & \frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70}\\
0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0
\end{bmatrix}$$

$$\mathbb{M}_{4}^{(1,-1;a)}(A_{1u},1)=\begin{bmatrix}
0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0\\
-\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{28} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0
\end{bmatrix}$$

| | | |
|---|---|---|
| 779 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 780 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A_{2u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 781 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 782 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |
| 783 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

$\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{15}i}{168} & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & \frac{i}{7} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{15}i}{168} & 0 & \frac{\sqrt{6}i}{336} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{3i}{28} & 0\\
0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{56} & 0 & \frac{5\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{13i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{6}i}{336} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{13i}{56} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{15}i}{84} & 0 & -\frac{\sqrt{35}i}{56}
\end{bmatrix}$$

$\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{15}}{168} & 0 & 0 & 0 & 0 & \frac{3}{28} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{5}}{56} & 0 & -\frac{\sqrt{10}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & \frac{1}{7} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{15}}{168} & 0 & \frac{\sqrt{6}}{336} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{3}{28} & 0\\
0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & \frac{5\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & \frac{13}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & -\frac{\sqrt{6}}{336} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{13}{56} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}}{84} & 0 & -\frac{\sqrt{35}}{56}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

784 symmetry $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$

$\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

785 symmetry $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

786 symmetry $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$

$$\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,3) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{i}{56} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{i}{56} & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} \\ 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{9i}{56} & 0 & 0 & 0 & \frac{17\sqrt{3}i}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & \frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & \frac{9i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,3) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\ 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{9}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{9}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$$

787 symmetry $-\frac{5x^6}{16} - \frac{15x^4y^2}{16} + \frac{45x^4z^2}{8} - \frac{15x^2y^4}{16} + \frac{45x^2y^2z^2}{4} - \frac{15x^2z^4}{2} - \frac{5y^6}{16} + \frac{45y^4z^2}{8} - \frac{15y^2z^4}{2} + z^6$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 788 | $\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)$ | $\frac{\sqrt{462}x^6}{32}-\frac{15\sqrt{462}x^4y^2}{32}+\frac{15\sqrt{462}x^2y^4}{32}-\frac{\sqrt{462}y^6}{32}$ |
| 789 | $\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)$ | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 790 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |
| 791 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

$\mathbb{M}_6^{(1,-1;a)}(A_{1u},3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_6^{(1,-1;a)}(A_{2u},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 792 | symmetry | $\frac{15\sqrt{154}x^4yz}{16} - \frac{15\sqrt{154}x^2y^3z}{8} + \frac{3\sqrt{154}y^5z}{16}$ |
| 793 | symmetry | $\frac{3\sqrt{154}x^5z}{16} - \frac{15\sqrt{154}x^3y^2z}{8} + \frac{15\sqrt{154}xy^4z}{16}$ |

$\mathbb{M}_6^{(1,-1;a)}(A_{2u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 794 | symmetry | $\frac{5\sqrt{21}x^4yz}{8}+\frac{5\sqrt{21}x^2y^3z}{4}-\frac{5\sqrt{21}x^2yz^3}{2}+\frac{5\sqrt{21}y^5z}{8}-\frac{5\sqrt{21}y^3z^3}{2}+\sqrt{21}yz^5$ |
| 795 | symmetry | $-\frac{5\sqrt{21}x^5z}{8}-\frac{5\sqrt{21}x^3y^2z}{4}+\frac{5\sqrt{21}x^3z^3}{2}-\frac{5\sqrt{21}xy^4z}{8}+\frac{5\sqrt{21}xy^2z^3}{2}-\sqrt{21}xz^5$ |

$\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{21}}{12}&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{15}}{12}\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{15}}{12}&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{21}}{12}&0&0&0&0&0&0
\end{bmatrix}$$

$\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{22}i}{264}&0&-\frac{\sqrt{462}i}{264}&0&0&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{770}i}{264}&0&\frac{5\sqrt{154}i}{264}&0&0&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{1155}i}{132}&0&-\frac{5\sqrt{77}i}{132}&0&0&0\\
0&0&0&0&0&0&0&0&0&\frac{5\sqrt{77}i}{132}&0&\frac{\sqrt{1155}i}{132}&0&0\\
0&0&0&0&0&0&0&0&0&0&-\frac{5\sqrt{154}i}{264}&0&-\frac{\sqrt{770}i}{264}&0\\
0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{462}i}{264}&0&\frac{\sqrt{22}i}{264}
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 796 | symmetry | $\frac{3\sqrt{7}x^5y}{4} - \frac{15\sqrt{7}x^3yz^2}{2} - \frac{3\sqrt{7}xy^5}{4} + \frac{15\sqrt{7}xy^3z^2}{2}$ |
| 797 | symmetry | $-\frac{3\sqrt{7}x^6}{16} + \frac{15\sqrt{7}x^4y^2}{16} + \frac{15\sqrt{7}x^4z^2}{8} + \frac{15\sqrt{7}x^2y^4}{16} - \frac{45\sqrt{7}x^2y^2z^2}{4} - \frac{3\sqrt{7}y^6}{16} + \frac{15\sqrt{7}y^4z^2}{8}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 798 | symmetry | $\frac{\sqrt{210}x^5y}{16}+\frac{\sqrt{210}x^3y^3}{8}-\sqrt{210}x^3yz^2+\frac{\sqrt{210}xy^5}{16}-\sqrt{210}xy^3z^2+\sqrt{210}xyz^4$ |
| 799 | symmetry | $\frac{\sqrt{210}x^6}{32}+\frac{\sqrt{210}x^4y^2}{32}-\frac{\sqrt{210}x^4z^2}{2}-\frac{\sqrt{210}x^2y^4}{32}+\frac{\sqrt{210}x^2z^4}{2}-\frac{\sqrt{210}y^6}{32}+\frac{\sqrt{210}y^4z^2}{2}-\frac{\sqrt{210}y^2z^4}{2}$ |

$\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{154}}{44}&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2310}}{132}&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{165}}{66}\\
0&0&0&0&0&0&-\frac{\sqrt{165}}{66}&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{2310}}{132}&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{154}}{44}&0&0&0&0&0
\end{bmatrix}$$

$\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,4)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&-\frac{\sqrt{77}i}{66}&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{11}i}{66}&0&0&0&\frac{\sqrt{385}i}{66}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{154}i}{66}&0&0&0&-\frac{\sqrt{462}i}{66}&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{462}i}{66}&0&0&0&\frac{\sqrt{154}i}{66}&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{385}i}{66}&0&0&0&-\frac{\sqrt{11}i}{66}\\
0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{77}i}{66}&0&0&0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 800 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,4)$

$$\begin{bmatrix}
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{77}}{66}&0&0&0&0\\
0&0&0&0&0&0&-\frac{\sqrt{11}}{66}&0&0&0&-\frac{\sqrt{385}}{66}&0&0&0\\
0&0&0&0&0&0&0&\frac{\sqrt{154}}{66}&0&0&0&\frac{\sqrt{462}}{66}&0&0\\
0&0&0&0&0&0&0&0&-\frac{\sqrt{462}}{66}&0&0&0&-\frac{\sqrt{154}}{66}&0\\
0&0&0&0&0&0&0&0&0&\frac{\sqrt{385}}{66}&0&0&0&\frac{\sqrt{11}}{66}\\
0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{77}}{66}&0&0&0
\end{bmatrix}$$

| No. | multipole | matrix |
|---|---|---|
| 801 | symmetry | $\sqrt{3}yz$ |

$\mathbb{M}_{2}^{(1,0;a)}(A_{1u})$

$$\begin{bmatrix}
0&-\frac{\sqrt{15}}{35}&0&0&0&0&0&0&\frac{\sqrt{6}}{14}&0&0&0&0&0\\
0&0&-\frac{\sqrt{10}}{70}&0&0&0&0&0&0&\frac{3\sqrt{30}}{70}&0&0&0&0\\
0&0&0&\frac{\sqrt{10}}{70}&0&0&0&0&0&0&\frac{3\sqrt{30}}{70}&0&0&0\\
0&0&0&0&\frac{\sqrt{15}}{35}&0&0&0&0&0&0&\frac{\sqrt{6}}{14}&0&0\\
\frac{\sqrt{15}}{14}&0&0&0&0&0&0&\frac{\sqrt{10}}{28}&0&0&0&0&0&0\\
0&-\frac{\sqrt{15}}{70}&0&0&0&0&0&0&\frac{\sqrt{6}}{28}&0&0&0&0&0\\
0&0&-\frac{2\sqrt{15}}{35}&0&0&0&0&0&0&\frac{\sqrt{5}}{70}&0&0&0&0\\
0&0&0&-\frac{2\sqrt{15}}{35}&0&0&0&0&0&0&-\frac{\sqrt{5}}{70}&0&0&0\\
0&0&0&0&-\frac{\sqrt{15}}{70}&0&0&0&0&0&0&-\frac{\sqrt{6}}{28}&0&0\\
0&0&0&0&0&\frac{\sqrt{15}}{14}&0&0&0&0&0&0&-\frac{\sqrt{10}}{28}&0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 802 | symmetry | $-\sqrt{3}xz$ |
| 803 | symmetry | $\sqrt{3}xy$ |

$\mathbb{M}_{2,1}^{(1,0;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{i}{14} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{210} & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 \\
0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & -\frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & -\frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & \frac{\sqrt{6}i}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{168}
\end{bmatrix}$$

$\mathbb{M}_{2,2}^{(1,0;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\
0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\
0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 804 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |
| 805 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

$\mathbb{M}_{2,1}^{(1,0;a)}(E_u, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{2,2}^{(1,0;a)}(E_u, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\
0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{30}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 806 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| 807 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$\mathbb{M}_4^{(1,0;a)}(A_{1u},1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{2}}{140} & 0 & 0 \\
-\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0
\end{bmatrix}$$

$\mathbb{M}_4^{(1,0;a)}(A_{1u},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{35}}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{560} & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 \\
-\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{100} \\
0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & \frac{3\sqrt{30}}{400} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 \\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

| | | |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(A_{2u})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & -\frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 \\ -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} \\ 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & \frac{3\sqrt{30}i}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 \\ -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{200} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 808 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |
| | $\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{560} & 0 & -\frac{i}{56} & 0 & 0 & 0 & 0 & \frac{27\sqrt{15}i}{700} & 0 & \frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{175} & 0 & -\frac{9i}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & -\frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & -\frac{9i}{140} & 0 & -\frac{9\sqrt{15}i}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{56} & 0 & \frac{\sqrt{10}i}{560} & 0 & 0 & 0 & 0 & \frac{9\sqrt{3}i}{70} & 0 & \frac{27\sqrt{15}i}{700} & 0 \\ 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{280} & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{70} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{15}i}{1400} & 0 & -\frac{\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{2800} & 0 & -\frac{\sqrt{6}i}{80} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & -\frac{\sqrt{10}i}{2800} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{140} & 0 & \frac{13\sqrt{15}i}{1400} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{21}i}{280} \end{bmatrix}$ |
| 809 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 810 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |
| 811 | symmetry | $\frac{\sqrt{35}x^4}{8}-\frac{3\sqrt{35}x^2y^2}{4}+\frac{\sqrt{35}y^4}{8}$ |

$$\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)=\begin{bmatrix}
\frac{\sqrt{10}}{560} & 0 & -\frac{1}{56} & 0 & 0 & 0 & 0 & -\frac{27\sqrt{15}}{700} & 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{6}}{112} & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & \frac{9\sqrt{15}}{175} & 0 & -\frac{9}{140} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{3}}{56} & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & \frac{9}{140} & 0 & -\frac{9\sqrt{15}}{175} & 0 & 0\\
0 & 0 & 0 & -\frac{1}{56} & 0 & \frac{\sqrt{10}}{560} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{3}}{70} & 0 & \frac{27\sqrt{15}}{700} & 0\\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{280} & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{10}}{70} & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & \frac{13\sqrt{15}}{1400} & 0 & -\frac{\sqrt{3}}{140} & 0 & 0 & 0 & 0\\
0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{2800} & 0 & -\frac{\sqrt{6}}{80} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{10}}{2800} & 0 & 0\\
0 & 0 & 0 & \frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{140} & 0 & \frac{13\sqrt{15}}{1400} & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{21}}{280}
\end{bmatrix}$$

$$\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{700} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{100}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{350} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{100}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{350} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 812 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |
| 813 | symmetry | $\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$ |

$\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{350} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{350} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{4,1}^{(1,0;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & \frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\
\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & -\frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{\sqrt{15}i}{280} & 0 & 0 & \frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 \\
0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{1400} \\
0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\
\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & \frac{17\sqrt{5}i}{1400} & 0 & 0 \\
0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{15}i}{1400} & 0 \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{1400} \\
0 & 0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 9

| No. | multipole | matrix |
|---|---|---|

814 symmetry 1

$\mathbb{M}_{4,2}^{(1,0;a)}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & \frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{280} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{280} & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 \\
0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}}{1400} \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & \frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}}{1400} \\
0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}_{0}^{(1,1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

815 symmetry $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{M}_2^{(1,1;a)}(A_{1u})$

$$\begin{bmatrix}
0 & \frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\
\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{105} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0
\end{bmatrix}$$

816 symmetry $\sqrt{3}yz$

$\mathbb{M}_{2,1}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
-\frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\
0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42}
\end{bmatrix}$$

817 symmetry $-\sqrt{3}xz$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

$\mathbb{M}_{2,2}^{(1,1;a)}(E_u,1)$

$$\begin{bmatrix}
\frac{2\sqrt{15}}{35} & 0 & -\frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}}{7} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 \\
0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{15}}{70} & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & \frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & \frac{1}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & \frac{\sqrt{15}}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & -\frac{\sqrt{10}}{210} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{14}}{42}
\end{bmatrix}$$

818 symmetry $\sqrt{3}xy$

$\mathbb{M}_{2,1}^{(1,1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{280} & 0 & 0 & 0 \\
-\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 \\
0 & -\frac{8i}{35} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{280} & 0 \\
0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} \\
0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 \\
0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 \\
0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & \frac{\sqrt{70}i}{105} \\
0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0
\end{bmatrix}$$

819 symmetry $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

820 symmetry $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$

$\mathbb{M}_{2,2}^{(1,1;a)}(E_u,2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\
-\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\
0 & -\frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\
0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\
0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\
0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\
0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\
0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{105} & 0 & 0 & 0
\end{bmatrix}$$

821 symmetry $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$

$\mathbb{M}_4^{(1,1;a)}(A_{1u},1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{330}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{70} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{330}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{70} & 0 & 0 \\
-\frac{\sqrt{330}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{33}}{1155} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{385} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{33}}{1155} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

822 symmetry $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$

$\mathbb{M}_4^{(1,1;a)}(A_{1u}, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{70} & \frac{\sqrt{330}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 \\
\frac{\sqrt{385}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{300} \\
0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & -\frac{3\sqrt{55}}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 \\
-\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{550} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 & 0
\end{bmatrix}$$

823 symmetry $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$

$\mathbb{M}_4^{(1,1;a)}(A_{2u})$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{70} & \frac{\sqrt{330}i}{300} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 \\
\frac{\sqrt{385}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{300} \\
0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{770} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1650} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & -\frac{3\sqrt{55}i}{550} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{3850} & 0 \\
-\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{550} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

824 symmetry $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$

$$\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)=\begin{bmatrix}
\frac{\sqrt{165}i}{210} & 0 & \frac{\sqrt{66}i}{42} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{700} & 0 & -\frac{\sqrt{22}i}{70} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{11}i}{14} & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{175} & 0 & \frac{\sqrt{66}i}{420} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{22}i}{14} & 0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{420} & 0 & \frac{\sqrt{110}i}{175} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{66}i}{42} & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{70} & 0 & -\frac{3\sqrt{110}i}{700} & 0\\
0 & \frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{770} & 0 & -\frac{\sqrt{66}i}{231} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{165}i}{210} & 0 & -\frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & \frac{13\sqrt{110}i}{3850} & 0 & \frac{\sqrt{22}i}{385} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{11550} & 0 & \frac{\sqrt{11}i}{110} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{110} & 0 & \frac{\sqrt{165}i}{11550} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{66}i}{84} & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{385} & 0 & -\frac{13\sqrt{110}i}{3850} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{231} & 0 & \frac{\sqrt{154}i}{770}
\end{bmatrix}$$

825 symmetry $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$

$$\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)=\begin{bmatrix}
-\frac{\sqrt{165}}{210} & 0 & \frac{\sqrt{66}}{42} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{700} & 0 & -\frac{\sqrt{22}}{70} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{11}}{14} & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{175} & 0 & \frac{\sqrt{66}}{420} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{22}}{14} & 0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{420} & 0 & \frac{\sqrt{110}}{175} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}}{42} & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{70} & 0 & -\frac{3\sqrt{110}}{700} & 0\\
0 & \frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{770} & 0 & -\frac{\sqrt{66}}{231} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{165}}{210} & 0 & -\frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{110}}{3850} & 0 & \frac{\sqrt{22}}{385} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{11550} & 0 & \frac{\sqrt{11}}{110} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{110} & 0 & \frac{\sqrt{165}}{11550} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{66}}{84} & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{385} & 0 & -\frac{13\sqrt{110}}{3850} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{231} & 0 & \frac{\sqrt{154}}{770}
\end{bmatrix}$$

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| 826 | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ | $\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}i}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}i}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 827 | $\frac{\sqrt{5}x^3y}{2} + \frac{\sqrt{5}xy^3}{2} - 3\sqrt{5}xyz^2$ | $\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|

828 symmetry

$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$

$\mathbb{M}^{(1,1;a)}_{4,1}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{700} & 0 & 0 & 0 & \frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{70} & 0 & 0 & 0 & \frac{\sqrt{22}i}{14} & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & \frac{\sqrt{55}i}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{70} & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 & 0 & -\frac{11\sqrt{165}i}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & \frac{\sqrt{385}i}{700} \\
0 & 0 & -\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{33}i}{84} & 0 & 0 & \frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & \frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & \frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & -\frac{17\sqrt{330}i}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{140} & 0 & 0 & -\frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & -\frac{9\sqrt{110}i}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{385} & 0 & 0 & 0 & \frac{3\sqrt{385}i}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0
\end{bmatrix}$$

$\mathbb{M}^{(1,1;a)}_{4,2}(E_u,3)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & \frac{\sqrt{110}}{70} & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}}{700} \\
0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & \frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0
\end{bmatrix}$$

bra: $= \langle \frac{5}{2}, \frac{5}{2}; f|, \langle \frac{5}{2}, \frac{3}{2}; f|, \langle \frac{5}{2}, \frac{1}{2}; f|, \langle \frac{5}{2}, -\frac{1}{2}; f|, \langle \frac{5}{2}, -\frac{3}{2}; f|, \langle \frac{5}{2}, -\frac{5}{2}; f|, \langle \frac{7}{2}, \frac{7}{2}; f|, \langle \frac{7}{2}, \frac{5}{2}; f|, \langle \frac{7}{2}, \frac{3}{2}; f|, \langle \frac{7}{2}, \frac{1}{2}; f|, \langle \frac{7}{2}, -\frac{1}{2}; f|, \langle \frac{7}{2}, -\frac{3}{2}; f|, \langle \frac{7}{2}, -\frac{5}{2}; f|, \langle \frac{7}{2}, -\frac{7}{2}; f|$

ket: $= |\frac{5}{2}, \frac{5}{2}; f\rangle, |\frac{5}{2}, \frac{3}{2}; f\rangle, |\frac{5}{2}, \frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{1}{2}; f\rangle, |\frac{5}{2}, -\frac{3}{2}; f\rangle, |\frac{5}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{7}{2}; f\rangle, |\frac{7}{2}, \frac{5}{2}; f\rangle, |\frac{7}{2}, \frac{3}{2}; f\rangle, |\frac{7}{2}, \frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{1}{2}; f\rangle, |\frac{7}{2}, -\frac{3}{2}; f\rangle, |\frac{7}{2}, -\frac{5}{2}; f\rangle, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 10: (f,f) block.

| No. | multipole | matrix |
|---|---|---|
| 829 | symmetry | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 830 | symmetry | $\mathbb{Q}_0^{(a)}(A_{1g})$ | $\begin{bmatrix} \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \end{bmatrix}$ | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 831 | symmetry | $\sqrt{3}yz$ |

$\mathbb{Q}_2^{(a)}(A_{1g})$

$$\begin{bmatrix}
-\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 832 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{Q}_{2,1}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}i}{98} & 0 & \frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}i}{588} & 0 & -\frac{\sqrt{105}i}{588} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{196} & 0 & -\frac{5\sqrt{3}i}{84} \\
-\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & \frac{5\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{7}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{147} & 0 & \frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{21}i}{588} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{588} & 0 & -\frac{5\sqrt{42}i}{147} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{147} & 0 & -\frac{5\sqrt{2}i}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 833 | symmetry | $\sqrt{3}xy$ |

$\mathbb{Q}_{2,2}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{70}}{98} & 0 & \frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{588} & 0 & \frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{70}}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{11\sqrt{21}}{588} & 0 & -\frac{\sqrt{105}}{588} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{196} & 0 & -\frac{5\sqrt{3}}{84} \\
-\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{7}}{196} & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 \\
0 & \frac{11\sqrt{21}}{588} & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & \frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & -\frac{5\sqrt{42}}{147} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & -\frac{5\sqrt{2}}{28} \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 834 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$\mathbb{Q}_{2,1}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 \\
0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 \\
0 & 0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} \\
0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 & 0 \\
-\frac{\sqrt{105}i}{294} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 \\
0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{294} & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 \\
0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 835 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$\mathbb{Q}_{2,2}^{(a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 \\
0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 \\
0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 \\
-\frac{\sqrt{105}}{294} & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 \\
0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\
0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 836 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | | $\mathbb{Q}_4^{(a)}(A_{1g},1)$ |
| | | $\begin{bmatrix} \frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{39\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{77}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{77}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{77}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{39\sqrt{77}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 837 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$\mathbb{Q}_4^{(a)}(A_{1g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{14} & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 \\
\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 \\
\frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{154} \\
0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{154} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 838 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |

$\mathbb{Q}_4^{(a)}(A_{2g})$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{14} & -\frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{308} & 0 \\
\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}i}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & \frac{3\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 \\
\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}i}{154} \\
0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{154} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 839 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |

$$\mathbb{Q}_{4,1}^{(a)}(E_g,1)=\begin{bmatrix}
0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{154} & 0 & -\frac{5\sqrt{385}i}{539} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{154}i}{98} & 0 & \frac{\sqrt{385}i}{98} & 0 & 0 & 0 & 0 & \frac{13\sqrt{231}i}{1078} & 0 & \frac{\sqrt{1155}i}{539} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{385}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{2156} & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & \frac{\sqrt{154}i}{2156} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{385}i}{98} & 0 & \frac{\sqrt{154}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{539} & 0 & -\frac{13\sqrt{231}i}{1078} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{385}i}{539} & 0 & \frac{\sqrt{165}i}{154}\\
\frac{\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{154} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{13\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{154} & 0 & \frac{3\sqrt{2310}i}{1078} & 0 & 0 & 0 & 0 & 0\\
\frac{5\sqrt{385}i}{539} & 0 & \frac{\sqrt{154}i}{2156} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2310}i}{1078} & 0 & \frac{9\sqrt{462}i}{1078} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{1155}i}{539} & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{462}i}{1078} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & \frac{\sqrt{1155}i}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{462}i}{1078} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{154}i}{2156} & 0 & -\frac{5\sqrt{385}i}{539} & 0 & 0 & 0 & 0 & \frac{9\sqrt{462}i}{1078} & 0 & -\frac{3\sqrt{2310}i}{1078} & 0\\
0 & 0 & 0 & 0 & \frac{13\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}i}{1078} & 0 & \frac{3\sqrt{110}i}{154}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{154} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 840 | symmetry | $\mathbb{Q}_{4,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{154} & 0 & -\frac{5\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{154}}{98} & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{231}}{1078} & 0 & \frac{\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{2156} & 0 & \frac{\sqrt{2310}}{308} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{308} & 0 & \frac{\sqrt{154}}{2156} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{385}}{98} & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{539} & 0 & -\frac{13\sqrt{231}}{1078} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{539} & 0 & \frac{\sqrt{165}}{154} \\ \frac{\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{154} & 0 & \frac{3\sqrt{2310}}{1078} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{385}}{539} & 0 & \frac{\sqrt{154}}{2156} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{1078} & 0 & \frac{9\sqrt{462}}{1078} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{539} & 0 & \frac{\sqrt{2310}}{308} & 0 & 0 & 0 & 0 & \frac{9\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{308} & 0 & \frac{\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{462}}{1078} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{154}}{2156} & 0 & -\frac{5\sqrt{385}}{539} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{462}}{1078} & 0 & -\frac{3\sqrt{2310}}{1078} & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2310}}{1078} & 0 & \frac{3\sqrt{110}}{154} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{154} & 0 \end{bmatrix}$ | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 841 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

$\mathbb{Q}_{4,1}^{(a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{66}i}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{77} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 842 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

$$\mathbb{Q}_{4,2}^{(a)}(E_g,2)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{66}}{77} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{77}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{66}}{77} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{2\sqrt{66}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154}\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{2\sqrt{66}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 843 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$$\mathbb{Q}_{4,1}^{(a)}(E_g,3) = \begin{bmatrix}
0 & 0 & \frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & \frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 \\
-\frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & \frac{3\sqrt{154}i}{196} & 0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 \\
0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & \frac{3\sqrt{66}i}{154} \\
0 & 0 & 0 & -\frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{539} & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 & 0 \\
-\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{539} & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 \\
0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} \\
0 & 0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 844 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |

$$\mathbb{Q}_{4,2}^{(a)}(E_g,3)\left[\begin{array}{ccccccccccccc}
0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0\\
-\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0\\
0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0\\
0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154}\\
0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & 0\\
0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0\\
-\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0\\
0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0\\
0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154}\\
0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 845 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |

$\mathbb{Q}_6^{(a)}(A_{1g},1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 846 | $\mathbb{Q}_6^{(a)}(A_{1g},2)$ | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 847 | symmetry | $\mathbb{Q}_6^{(a)}(A_{1g},3)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 \\
-\frac{\sqrt{66}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} \\
0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 848 | symmetry | $\mathbb{Q}_6^{(a)}(A_{2g},1)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{9\sqrt{210}x^4yz}{16}-\frac{3\sqrt{210}x^2y^3z}{8}+\frac{3\sqrt{210}x^2yz^3}{2}+\frac{3\sqrt{210}y^5z}{16}-\frac{\sqrt{210}y^3z^3}{2}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 849 | symmetry | $\frac{15\sqrt{154}x^4yz}{16} - \frac{15\sqrt{154}x^2y^3z}{8} + \frac{3\sqrt{154}y^5z}{16}$ |

$\mathbb{Q}_6^{(a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} & -\frac{\sqrt{77}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 \\
-\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{77} \\
0 & \frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 850 | symmetry | $\mathbb{Q}_{6,1}^{(a)}(E_g,1)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{3\sqrt{154}x^5z}{16} - \frac{15\sqrt{154}x^3y^2z}{8} + \frac{15\sqrt{154}xy^4z}{16}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 851 | $\mathbb{Q}_{6,2}^{(a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\frac{5\sqrt{21}x^4yz}{8}+\frac{5\sqrt{21}x^2y^3z}{4}-\frac{5\sqrt{21}x^2yz^3}{2}+\frac{5\sqrt{21}y^5z}{8}-\frac{5\sqrt{21}y^3z^3}{2}+\sqrt{21}yz^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 852 | symmetry | $-\frac{5\sqrt{21}x^5z}{8}-\frac{5\sqrt{21}x^3y^2z}{4}+\frac{5\sqrt{21}x^3z^3}{2}-\frac{5\sqrt{21}xy^4z}{8}+\frac{5\sqrt{21}xy^2z^3}{2}-\sqrt{21}xz^5$ |

$$\mathbb{Q}_{6,1}^{(a)}(E_g,2)\left[\begin{array}{cccccccccccccc}
0&0&0&0&0&0&\frac{\sqrt{231}i}{924}&0&\frac{\sqrt{11}i}{44}&0&0&0&0&0\\
0&0&0&0&0&0&0&-\frac{\sqrt{165}i}{132}&0&-\frac{5\sqrt{33}i}{132}&0&0&0&0\\
0&0&0&0&0&0&0&0&\frac{\sqrt{110}i}{44}&0&\frac{5\sqrt{66}i}{132}&0&0&0\\
0&0&0&0&0&0&0&0&0&-\frac{5\sqrt{66}i}{132}&0&-\frac{\sqrt{110}i}{44}&0&0\\
0&0&0&0&0&0&0&0&0&0&\frac{5\sqrt{33}i}{132}&0&\frac{\sqrt{165}i}{132}&0\\
0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{11}i}{44}&0&-\frac{\sqrt{231}i}{924}\\
-\frac{\sqrt{231}i}{924}&0&0&0&0&0&0&\frac{\sqrt{154}i}{308}&0&0&0&0&0&0\\
0&\frac{\sqrt{165}i}{132}&0&0&0&0&-\frac{\sqrt{154}i}{308}&0&-\frac{\sqrt{66}i}{66}&0&0&0&0&0\\
-\frac{\sqrt{11}i}{44}&0&-\frac{\sqrt{110}i}{44}&0&0&0&0&\frac{\sqrt{66}i}{66}&0&\frac{\sqrt{330}i}{132}&0&0&0&0\\
0&\frac{5\sqrt{33}i}{132}&0&\frac{5\sqrt{66}i}{132}&0&0&0&0&-\frac{\sqrt{330}i}{132}&0&0&0&0&0\\
0&0&-\frac{5\sqrt{66}i}{132}&0&-\frac{5\sqrt{33}i}{132}&0&0&0&0&0&0&-\frac{\sqrt{330}i}{132}&0&0\\
0&0&0&\frac{\sqrt{110}i}{44}&0&\frac{\sqrt{11}i}{44}&0&0&0&0&\frac{\sqrt{330}i}{132}&0&\frac{\sqrt{66}i}{66}&0\\
0&0&0&0&-\frac{\sqrt{165}i}{132}&0&0&0&0&0&0&-\frac{\sqrt{66}i}{66}&0&-\frac{\sqrt{154}i}{308}\\
0&0&0&0&0&\frac{\sqrt{231}i}{924}&0&0&0&0&0&0&\frac{\sqrt{154}i}{308}&0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 853 | symmetry | $\frac{3\sqrt{7}x^5y}{4} - \frac{15\sqrt{7}x^3yz^2}{2} - \frac{3\sqrt{7}xy^5}{4} + \frac{15\sqrt{7}xy^3z^2}{2}$ |

$$\mathbb{Q}_{6,2}^{(a)}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{924} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & -\frac{5\sqrt{33}}{132} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{44} & 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{132} & 0 & -\frac{\sqrt{110}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{132} & 0 & \frac{\sqrt{165}}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{231}}{924} \\
-\frac{\sqrt{231}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{11}}{44} & 0 & -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{33}}{132} & 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5\sqrt{66}}{132} & 0 & -\frac{5\sqrt{33}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{110}}{44} & 0 & \frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & \frac{\sqrt{66}}{66} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & -\frac{\sqrt{154}}{308} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{308} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 854 | symmetry | $\mathbb{Q}_{6,1}^{(a)}(E_g,3)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{3\sqrt{7}x^6}{16}+\frac{15\sqrt{7}x^4y^2}{16}+\frac{15\sqrt{7}x^4z^2}{8}+\frac{15\sqrt{7}x^2y^4}{16}-\frac{45\sqrt{7}x^2y^2z^2}{4}-\frac{3\sqrt{7}y^6}{16}+\frac{15\sqrt{7}y^4z^2}{8}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 855 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

$\mathbb{Q}_{6,2}^{(a)}(E_g, 3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{308} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 856 | symmetry | $\frac{\sqrt{210}x^6}{32}+\frac{\sqrt{210}x^4y^2}{32}-\frac{\sqrt{210}x^4z^2}{2}-\frac{\sqrt{210}x^2y^4}{32}+\frac{\sqrt{210}x^2z^4}{2}-\frac{\sqrt{210}y^6}{32}+\frac{\sqrt{210}y^4z^2}{2}-\frac{\sqrt{210}y^2z^4}{2}$ |

$$\mathbb{Q}_{6,1}^{(a)}(E_g,4)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & \frac{\sqrt{462}i}{462}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0\\
0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & 0\\
-\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & \frac{\sqrt{330}i}{66} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0\\
0 & \frac{\sqrt{330}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0\\
0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462}\\
0 & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 857 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{Q}_{6,2}^{(a)}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{462} & 0 & 0 & 0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & -\frac{\sqrt{462}}{462}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & 0\\
0 & \frac{\sqrt{462}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0\\
-\frac{\sqrt{66}}{66} & 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0\\
0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0\\
0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{66} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462}\\
0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 858 | $\mathbb{Q}_2^{(1,-1;a)}(A_{1g})$ | $\begin{bmatrix} -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{bmatrix}$ | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 859 | symmetry | $-\sqrt{3}xz$ |

$\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{49} & 0 & \frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}i}{49} & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}i}{392} & 0 & -\frac{\sqrt{210}i}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & -\frac{5\sqrt{6}i}{56} \\
-\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{15\sqrt{14}i}{392} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{49} & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{42}i}{392} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & \frac{2\sqrt{21}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{49} & 0 & \frac{3i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 860 | symmetry | $\sqrt{3}xy$ |

$$\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,1)$$

$$\begin{bmatrix}
0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & \frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}}{49} & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{392} & 0 & \frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & \frac{11\sqrt{42}}{392} & 0 & -\frac{\sqrt{210}}{392} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{392} & 0 & -\frac{5\sqrt{6}}{56} \\
-\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\
\frac{15\sqrt{14}}{392} & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{11\sqrt{42}}{392} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & \frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & \frac{2\sqrt{21}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & \frac{3}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 861 | symmetry | $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 \\ 0 & 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{196} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 862 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 \\
0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} \\
0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{196} & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{\sqrt{3}}{14} \\
0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 863 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)$ | $\begin{bmatrix} \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{21}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 864 | $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 \\ \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 \\ \frac{5}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 \\ \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 \\ \frac{5i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} \\ 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 865 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4} - \frac{3\sqrt{10}y^3z}{4} + \sqrt{10}yz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 866 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |

$\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{5\sqrt{105}i}{294} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{42}i}{147} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{196} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{1176} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{1176} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{105}i}{147} & 0 & \frac{\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & -\frac{13\sqrt{7}i}{196} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}i}{294} & 0 & \frac{\sqrt{5}i}{28}\\
\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{13\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0\\
\frac{5\sqrt{105}i}{294} & 0 & \frac{\sqrt{42}i}{1176} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & -\frac{3\sqrt{14}i}{49} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{42}i}{1176} & 0 & -\frac{5\sqrt{105}i}{294} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0\\
0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & -\frac{\sqrt{30}i}{21}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 867 | symmetry | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{147} & 0 & \frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{7}}{196} & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{1176} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{1176} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & \frac{\sqrt{42}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & -\frac{13\sqrt{7}}{196} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & \frac{\sqrt{5}}{28} \\ \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{105}}{294} & 0 & \frac{\sqrt{42}}{1176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{98} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{1176} & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & -\frac{\sqrt{30}}{21} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 \end{bmatrix}$ | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 868 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

$\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 869 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

$\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 870 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$$\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,3) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 \\ -\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 \\ 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} \\ 0 & 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 871 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |
| | $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 \\ -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 \\ 0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{5}}{7} \\ 0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 872 | $\mathbb{Q}_6^{(1,-1;a)}(A_{1g},1)$ | symmetry |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{462} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{77}}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154}
\end{bmatrix}$$

$$\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 873 | $\mathbb{Q}_6^{(1,-1;a)}(A_{1g},2)$ | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 874 | $\mathbb{Q}_6^{(1,-1;a)}(A_{1g},3)$ | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 \\ -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} \\ 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} & 0 & 0 & 0 \end{bmatrix}$ | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 875 | $\mathbb{Q}_6^{(1,-1;a)}(A_{2g},1)$ symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 876 | symmetry | $\frac{15\sqrt{154}x^4yz}{16} - \frac{15\sqrt{154}x^2y^3z}{8} + \frac{3\sqrt{154}y^5z}{16}$ |

$\mathbb{Q}_6^{(1,-1;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & \frac{\sqrt{462}i}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 \\
-\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{77} \\
0 & \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 877 | $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\frac{3\sqrt{154}x^5z}{16} - \frac{15\sqrt{154}x^3y^2z}{8} + \frac{15\sqrt{154}xy^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 878 | symmetry | $\frac{5\sqrt{21}x^4yz}{8}+\frac{5\sqrt{21}x^2y^3z}{4}-\frac{5\sqrt{21}x^2yz^3}{2}+\frac{5\sqrt{21}y^5z}{8}-\frac{5\sqrt{21}y^3z^3}{2}+\sqrt{21}yz^5$ |
| | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 879 | symmetry | $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{1848} & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{264} & 0 & -\frac{5\sqrt{22}i}{264} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & \frac{5\sqrt{11}i}{132} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{132} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{264} & 0 & \frac{\sqrt{110}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{154}i}{1848} \\
-\frac{\sqrt{154}i}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{110}i}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{66}i}{264} & 0 & -\frac{\sqrt{165}i}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{22}i}{264} & 0 & \frac{5\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{11}i}{132} & 0 & -\frac{5\sqrt{22}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{165}i}{132} & 0 & \frac{\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 & -\frac{\sqrt{11}i}{11} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & \frac{\sqrt{231}i}{154} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} & 0
\end{bmatrix}$$

$$-\frac{5\sqrt{21}x^5z}{8} - \frac{5\sqrt{21}x^3y^2z}{4} + \frac{5\sqrt{21}x^3z^3}{2} - \frac{5\sqrt{21}xy^4z}{8} + \frac{5\sqrt{21}xy^2z^3}{2} - \sqrt{21}xz^5$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 880 | symmetry | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{1848} & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & -\frac{5\sqrt{22}}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{264} & 0 & \frac{\sqrt{110}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{154}}{1848} \\ -\frac{\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{264} & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{66}}{264} & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{22}}{264} & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & -\frac{5\sqrt{22}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & \frac{\sqrt{66}}{264} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & -\frac{\sqrt{11}}{11} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & \frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 \end{bmatrix}$ | $\frac{3\sqrt{7}x^5y}{4} - \frac{15\sqrt{7}x^3yz^2}{2} - \frac{3\sqrt{7}xy^5}{4} + \frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 881 | symmetry | $-\frac{3\sqrt{7}x^6}{16}+\frac{15\sqrt{7}x^4y^2}{16}+\frac{15\sqrt{7}x^4z^2}{8}+\frac{15\sqrt{7}x^2y^4}{16}-\frac{45\sqrt{7}x^2y^2z^2}{4}-\frac{3\sqrt{7}y^6}{16}+\frac{15\sqrt{7}y^4z^2}{8}$ |

$$\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154}\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 882 | symmetry | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 883 | symmetry | $\frac{\sqrt{210}x^6}{32}+\frac{\sqrt{210}x^4y^2}{32}-\frac{\sqrt{210}x^4z^2}{2}-\frac{\sqrt{210}x^2y^4}{32}+\frac{\sqrt{210}x^2z^4}{2}-\frac{\sqrt{210}y^6}{32}+\frac{\sqrt{210}y^4z^2}{2}-\frac{\sqrt{210}y^2z^4}{2}$ |

$$\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,4)=\begin{bmatrix}
0&0&0&0&0&0&0&0&0&\frac{\sqrt{11}i}{66}&0&0&0&0\\
0&0&0&0&0&0&\frac{\sqrt{77}i}{462}&0&0&0&-\frac{\sqrt{55}i}{66}&0&0&0\\
0&0&0&0&0&0&0&-\frac{\sqrt{22}i}{66}&0&0&0&\frac{\sqrt{66}i}{66}&0&0\\
0&0&0&0&0&0&0&0&\frac{\sqrt{66}i}{66}&0&0&0&-\frac{\sqrt{22}i}{66}&0\\
0&0&0&0&0&0&0&0&0&-\frac{\sqrt{55}i}{66}&0&0&0&\frac{\sqrt{77}i}{462}\\
0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{11}i}{66}&0&0&0\\
0&-\frac{\sqrt{77}i}{462}&0&0&0&0&0&0&-\frac{\sqrt{770}i}{154}&0&0&0&0&0\\
0&0&\frac{\sqrt{22}i}{66}&0&0&0&0&0&0&\frac{\sqrt{66}i}{22}&0&0&0&0\\
0&0&0&-\frac{\sqrt{66}i}{66}&0&0&\frac{\sqrt{770}i}{154}&0&0&0&-\frac{\sqrt{22}i}{22}&0&0&0\\
-\frac{\sqrt{11}i}{66}&0&0&0&\frac{\sqrt{55}i}{66}&0&0&-\frac{\sqrt{66}i}{22}&0&0&0&-\frac{\sqrt{22}i}{22}&0&0\\
0&\frac{\sqrt{55}i}{66}&0&0&0&-\frac{\sqrt{11}i}{66}&0&0&\frac{\sqrt{22}i}{22}&0&0&0&\frac{\sqrt{66}i}{22}&0\\
0&0&-\frac{\sqrt{66}i}{66}&0&0&0&0&0&0&\frac{\sqrt{22}i}{22}&0&0&0&-\frac{\sqrt{770}i}{154}\\
0&0&0&\frac{\sqrt{22}i}{66}&0&0&0&0&0&0&-\frac{\sqrt{66}i}{22}&0&0&0\\
0&0&0&0&-\frac{\sqrt{77}i}{462}&0&0&0&0&0&0&\frac{\sqrt{770}i}{154}&0&0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 884 | $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{462} & 0 & 0 & 0 & \frac{\sqrt{55}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{66} & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & 0 & \frac{\sqrt{22}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & 0 & -\frac{\sqrt{77}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 \\ -\frac{\sqrt{11}}{66} & 0 & 0 & 0 & -\frac{\sqrt{55}}{66} & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & \frac{\sqrt{55}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & \frac{\sqrt{770}}{154} \\ 0 & 0 & 0 & \frac{\sqrt{22}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{154} & 0 & 0 \end{bmatrix}$ | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 885 | symmetry | $\mathbb{Q}_0^{(1,1;a)}(A_{1g})$ |

$$\begin{bmatrix}
-\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28}
\end{bmatrix}$$

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 886 | $\mathbb{Q}_2^{(1,1;a)}(A_{1g})$ | $\begin{bmatrix} \frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \end{bmatrix}$ |
| | symmetry | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 887 | symmetry | $-\sqrt{3}xz$ |

$$\mathbb{Q}_{2,1}^{(1,1;a)}(E_g,1)$$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{105}i}{98} & 0 & -\frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{294} & 0 & -\frac{\sqrt{70}i}{294} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{294} & 0 & -\frac{5\sqrt{2}i}{42} \\
-\frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}i}{294} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{14}i}{294} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{\sqrt{3}i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 888 | symmetry | $\sqrt{3}xy$ |

$$\mathbb{Q}_{2,2}^{(1,1;a)}(E_g,1)$$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{42} & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{105}}{98} & 0 & -\frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{294} & 0 & \frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{147} & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & 0 & \frac{\sqrt{105}}{147} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & \frac{11\sqrt{14}}{294} & 0 & -\frac{\sqrt{70}}{294} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{294} & 0 & -\frac{5\sqrt{2}}{42} \\
-\frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{42}}{294} & 0 & \frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{11\sqrt{14}}{294} & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}}{21} & 0 & \frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{147} & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{3}}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 889 | symmetry | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

$$\mathbb{Q}_{2,1}^{(1,1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 \\ \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 \\ 0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 \\ 0 & 0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} \\ 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 \\ 0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{i}{14} \\ 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 890 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

$\mathbb{Q}_{2,2}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 \\
\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 \\
0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 \\
0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} \\
0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{147} & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 \\
0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 \\
0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{1}{14} \\
0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 891 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

$\mathbb{Q}_4^{(1,1;a)}(A_{1g},1)$

$$\begin{bmatrix}
-\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{16\sqrt{231}}{1617} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2156} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{2156} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{2156} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2156} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6468} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 892 | $\mathbb{Q}_4^{(1,1;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{42} & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 \\ -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{231} & 0 \\ \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{231} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} \\ 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{231} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 893 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |

$\mathbb{Q}_4^{(1,1;a)}(A_{2g})$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{42} & -\frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{385} & 0 \\
-\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{231} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{231} & 0 \\
\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{231} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\
0 & \frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{231} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 894 | symmetry | $\frac{3\sqrt{10}x^3z}{4}+\frac{3\sqrt{10}xy^2z}{4}-\sqrt{10}xz^3$ |

$$\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)=\begin{bmatrix}
0 & \frac{\sqrt{1155}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} & 0 & -\frac{10\sqrt{462}i}{1617} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{1155}i}{147} & 0 & -\frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & \frac{13\sqrt{770}i}{2695} & 0 & \frac{2\sqrt{154}i}{539} & 0 & 0 & 0 & 0\\
0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{8085} & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{77} & 0 & \frac{\sqrt{1155}i}{8085} & 0 & 0\\
0 & 0 & 0 & -\frac{5\sqrt{462}i}{294} & 0 & -\frac{\sqrt{1155}i}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}i}{539} & 0 & -\frac{13\sqrt{770}i}{2695} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{462}i}{1617} & 0 & \frac{\sqrt{22}i}{77}\\
\frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{462} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{13\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{462} & 0 & \frac{5\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 & 0\\
\frac{10\sqrt{462}i}{1617} & 0 & \frac{\sqrt{1155}i}{8085} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{1078} & 0 & \frac{3\sqrt{385}i}{1078} & 0 & 0 & 0 & 0\\
0 & -\frac{2\sqrt{154}i}{539} & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1078} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{77}i}{77} & 0 & \frac{2\sqrt{154}i}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1078} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{8085} & 0 & -\frac{10\sqrt{462}i}{1617} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{1078} & 0 & -\frac{5\sqrt{77}i}{1078} & 0\\
0 & 0 & 0 & 0 & \frac{13\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{1078} & 0 & \frac{5\sqrt{33}i}{462}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{462} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 895 | symmetry | $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

$\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{1155}}{147} & 0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{770}}{2695} & 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{8085} & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & \frac{\sqrt{1155}}{8085} & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & -\frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}}{539} & 0 & -\frac{13\sqrt{770}}{2695} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{462}}{1617} & 0 & \frac{\sqrt{22}}{77} \\
\frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{462} & 0 & \frac{5\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 \\
-\frac{10\sqrt{462}}{1617} & 0 & \frac{\sqrt{1155}}{8085} & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{1078} & 0 & \frac{3\sqrt{385}}{1078} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{154}}{539} & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{77}}{77} & 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1078} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{1155}}{8085} & 0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1078} & 0 & -\frac{5\sqrt{77}}{1078} & 0 \\
0 & 0 & 0 & 0 & -\frac{13\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{1078} & 0 & \frac{5\sqrt{33}}{462} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{462} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 896 | symmetry | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

$$\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2) \left[\begin{array}{cccccccccccccc}
0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}i}{231} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 897 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

$\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}}{231} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{55}}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{308} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{33}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{308} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 898 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,3)$ =

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & \frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 \\
\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 \\
0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{98} & 0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 \\
0 & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{6\sqrt{55}i}{385} \\
0 & 0 & 0 & \frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}i}{539} & 0 & 0 & 0 \\
0 & -\frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 & 0 \\
-\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 \\
0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{539} & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 \\
0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} \\
0 & 0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 899 | symmetry | $z$ |

$$\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,3)$$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\
\frac{\sqrt{1155}}{98} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 \\
0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 \\
0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} \\
0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & 0 \\
0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 \\
-\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 \\
0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 \\
0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} \\
0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix | |
|---|---|---|---|
| 900 | symmetry | $\mathbb{G}_1^{(1,0;a)}(A_{2g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | | | $x$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 901 | symmetry | $\mathbb{G}_{1,1}^{(1,0;a)}(E_g)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{6}i}{8} \\
-\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*y*

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 902 | symmetry | $\mathbb{G}_{1,2}^{(1,0;a)}(E_g)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{210}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{6}}{8} \\ -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 10

| | | No. | multipole | matrix |
|---|---|---|---|---|
| | | 903 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| $\mathbb{G}_3^{(1,0;a)}(A_{1g})$ | | | | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 904 | symmetry | $\mathbb{G}_3^{(1,0;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 905 | symmetry | $\mathbb{G}_3^{(1,0;a)}(A_{2g},2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 906 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{12} & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & -\frac{\sqrt{5}i}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{7}i}{12} \\
\frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{i}{6} & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2}i}{24} & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 907 | symmetry | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{12} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{12} & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & -\frac{\sqrt{5}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{7}}{12} \\ \frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{6} & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 908 | symmetry | $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\sqrt{15}xyz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 909 | symmetry | $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{1}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 910 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{G}_5^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 911 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

$$\mathbb{G}_5^{(1,0;a)}(A_{2g},1)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 912 | $\mathbb{G}_5^{(1,0;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 913 | symmetry | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 914 | symmetry | $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{\sqrt{15}x^5}{8} + \frac{\sqrt{15}x^3y^2}{4} - \frac{3\sqrt{15}x^3z^2}{2} + \frac{\sqrt{15}xy^4}{8} - \frac{3\sqrt{15}xy^2z^2}{2} + \sqrt{15}xz^4$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 915 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

$$\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{70}i}{840} & 0 & \frac{13\sqrt{14}i}{168} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{105}i}{420} & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{84} & 0 & -\frac{11\sqrt{105}i}{420} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}i}{168} & 0 & \frac{23\sqrt{70}i}{840} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & -\frac{\sqrt{2}i}{24}\\
-\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{23\sqrt{70}i}{840} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{5\sqrt{42}i}{168} & 0 & -\frac{11\sqrt{105}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{13\sqrt{14}i}{168} & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{7}i}{84} & 0 & \frac{13\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{11\sqrt{105}i}{420} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & -\frac{23\sqrt{70}i}{840} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 916 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |

$$\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{70}}{840} & 0 & \frac{13\sqrt{14}}{168} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{105}}{420} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & -\frac{11\sqrt{105}}{420} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}}{168} & 0 & \frac{23\sqrt{70}}{840} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & -\frac{\sqrt{2}}{24}\\
-\frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{23\sqrt{70}}{840} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{42}}{168} & 0 & -\frac{11\sqrt{105}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{13\sqrt{14}}{168} & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{7}}{84} & 0 & \frac{13\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{11\sqrt{105}}{420} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{23\sqrt{70}}{840} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 917 | symmetry | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 918 | $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)$ | symmetry |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 919 | symmetry | $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,4)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 920 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

$\mathbb{G}_{5,2}^{(1,0;a)}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{60} & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & \frac{\sqrt{70}}{60} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 921 | symmetry | $\mathbb{T}_2^{(1,0;a)}(A_{1g})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\sqrt{3}yz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 922 | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{24} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & \frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{12} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{70}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{2}}{24} \\ \frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{12} & 0 & -\frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $-\sqrt{3}xz$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 923 | symmetry | $\sqrt{3}xy$ |

$$\mathbb{T}_{2,2}^{(1,0;a)}(E_g,1)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{24} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & -\frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{168} & 0 & \frac{\sqrt{70}i}{168} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & \frac{5\sqrt{2}i}{24} \\
-\frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{42}i}{168} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{11\sqrt{14}i}{168} & 0 & \frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}i}{12} & 0 & \frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 924 | symmetry | $\mathbb{T}_{2,1}^{(1,0;a)}(E_g,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{84} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 925 | symmetry | $\mathbb{T}_{2,2}^{(1,0;a)}(E_g,2)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{35}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 926 | symmetry | $\mathbb{T}_4^{(1,0;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}i}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 927 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

$\mathbb{T}_4^{(1,0;a)}(A_{1g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{440} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}i}{440} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 928 | symmetry | $-\frac{3\sqrt{10}x^2yz}{4}-\frac{3\sqrt{10}y^3z}{4}+\sqrt{10}yz^3$ |

$$\mathbb{T}_4^{(1,0;a)}(A_{2g})=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}}{220} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{440} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{440}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{4,1}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{1155}}{1540} & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{3080} & 0 & \frac{\sqrt{462}}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 & \frac{\sqrt{770}}{3080} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{154} & 0 & -\frac{13\sqrt{1155}}{1540} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & \frac{\sqrt{33}}{44} \\ -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{77}}{154} & 0 & -\frac{\sqrt{770}}{3080} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}}{154} & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{88} & 0 & -\frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}}{3080} & 0 & \frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 929 | symmetry | $\frac{3\sqrt{10}x^3z}{4} + \frac{3\sqrt{10}xy^2z}{4} - \sqrt{10}xz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 930 | symmetry | $\mathbb{T}^{(1,0;a)}_{4,2}(E_g,1)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{1155}i}{1540} & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{3080} & 0 & -\frac{\sqrt{462}i}{88} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{88} & 0 & -\frac{\sqrt{770}i}{3080} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & \frac{13\sqrt{1155}i}{1540} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & -\frac{\sqrt{33}i}{44} \\ \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{77}i}{154} & 0 & \frac{\sqrt{770}i}{3080} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{154} & 0 & \frac{\sqrt{462}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}i}{88} & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}i}{3080} & 0 & -\frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{\sqrt{35}x^3y}{2}+\frac{\sqrt{35}xy^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 931 | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{\sqrt{35}x^4}{8} - \frac{3\sqrt{35}x^2y^2}{4} + \frac{\sqrt{35}y^4}{8}$ |

*continued ...*

$\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 2)$

Table 10

| No. | multipole | matrix |
|---|---|---|
| 932 | symmetry | $\frac{\sqrt{5}x^3y}{2}+\frac{\sqrt{5}xy^3}{2}-3\sqrt{5}xyz^2$ |

$\mathbb{T}^{(1,0;a)}_{4,2}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{55} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{110} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{55} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}i}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{1155}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 933 | symmetry | $\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$ |

$$\mathbb{T}_{4,1}^{(1,0;a)}(E_g,3) \left[\begin{array}{cccccccccccccc}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{220} & 0 & 0 & 0 & \frac{\sqrt{462}}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1540} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & -\frac{9\sqrt{1155}}{1540} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{308} & 0 & 0 & 0 & \frac{3\sqrt{330}}{220} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{330}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2310}}{154} & 0 & 0 & 0 & \frac{\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{462}}{308} & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 934 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |

$$\mathbb{T}^{(1,0;a)}_{4,2}(E_g,3)\left[\begin{array}{ccccccccccccc}
0&0&0&0&0&0&0&0&0&\frac{\sqrt{2310}i}{154}&0&0&0&0\\
0&0&0&0&0&0&-\frac{3\sqrt{330}i}{220}&0&0&0&\frac{\sqrt{462}i}{308}&0&0&0\\
0&0&0&0&0&0&0&\frac{9\sqrt{1155}i}{1540}&0&0&0&-\frac{17\sqrt{385}i}{1540}&0&0\\
0&0&0&0&0&0&0&0&\frac{17\sqrt{385}i}{1540}&0&0&0&-\frac{9\sqrt{1155}i}{1540}&0\\
0&0&0&0&0&0&0&0&0&-\frac{\sqrt{462}i}{308}&0&0&0&\frac{3\sqrt{330}i}{220}\\
0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2310}i}{154}&0&0&0\\
0&\frac{3\sqrt{330}i}{220}&0&0&0&0&0&0&0&0&0&0&0&0\\
0&0&-\frac{9\sqrt{1155}i}{1540}&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&-\frac{17\sqrt{385}i}{1540}&0&0&0&0&0&0&0&0&0&0\\
-\frac{\sqrt{2310}i}{154}&0&0&0&\frac{\sqrt{462}i}{308}&0&0&0&0&0&0&0&0&0\\
0&-\frac{\sqrt{462}i}{308}&0&0&0&\frac{\sqrt{2310}i}{154}&0&0&0&0&0&0&0&0\\
0&0&\frac{17\sqrt{385}i}{1540}&0&0&0&0&0&0&0&0&0&0&0\\
0&0&0&\frac{9\sqrt{1155}i}{1540}&0&0&0&0&0&0&0&0&0&0\\
0&0&0&0&-\frac{3\sqrt{330}i}{220}&0&0&0&0&0&0&0&0&0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 935 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |
| | $\mathbb{T}_6^{(1,0;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A_{1g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 936 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 937 | symmetry | $\mathbb{T}_6^{(1,0;a)}(A_{1g},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 938 | $\mathbb{T}_6^{(1,0;a)}(A_{2g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 939 | $\mathbb{T}_6^{(1,0;a)}(A_{2g},2)$ | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\frac{15\sqrt{154}x^4yz}{16} - \frac{15\sqrt{154}x^2y^3z}{8} + \frac{3\sqrt{154}y^5z}{16}$ |

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 940 | $\mathbb{T}^{(1,0;a)}_{6,1}(E_g,1)$ | symmetry | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $\frac{3\sqrt{154}x^5z}{16} - \frac{15\sqrt{154}x^3y^2z}{8} + \frac{15\sqrt{154}xy^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}^{(1,0;a)}_{6,2}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 941 | symmetry | $\frac{5\sqrt{21}x^4yz}{8} + \frac{5\sqrt{21}x^2y^3z}{4} - \frac{5\sqrt{21}x^2yz^3}{2} + \frac{5\sqrt{21}y^5z}{8} - \frac{5\sqrt{21}y^3z^3}{2} + \sqrt{21}yz^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 942 | symmetry | $-\frac{5\sqrt{21}x^5z}{8}-\frac{5\sqrt{21}x^3y^2z}{4}+\frac{5\sqrt{21}x^3z^3}{2}-\frac{5\sqrt{21}xy^4z}{8}+\frac{5\sqrt{21}xy^2z^3}{2}-\sqrt{21}xz^5$ |

$$\mathbb{T}^{(1,0;a)}_{6,1}(E_g,2)=\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{264} & 0 & \frac{\sqrt{462}}{264} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{264} & 0 & -\frac{5\sqrt{154}}{264} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{132} & 0 & \frac{5\sqrt{77}}{132} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{132} & 0 & -\frac{\sqrt{1155}}{132} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{264} & 0 & \frac{\sqrt{770}}{264} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{264} & 0 & -\frac{\sqrt{22}}{264}\\
\frac{\sqrt{22}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{770}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{462}}{264} & 0 & \frac{\sqrt{1155}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{5\sqrt{154}}{264} & 0 & -\frac{5\sqrt{77}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & \frac{5\sqrt{77}}{132} & 0 & \frac{5\sqrt{154}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{1155}}{132} & 0 & -\frac{\sqrt{462}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{770}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 943 | symmetry | $\mathbb{T}^{(1,0;a)}_{6,2}(E_g,2)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{264} & 0 & -\frac{\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 & \frac{5\sqrt{154}i}{264} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{132} & 0 & -\frac{5\sqrt{77}i}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{132} & 0 & \frac{\sqrt{1155}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{264} & 0 & -\frac{\sqrt{770}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{264} & 0 & \frac{\sqrt{22}i}{264} \\ -\frac{\sqrt{22}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{462}i}{264} & 0 & -\frac{\sqrt{1155}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{154}i}{264} & 0 & \frac{5\sqrt{77}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{77}i}{132} & 0 & -\frac{5\sqrt{154}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}i}{132} & 0 & \frac{\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ $\frac{3\sqrt{7}x^5y}{4} - \frac{15\sqrt{7}x^3yz^2}{2} - \frac{3\sqrt{7}xy^5}{4} + \frac{15\sqrt{7}xy^3z^2}{2}$ |

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 944 | symmetry | $-\frac{3\sqrt{7}x^6}{16}+\frac{15\sqrt{7}x^4y^2}{16}+\frac{15\sqrt{7}x^4z^2}{8}+\frac{15\sqrt{7}x^2y^4}{16}-\frac{45\sqrt{7}x^2y^2z^2}{4}-\frac{3\sqrt{7}y^6}{16}+\frac{15\sqrt{7}y^4z^2}{8}$ |

$$\mathbb{T}^{(1,0;a)}_{6,1}(E_g,3)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 945 | symmetry | $\frac{\sqrt{210}x^5y}{16} + \frac{\sqrt{210}x^3y^3}{8} - \sqrt{210}x^3yz^2 + \frac{\sqrt{210}xy^5}{16} - \sqrt{210}xy^3z^2 + \sqrt{210}xyz^4$ |

$\mathbb{T}^{(1,0;a)}_{6,2}(E_g, 3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 946 | symmetry | $\frac{\sqrt{210}x^6}{32}+\frac{\sqrt{210}x^4y^2}{32}-\frac{\sqrt{210}x^4z^2}{2}-\frac{\sqrt{210}x^2y^4}{32}+\frac{\sqrt{210}x^2z^4}{2}-\frac{\sqrt{210}y^6}{32}+\frac{\sqrt{210}y^4z^2}{2}-\frac{\sqrt{210}y^2z^4}{2}$ |
| | $\mathbb{T}^{(1,0;a)}_{6,1}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 947 | symmetry | $z$ |

$\mathbb{T}^{(1,0;a)}_{6,2}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{66} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{66} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & \frac{\sqrt{462}i}{66} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{66} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 & \frac{\sqrt{11}i}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{66} & 0 & 0 & 0 \\
0 & \frac{\sqrt{11}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{462}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{77}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 & \frac{\sqrt{77}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 948 | symmetry | $x$ |

$\mathbb{M}_1^{(a)}(A_{2g})$

$$\begin{bmatrix}
\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{28}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 949 | symmetry | $y$ |

$\mathbb{M}_{1,1}^{(a)}(E_g)$

$$\begin{bmatrix}
0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{196} & 0 & \frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{7}}{49} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & \frac{\sqrt{42}}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{196} & 0 & \frac{\sqrt{70}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{196} & 0 & \frac{\sqrt{105}}{196} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{196} & 0 & \frac{\sqrt{3}}{28} \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{196} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{196} & 0 & -\frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & -\frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{42}}{98} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{2}}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 950 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$\mathbb{M}_{1,2}^{(a)}(E_g)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}i}{49} & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{196} & 0 & -\frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & -\frac{\sqrt{42}i}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}i}{49} & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{196} & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{196} & 0 & -\frac{\sqrt{105}i}{196} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{196} & 0 & -\frac{\sqrt{3}i}{28} \\
\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & -\frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{98} & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & -\frac{3\sqrt{14}i}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{196} & 0 & \frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{49} & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & -\frac{3\sqrt{42}i}{98} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{98} & 0 & -\frac{3\sqrt{2}i}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 951 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |

$\mathbb{M}_3^{(a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 \\
-\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\
0 & -\frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 \\
-\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} \\
0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 952 | symmetry | $\mathbb{M}_3^{(a)}(A_{2g},1)$ | $\begin{bmatrix} -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 953 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |
| | $\mathbb{M}_3^{(a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\ -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 954 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$\mathbb{M}^{(a)}_{3,1}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}}{14} & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\
0 & \frac{1}{14} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2}}{7} & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & \frac{\sqrt{15}}{42} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & -\frac{\sqrt{21}}{42} \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{1}{7} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{21} & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{84} & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & \frac{\sqrt{30}}{42} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & -\frac{\sqrt{6}}{42} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{14}}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 955 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{M}_{3,2}^{(a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}i}{14} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & \frac{i}{14} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{i}{14} & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & -\frac{\sqrt{15}i}{42} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & \frac{\sqrt{21}i}{42} \\
-\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{i}{7} & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{21} & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{84} & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{i}{7} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & \frac{\sqrt{14}i}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 956 | symmetry | $\sqrt{15}xyz$ |

$\mathbb{M}^{(a)}_{3,1}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 \\
\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\
0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\
0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{14} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 \\
\frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 \\
0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{3}{14} & 0 \\
0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} \\
0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 957 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$$\mathbb{M}_{3,2}^{(a)}(E_g,2)=\begin{bmatrix}
0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0\\
-\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & \frac{i}{14} & 0 & 0\\
0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0\\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42}\\
0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0\\
0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{i}{14} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0\\
-\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0\\
0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{3i}{14} & 0\\
0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42}\\
0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 958 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

$\mathbb{M}_5^{(a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 \\
\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 \\
\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} \\
0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 959 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(a)}(A_{2g},1)$ | $\begin{bmatrix} \frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{42}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 960 | symmetry | $\mathbb{M}_5^{(a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 \end{bmatrix}$ | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 961 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |

$\mathbb{M}^{(a)}_{5,1}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 962 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |

$\mathbb{M}^{(a)}_{5,2}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 963 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

$$\mathbb{M}^{(a)}_{5,1}(E_g,2)=\begin{bmatrix}
0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & \frac{5\sqrt{35}}{196} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{14}}{98} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & \frac{23\sqrt{21}}{588} & 0 & -\frac{13\sqrt{105}}{588} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{35}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{196} & 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{70}}{49} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{588} & 0 & \frac{11\sqrt{14}}{196} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & \frac{13\sqrt{105}}{588} & 0 & -\frac{23\sqrt{21}}{588} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{196} & 0 & \frac{\sqrt{15}}{84}\\
-\frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{23\sqrt{21}}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0\\
\frac{5\sqrt{35}}{196} & 0 & -\frac{11\sqrt{14}}{196} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & \frac{\sqrt{42}}{588} & 0 & 0 & 0 & 0\\
0 & -\frac{13\sqrt{105}}{588} & 0 & -\frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{588} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0\\
0 & 0 & \frac{\sqrt{210}}{588} & 0 & \frac{13\sqrt{105}}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & \frac{\sqrt{42}}{588} & 0 & 0\\
0 & 0 & 0 & \frac{11\sqrt{14}}{196} & 0 & -\frac{5\sqrt{35}}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{588} & 0 & -\frac{2\sqrt{210}}{147} & 0\\
0 & 0 & 0 & 0 & -\frac{23\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & \frac{\sqrt{10}}{28}\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 964 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |

$\mathbb{M}_{5,2}^{(a)}(E_g,2)$ =

$$\begin{bmatrix}
0 & -\frac{\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & -\frac{5\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{14}i}{98} & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & \frac{23\sqrt{21}i}{588} & 0 & \frac{13\sqrt{105}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{196} & 0 & -\frac{\sqrt{210}i}{588} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}i}{49} & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{588} & 0 & -\frac{11\sqrt{14}i}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & \frac{13\sqrt{105}i}{588} & 0 & \frac{23\sqrt{21}i}{588} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{196} & 0 & -\frac{\sqrt{15}i}{84} \\
\frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{23\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{2\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{35}i}{196} & 0 & \frac{11\sqrt{14}i}{196} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}i}{147} & 0 & -\frac{\sqrt{42}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{105}i}{588} & 0 & \frac{\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{588} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{210}i}{588} & 0 & -\frac{13\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & -\frac{\sqrt{42}i}{588} & 0 & 0 \\
0 & 0 & 0 & \frac{11\sqrt{14}i}{196} & 0 & \frac{5\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{588} & 0 & \frac{2\sqrt{210}i}{147} & 0 \\
0 & 0 & 0 & 0 & -\frac{23\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}i}{147} & 0 & -\frac{\sqrt{10}i}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 965 | symmetry | $\mathbb{M}^{(a)}_{5,1}(E_g,3)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 966 | $\mathbb{M}^{(a)}_{5,2}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 967 | symmetry | $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |

$\mathbb{M}^{(a)}_{5,1}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0\\
-\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0\\
0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0\\
0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & \frac{\sqrt{21}}{42}\\
0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0\\
0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0\\
-\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0\\
0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0\\
0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{\sqrt{210}}{84}\\
0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 968 | symmetry | $z$ |

$\mathbb{M}^{(a)}_{5,2}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 \\
0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} \\
0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 \\
\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} \\
0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 969 | symmetry | $x$ |

$\mathbb{M}_1^{(1,-1;a)}(A_{2g})$

$$\begin{bmatrix}
-\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 970 | symmetry | $y$ |

$$\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)$$

$$\begin{bmatrix}
0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & -\frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & -\frac{\sqrt{105}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{49} & 0 & -\frac{\sqrt{3}}{7} \\
\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}}{49} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{2\sqrt{14}}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{\sqrt{42}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{2}}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 971 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)$

$$\begin{bmatrix}
0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & \frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{98} & 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & \frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{7}i}{49} & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{14}i}{98} & 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{49} & 0 & \frac{\sqrt{105}i}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{49} & 0 & \frac{\sqrt{3}i}{7} \\
-\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{49} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & -\frac{\sqrt{210}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{49} & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & -\frac{2\sqrt{14}i}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & -\frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{49} & 0 & -\frac{\sqrt{210}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & -\frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & -\frac{\sqrt{42}i}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & -\frac{\sqrt{2}i}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 972 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |

$\mathbb{M}_3^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 \\
\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} \\
0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{49} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{147} & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{49} & 0 \\
\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} \\
0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{49} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 973 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 974 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |

$\mathbb{M}_3^{(1,-1;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 \\
\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} \\
0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 \\
\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} \\
0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 975 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{49} & 0 & -\frac{\sqrt{210}}{490} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{147} & 0 & \frac{2\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}}{490} & 0 & -\frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{147} & 0 & -\frac{\sqrt{35}}{147} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{105}}{245} & 0 & -\frac{\sqrt{210}}{490} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}}{490} & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{147} & 0 & -\frac{5\sqrt{14}}{147} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & \frac{\sqrt{10}}{21} \\
-\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{14}}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{210}}{147} & 0 & \frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{70}}{147} & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}}{147} & 0 & -\frac{2\sqrt{70}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & -\frac{\sqrt{35}}{98} & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & -\frac{\sqrt{15}}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 976 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & -\frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & -\frac{2\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{49} & 0 & \frac{\sqrt{210}i}{490} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{147} & 0 & -\frac{2\sqrt{70}i}{147} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{490} & 0 & \frac{2\sqrt{105}i}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{147} & 0 & \frac{\sqrt{35}i}{147} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{105}i}{245} & 0 & \frac{\sqrt{210}i}{490} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{147} & 0 & \frac{5\sqrt{21}i}{147} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{490} & 0 & -\frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}i}{147} & 0 & \frac{5\sqrt{14}i}{147} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}i}{147} & 0 & -\frac{\sqrt{10}i}{21} \\
\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{14}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{210}i}{147} & 0 & -\frac{5\sqrt{21}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{70}i}{147} & 0 & -\frac{\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}i}{147} & 0 & \frac{2\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{21}i}{147} & 0 & \frac{2\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & \frac{\sqrt{35}i}{98} & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{15}i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 977 | symmetry | $\sqrt{15}xyz$ |

$\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & -\frac{10\sqrt{7}}{147} & 0 & 0 & 0 \\
-\frac{\sqrt{105}}{98} & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 \\
0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{7}}{147} & 0 & 0 & 0 & \frac{2\sqrt{5}}{21} \\
0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 \\
0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 \\
-\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & -\frac{10\sqrt{7}}{147} & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 \\
0 & -\frac{10\sqrt{7}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 \\
0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} \\
0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 978 | symmetry | $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)$ $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{21} \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 \\ \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 \\ 0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 \end{bmatrix}$ $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 979 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

$\mathbb{M}_5^{(1,-1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{385} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{231} & \frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{385} & 0 \\
-\frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{385} \\
0 & \frac{5\sqrt{22}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{77} & \frac{\sqrt{770}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{154} & 0 \\
-\frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{77} \\
0 & \frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 980 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)$ | $\begin{bmatrix} -\frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{385}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{385}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154} \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 981 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |

$\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{77} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{385} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{231} & \frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{385} & 0 \\
-\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{385} \\
0 & \frac{5\sqrt{22}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & \frac{\sqrt{770}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 \\
-\frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} \\
0 & \frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 982 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |

$\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{77} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{231}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 983 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |

$$\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 984 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

$$\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)=\begin{bmatrix}
0 & -\frac{\sqrt{1155}}{3234} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{154} & 0 & -\frac{5\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{1155}}{3234} & 0 & \frac{5\sqrt{462}}{3234} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{770}}{5390} & 0 & \frac{13\sqrt{154}}{1078} & 0 & 0 & 0 & 0\\
0 & \frac{5\sqrt{462}}{3234} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{245} & 0 & -\frac{\sqrt{77}}{539} & 0 & 0 & 0\\
0 & 0 & -\frac{5\sqrt{231}}{1617} & 0 & \frac{5\sqrt{462}}{3234} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{539} & 0 & -\frac{\sqrt{1155}}{245} & 0 & 0\\
0 & 0 & 0 & \frac{5\sqrt{462}}{3234} & 0 & -\frac{\sqrt{1155}}{3234} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{154}}{1078} & 0 & \frac{23\sqrt{770}}{5390} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{1078} & 0 & -\frac{\sqrt{22}}{154}\\
\frac{\sqrt{22}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{23\sqrt{770}}{5390} & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & -\frac{20\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{462}}{1078} & 0 & \frac{\sqrt{1155}}{245} & 0 & 0 & 0 & 0 & -\frac{20\sqrt{77}}{539} & 0 & \frac{\sqrt{385}}{1078} & 0 & 0 & 0 & 0\\
0 & \frac{13\sqrt{154}}{1078} & 0 & \frac{\sqrt{77}}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1078} & 0 & \frac{10\sqrt{231}}{539} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{77}}{539} & 0 & -\frac{13\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & \frac{10\sqrt{231}}{539} & 0 & \frac{\sqrt{385}}{1078} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{1155}}{245} & 0 & \frac{5\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1078} & 0 & -\frac{20\sqrt{77}}{539} & 0\\
0 & 0 & 0 & 0 & \frac{23\sqrt{770}}{5390} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{77}}{539} & 0 & \frac{5\sqrt{33}}{154}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 985 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |

$\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & \frac{\sqrt{1155}i}{3234} & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{154} & 0 & \frac{5\sqrt{462}i}{1078} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{1155}i}{3234} & 0 & -\frac{5\sqrt{462}i}{3234} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{770}i}{5390} & 0 & -\frac{13\sqrt{154}i}{1078} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{462}i}{3234} & 0 & \frac{5\sqrt{231}i}{1617} & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{245} & 0 & \frac{\sqrt{77}i}{539} & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{231}i}{1617} & 0 & -\frac{5\sqrt{462}i}{3234} & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{539} & 0 & \frac{\sqrt{1155}i}{245} & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{462}i}{3234} & 0 & \frac{\sqrt{1155}i}{3234} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{154}i}{1078} & 0 & -\frac{23\sqrt{770}i}{5390} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{1078} & 0 & \frac{\sqrt{22}i}{154} \\
-\frac{\sqrt{22}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{23\sqrt{770}i}{5390} & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & \frac{20\sqrt{77}i}{539} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{462}i}{1078} & 0 & -\frac{\sqrt{1155}i}{245} & 0 & 0 & 0 & 0 & -\frac{20\sqrt{77}i}{539} & 0 & -\frac{\sqrt{385}i}{1078} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{154}i}{1078} & 0 & -\frac{\sqrt{77}i}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{1078} & 0 & -\frac{10\sqrt{231}i}{539} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{77}i}{539} & 0 & \frac{13\sqrt{154}i}{1078} & 0 & 0 & 0 & 0 & \frac{10\sqrt{231}i}{539} & 0 & -\frac{\sqrt{385}i}{1078} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{245} & 0 & -\frac{5\sqrt{462}i}{1078} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{1078} & 0 & \frac{20\sqrt{77}i}{539} & 0 \\
0 & 0 & 0 & 0 & \frac{23\sqrt{770}i}{5390} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{77}i}{539} & 0 & -\frac{5\sqrt{33}i}{154} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 986 | symmetry | $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 987 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

$\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}i}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 988 | symmetry | $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |

$\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,4)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & -\frac{\sqrt{770}}{385} & 0 & 0 & 0 & -\frac{2\sqrt{22}}{77} & 0 & 0 & 0 \\
\frac{\sqrt{330}}{924} & 0 & 0 & 0 & \frac{5\sqrt{66}}{924} & 0 & 0 & \frac{8\sqrt{55}}{385} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 \\
0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & 0 & \frac{8\sqrt{55}}{385} & 0 \\
0 & 0 & \frac{5\sqrt{66}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{22}}{77} & 0 & 0 & 0 & -\frac{\sqrt{770}}{385} \\
0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 & 0 \\
\frac{\sqrt{110}}{77} & 0 & 0 & 0 & -\frac{2\sqrt{22}}{77} & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 \\
0 & -\frac{2\sqrt{22}}{77} & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 \\
0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & \frac{5\sqrt{77}}{154} \\
0 & 0 & 0 & \frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 989 | symmetry | $\frac{3\sqrt{6006}x^5yz}{16} - \frac{5\sqrt{6006}x^3y^3z}{8} + \frac{3\sqrt{6006}xy^5z}{16}$ |

$\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 4)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & \frac{\sqrt{770}i}{385} & 0 & 0 & 0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & 0 \\
-\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 \\
0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 \\
0 & 0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{385} \\
0 & 0 & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 \\
0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 \\
0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} \\
0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{154} & 0 & 0
\end{bmatrix}$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A_{1g},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 990 | symmetry | $\frac{9\sqrt{21}x^6y}{32}+\frac{15\sqrt{21}x^4y^3}{32}-\frac{45\sqrt{21}x^4yz^2}{8}+\frac{3\sqrt{21}x^2y^5}{32}-\frac{15\sqrt{21}x^2y^3z^2}{4}+\frac{15\sqrt{21}x^2yz^4}{2}-\frac{3\sqrt{21}y^7}{32}+\frac{15\sqrt{21}y^5z^2}{8}-\frac{5\sqrt{21}y^3z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 991 | symmetry | $-\frac{35x^6z}{16}-\frac{105x^4y^2z}{16}+\frac{105x^4z^3}{8}-\frac{105x^2y^4z}{16}+\frac{105x^2y^2z^3}{4}-\frac{21x^2z^5}{2}-\frac{35y^6z}{16}+\frac{105y^4z^3}{8}-\frac{21y^2z^5}{2}+z^7$ |
| | $\mathbb{M}_7^{(1,-1;a)}(A_{1g},2)$ | $\begin{bmatrix} 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&\frac{\sqrt{1430}i}{286}&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&-\frac{\sqrt{10010}i}{286}&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&0&\frac{3\sqrt{2002}i}{286}&0&0 \\ 0&0&0&0&0&0 & -\frac{\sqrt{1430}i}{286}&0&0&0&0&0&-\frac{\sqrt{10010}i}{286}&0 \\ 0&0&0&0&0&0 & 0&\frac{\sqrt{10010}i}{286}&0&0&0&0&0&\frac{\sqrt{1430}i}{286} \\ 0&0&0&0&0&0 & 0&0&-\frac{3\sqrt{2002}i}{286}&0&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&\frac{\sqrt{10010}i}{286}&0&0&0&0 \\ 0&0&0&0&0&0 & 0&0&0&0&-\frac{\sqrt{1430}i}{286}&0&0&0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 992 | symmetry | $\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{572} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{35\sqrt{858}}{1716} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{35\sqrt{858}}{1716} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{572} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{1716} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716}
\end{bmatrix}$$

$$\frac{\sqrt{6006}x^6z}{32} - \frac{15\sqrt{6006}x^4y^2z}{32} + \frac{15\sqrt{6006}x^2y^4z}{32} - \frac{\sqrt{6006}y^6z}{32}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 993 | symmetry | $\frac{3\sqrt{21}x^7}{32} - \frac{3\sqrt{21}x^5y^2}{32} - \frac{15\sqrt{21}x^5z^2}{8} - \frac{15\sqrt{21}x^3y^4}{32} + \frac{15\sqrt{21}x^3y^2z^2}{4} + \frac{5\sqrt{21}x^3z^4}{2} - \frac{9\sqrt{21}xy^6}{32} + \frac{45\sqrt{21}xy^4z^2}{8} - \frac{15\sqrt{21}xy^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 994 | symmetry | $\mathbb{M}_7^{(1,-1;a)}(A_{2g},3)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{286} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{286} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{\sqrt{429}x^7}{32} - \frac{21\sqrt{429}x^5y^2}{32} + \frac{35\sqrt{429}x^3y^4}{32} - \frac{7\sqrt{429}xy^6}{32}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 995 | symmetry | $\frac{7\sqrt{429}x^6y}{32} - \frac{35\sqrt{429}x^4y^3}{32} + \frac{21\sqrt{429}x^2y^5}{32} - \frac{\sqrt{429}y^7}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 996 | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)$ | symmetry |

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$-\frac{\sqrt{231}x^7}{32}+\frac{9\sqrt{231}x^5y^2}{32}+\frac{3\sqrt{231}x^5z^2}{8}+\frac{5\sqrt{231}x^3y^4}{32}-\frac{15\sqrt{231}x^3y^2z^2}{4}-\frac{5\sqrt{231}xy^6}{32}+\frac{15\sqrt{231}xy^4z^2}{8}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 997 | symmetry | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}}{26} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{182}}{26} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}}{26} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}}{26} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{182}}{26} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}}{26} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\frac{5\sqrt{231}x^6y}{32} - \frac{5\sqrt{231}x^4y^3}{32} - \frac{15\sqrt{231}x^4yz^2}{8} - \frac{9\sqrt{231}x^2y^5}{32} + \frac{15\sqrt{231}x^2y^3z^2}{4} + \frac{\sqrt{231}y^7}{32} - \frac{3\sqrt{231}y^5z^2}{8}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 998 | symmetry | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}i}{26} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{182}i}{26} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}i}{26} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{78}i}{26} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{182}i}{26} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{78}i}{26} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{5\sqrt{7}x^7}{32}-\frac{15\sqrt{7}x^5y^2}{32}+\frac{15\sqrt{7}x^5z^2}{4}-\frac{15\sqrt{7}x^3y^4}{32}+\frac{15\sqrt{7}x^3y^2z^2}{2}-\frac{15\sqrt{7}x^3z^4}{2}-\frac{5\sqrt{7}xy^6}{32}+\frac{15\sqrt{7}xy^4z^2}{4}-\frac{15\sqrt{7}xy^2z^4}{2}+2\sqrt{7}xz^6$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 999 | symmetry | $-\frac{5\sqrt{7}x^6y}{32}-\frac{15\sqrt{7}x^4y^3}{32}+\frac{15\sqrt{7}x^4yz^2}{4}-\frac{15\sqrt{7}x^2y^5}{32}+\frac{15\sqrt{7}x^2y^3z^2}{2}-\frac{15\sqrt{7}x^2yz^4}{2}-\frac{5\sqrt{7}y^7}{32}+\frac{15\sqrt{7}y^5z^2}{4}-\frac{15\sqrt{7}y^3z^4}{2}+2\sqrt{7}yz^6$ | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&-\frac{\sqrt{858}}{858}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-\frac{\sqrt{858}}{858}&0&\frac{\sqrt{2002}}{286}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{2002}}{286}&0&-\frac{\sqrt{10010}}{286}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{10010}}{286}&0&\frac{5\sqrt{6006}}{858}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{5\sqrt{6006}}{858}&0&-\frac{\sqrt{10010}}{286}&0&0 \\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{10010}}{286}&0&\frac{\sqrt{2002}}{286}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{2002}}{286}&0&-\frac{\sqrt{858}}{858} \\ 0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{858}}{858}&0 \end{bmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)$ | $\frac{3\sqrt{231}x^6z}{16}-\frac{15\sqrt{231}x^4y^2z}{16}-\frac{5\sqrt{231}x^4z^3}{8}-\frac{15\sqrt{231}x^2y^4z}{16}+\frac{15\sqrt{231}x^2y^2z^3}{4}+\frac{3\sqrt{231}y^6z}{16}-\frac{5\sqrt{231}y^4z^3}{8}$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&\frac{\sqrt{858}i}{858}&0&0&0&0&0&0\\ 0&0&0&0&0&0&-\frac{\sqrt{858}i}{858}&0&-\frac{\sqrt{2002}i}{286}&0&0&0&0&0\\ 0&0&0&0&0&0&0&\frac{\sqrt{2002}i}{286}&0&\frac{\sqrt{10010}i}{286}&0&0&0&0\\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{10010}i}{286}&0&-\frac{5\sqrt{6006}i}{858}&0&0&0\\ 0&0&0&0&0&0&0&0&0&\frac{5\sqrt{6006}i}{858}&0&\frac{\sqrt{10010}i}{286}&0&0\\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{10010}i}{286}&0&-\frac{\sqrt{2002}i}{286}&0\\ 0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{2002}i}{286}&0&\frac{\sqrt{858}i}{858}\\ 0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{858}i}{858}&0 \end{bmatrix}$ |
| 1000 | symmetry | |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1001 | symmetry | $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 \end{bmatrix}$ $\frac{3\sqrt{231}x^5yz}{4} - \frac{5\sqrt{231}x^3yz^3}{2} - \frac{3\sqrt{231}xy^5z}{4} + \frac{5\sqrt{231}xy^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1002 | symmetry | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}i}{52} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}i}{52} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}i}{52} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}i}{52} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}i}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}i}{52} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}i}{52} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}i}{52} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$-\frac{15\sqrt{42}x^6z}{32} - \frac{15\sqrt{42}x^4y^2z}{32} + \frac{5\sqrt{42}x^4z^3}{2} + \frac{15\sqrt{42}x^2y^4z}{32} - \frac{3\sqrt{42}x^2z^5}{2} + \frac{15\sqrt{42}y^6z}{32} - \frac{5\sqrt{42}y^4z^3}{2} + \frac{3\sqrt{42}y^2z^5}{2}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1003 | symmetry | $\frac{15\sqrt{42}x^5yz}{16} + \frac{15\sqrt{42}x^3y^3z}{8} - 5\sqrt{42}x^3yz^3 + \frac{15\sqrt{42}xy^5z}{16} - 5\sqrt{42}xy^3z^3 + 3\sqrt{42}xyz^5$ |

$\mathbb{M}_{7,1}^{(1,-1;a)}(E_g, 5)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}}{286} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}}{286} & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1004 | symmetry | $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,5)$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{429}i}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5005}i}{286} & 0 & 0 \\ -\frac{\sqrt{429}i}{286} & 0 & 0 & 0 & \frac{\sqrt{15015}i}{286} & 0 \\ 0 & \frac{\sqrt{5005}i}{286} & 0 & 0 & 0 & -\frac{\sqrt{15015}i}{286} \\ 0 & 0 & -\frac{\sqrt{15015}i}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15015}i}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}i}{286} \end{bmatrix}$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{429}i}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5005}i}{286} & 0 & 0 \\ -\frac{\sqrt{429}i}{286} & 0 & 0 & 0 & \frac{\sqrt{15015}i}{286} & 0 \\ 0 & \frac{\sqrt{5005}i}{286} & 0 & 0 & 0 & -\frac{\sqrt{15015}i}{286} \\ 0 & 0 & -\frac{\sqrt{15015}i}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15015}i}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}i}{286} \end{bmatrix}$ | $z$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1005 | symmetry | $x$ |

$$\mathbb{M}_1^{(1,1;a)}(A_{2g}) \left[\begin{array}{cccccccccccccc}
\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42}
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|

1006 symmetry

$$\mathbb{M}_{1,1}^{(1,1;a)}(E_g) = \begin{bmatrix}
0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{21}}{49} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{392} & 0 & -\frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{210}}{245} & 0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{196} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{6\sqrt{105}}{245} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & -\frac{5\sqrt{21}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{210}}{245} & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{392} & 0 & -\frac{15\sqrt{14}}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{3\sqrt{10}}{56} \\
\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{392} & 0 & \frac{5\sqrt{21}}{196} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{70}}{392} & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & \frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{21}}{196} & 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{\sqrt{35}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{15}}{42} \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0
\end{bmatrix}$$

*y*

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1007 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

$$\mathbb{M}_{1,2}^{(1,1;a)}(E_g) \left[\begin{array}{cccccccccccccc}
0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{21}i}{49} & 0 & -\frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}i}{392} & 0 & \frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{210}i}{245} & 0 & -\frac{6\sqrt{105}i}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{196} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{6\sqrt{105}i}{245} & 0 & -\frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{196} & 0 & \frac{5\sqrt{21}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{210}i}{245} & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{392} & 0 & \frac{15\sqrt{14}i}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{3\sqrt{10}i}{56} \\
-\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{210}i}{392} & 0 & -\frac{5\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & \frac{5\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{70}i}{392} & 0 & -\frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{98} & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & -\frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & \frac{5\sqrt{7}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{21}i}{196} & 0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{98} & 0 & \frac{\sqrt{35}i}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & \frac{\sqrt{15}i}{42} \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0
\end{array}\right]$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1008 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |

$\mathbb{M}_3^{(1,1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{147} & -\frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{392} & 0 \\
-\frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{56} \\
0 & -\frac{2\sqrt{770}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{154}i}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{539} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{196} & \frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{154}i}{539} & 0 \\
\frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}i}{539} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} \\
0 & \frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{539} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1009 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

$\mathbb{M}_3^{(1,1;a)}(A_{2g},1)$

$$\begin{bmatrix}
-\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1010 | symmetry | $-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$ |

$\mathbb{M}_3^{(1,1;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{770}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{147} & -\frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{392} & 0 \\
-\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{56} \\
0 & -\frac{2\sqrt{770}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{196} & \frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 \\
\frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} \\
0 & \frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1011 | symmetry | $-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$ |

$$\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)=\begin{bmatrix}
0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0\\
-\frac{\sqrt{2310}}{147} & 0 & \frac{\sqrt{231}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{196} & 0 & \frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{231}}{147} & 0 & \frac{2\sqrt{462}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{392} & 0 & -\frac{\sqrt{154}}{392} & 0 & 0 & 0\\
0 & 0 & \frac{2\sqrt{462}}{147} & 0 & \frac{\sqrt{231}}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{392} & 0 & -\frac{\sqrt{2310}}{392} & 0 & 0\\
0 & 0 & 0 & \frac{\sqrt{231}}{147} & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{98} & 0 & -\frac{\sqrt{385}}{196} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & \frac{\sqrt{11}}{28}\\
-\frac{\sqrt{11}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{77} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{385}}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{77} & 0 & \frac{\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{231}}{98} & 0 & \frac{\sqrt{2310}}{392} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{539} & 0 & -\frac{\sqrt{770}}{539} & 0 & 0 & 0 & 0\\
0 & \frac{\sqrt{77}}{98} & 0 & \frac{\sqrt{154}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{539} & 0 & -\frac{2\sqrt{462}}{539} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{154}}{392} & 0 & -\frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{462}}{539} & 0 & -\frac{\sqrt{770}}{539} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{2310}}{392} & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{539} & 0 & \frac{\sqrt{154}}{539} & 0\\
0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{539} & 0 & \frac{\sqrt{66}}{77}\\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{77} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1012 | symmetry | $-\frac{\sqrt{15}x^2z}{2}+\frac{\sqrt{15}y^2z}{2}$ |

$\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)$

$$\begin{bmatrix}
0 & \frac{\sqrt{2310}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{231}i}{98} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{147} & 0 & -\frac{\sqrt{231}i}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{196} & 0 & -\frac{\sqrt{77}i}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{231}i}{147} & 0 & -\frac{2\sqrt{462}i}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{392} & 0 & \frac{\sqrt{154}i}{392} & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{462}i}{147} & 0 & -\frac{\sqrt{231}i}{147} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{392} & 0 & \frac{\sqrt{2310}i}{392} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{231}i}{147} & 0 & \frac{\sqrt{2310}i}{147} & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{98} & 0 & \frac{\sqrt{385}i}{196} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{98} & 0 & -\frac{\sqrt{11}i}{28} \\
\frac{\sqrt{11}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{385}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{77} & 0 & -\frac{\sqrt{154}i}{539} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{231}i}{98} & 0 & -\frac{\sqrt{2310}i}{392} & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{539} & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{77}i}{98} & 0 & -\frac{\sqrt{154}i}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & \frac{2\sqrt{462}i}{539} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{154}i}{392} & 0 & \frac{\sqrt{77}i}{98} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{462}i}{539} & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}i}{392} & 0 & \frac{\sqrt{231}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & -\frac{\sqrt{154}i}{539} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{539} & 0 & -\frac{\sqrt{66}i}{77} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{77} & 0
\end{bmatrix}$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1013 | symmetry | $\sqrt{15}xyz$ |

$\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)$

$$\begin{bmatrix}
0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}}{294} & 0 & 0 & \frac{\sqrt{22}}{28} & 0 & 0 & 0 & -\frac{\sqrt{770}}{196} & 0 & 0 & 0 \\
\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 \\
0 & \frac{\sqrt{2310}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 & 0 & \frac{\sqrt{77}}{196} & 0 \\
0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{196} & 0 & 0 & 0 & \frac{\sqrt{22}}{28} \\
0 & 0 & 0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & 0 \\
0 & \frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 & -\frac{\sqrt{55}}{77} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\
-\frac{\sqrt{154}}{98} & 0 & 0 & 0 & -\frac{\sqrt{770}}{196} & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 \\
0 & -\frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 \\
0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & \frac{\sqrt{55}}{77} \\
0 & 0 & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{77} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1014 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

$$\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)$$

$$\begin{bmatrix}
0 & 0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}i}{294} & 0 & 0 & -\frac{\sqrt{22}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 \\
-\frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{294} & 0 & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{294} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{294} & 0 & 0 & \frac{\sqrt{231}i}{196} & 0 & 0 & 0 & \frac{\sqrt{77}i}{196} & 0 \\
0 & 0 & \frac{\sqrt{2310}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & \frac{\sqrt{22}i}{28} \\
0 & 0 & 0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{98} & 0 & 0 & 0 \\
0 & \frac{\sqrt{22}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & \frac{\sqrt{55}i}{77} & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 \\
\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & \frac{3\sqrt{231}i}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{3\sqrt{231}i}{539} & 0 \\
0 & 0 & \frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{\sqrt{55}i}{77} \\
0 & 0 & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{77} & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1015 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

$\mathbb{M}_5^{(1,1;a)}(A_{1g})$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}i}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}i}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{77} & \frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{308} & 0 \\
\frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}i}{924} \\
0 & -\frac{\sqrt{1430}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2002}i}{1001} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}i}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{1001} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}i}{924} & -\frac{\sqrt{2002}i}{1001} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}i}{2002} & 0 \\
-\frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}i}{2002} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2002}i}{1001} \\
0 & \frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{1001} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}i}{2002} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}i}{1001} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1016 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

$\mathbb{M}_5^{(1,1;a)}(A_{2g},1)$

$$\begin{bmatrix}
\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1001}}{14014} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1001}}{14014} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{2002}
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1017 | symmetry | $\frac{3\sqrt{14}x^5}{16} - \frac{15\sqrt{14}x^3y^2}{8} + \frac{15\sqrt{14}xy^4}{16}$ |

$\mathbb{M}_5^{(1,1;a)}(A_{2g},2)$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{143}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}}{77} & \frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{308} & 0 \\
\frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{924} \\
0 & -\frac{\sqrt{1430}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{1001} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 \\
-\frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} \\
0 & \frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{1001} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1018 | symmetry | $-\frac{15\sqrt{14}x^4y}{16}+\frac{15\sqrt{14}x^2y^3}{8}-\frac{3\sqrt{14}y^5}{16}$ |

$$\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\left[\begin{array}{cccccccccccccc}
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}}{924} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{924}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{2002} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1001} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{2002}\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{1001} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{15015}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{2002} & 0 & 0 & 0 & 0 & 0
\end{array}\right]$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1019 | symmetry | $\frac{\sqrt{15}x^5}{8}+\frac{\sqrt{15}x^3y^2}{4}-\frac{3\sqrt{15}x^3z^2}{2}+\frac{\sqrt{15}xy^4}{8}-\frac{3\sqrt{15}xy^2z^2}{2}+\sqrt{15}xz^4$ |
| | | $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)$ |

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{286}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}i}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}i}{924} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{286}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15015}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}i}{2002} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}i}{1001} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}i}{2002} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{286}i}{1001} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15015}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}i}{2002} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1020 | symmetry | $\frac{\sqrt{15}x^4y}{8}+\frac{\sqrt{15}x^2y^3}{4}-\frac{3\sqrt{15}x^2yz^2}{2}+\frac{\sqrt{15}y^5}{8}-\frac{3\sqrt{15}y^3z^2}{2}+\sqrt{15}yz^4$ |

$$\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)$$

$$\begin{bmatrix}
0 & \frac{\sqrt{3003}}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{1848} & 0 & -\frac{5\sqrt{30030}}{12936} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{3003}}{539} & 0 & -\frac{\sqrt{30030}}{539} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{2002}}{12936} & 0 & \frac{13\sqrt{10010}}{12936} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{30030}}{539} & 0 & \frac{2\sqrt{15015}}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{588} & 0 & -\frac{\sqrt{5005}}{6468} & 0 & 0 & 0\\
0 & 0 & \frac{2\sqrt{15015}}{539} & 0 & -\frac{\sqrt{30030}}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{6468} & 0 & -\frac{\sqrt{3003}}{588} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{30030}}{539} & 0 & \frac{\sqrt{3003}}{539} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{10010}}{12936} & 0 & \frac{23\sqrt{2002}}{12936} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{30030}}{12936} & 0 & -\frac{\sqrt{1430}}{1848}\\
\frac{\sqrt{1430}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\frac{23\sqrt{2002}}{12936} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & \frac{4\sqrt{5005}}{7007} & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{30030}}{12936} & 0 & \frac{\sqrt{3003}}{588} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5005}}{7007} & 0 & -\frac{\sqrt{1001}}{14014} & 0 & 0 & 0 & 0\\
0 & \frac{13\sqrt{10010}}{12936} & 0 & \frac{\sqrt{5005}}{6468} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{14014} & 0 & -\frac{2\sqrt{15015}}{7007} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5005}}{6468} & 0 & -\frac{13\sqrt{10010}}{12936} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15015}}{7007} & 0 & -\frac{\sqrt{1001}}{14014} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3003}}{588} & 0 & \frac{5\sqrt{30030}}{12936} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{14014} & 0 & \frac{4\sqrt{5005}}{7007} & 0\\
0 & 0 & 0 & 0 & \frac{23\sqrt{2002}}{12936} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{5005}}{7007} & 0 & -\frac{\sqrt{2145}}{2002}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1021 | symmetry | $-\frac{3\sqrt{35}x^4z}{8}+\frac{9\sqrt{35}x^2y^2z}{4}-\frac{3\sqrt{35}y^4z}{8}$ |

$$\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)=\begin{bmatrix}
0 & -\frac{\sqrt{3003}i}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{1848} & 0 & \frac{5\sqrt{30030}i}{12936} & 0 & 0 & 0 & 0 & 0\\
\frac{\sqrt{3003}i}{539} & 0 & \frac{\sqrt{30030}i}{539} & 0 & 0 & 0 & 0 & -\frac{23\sqrt{2002}i}{12936} & 0 & -\frac{13\sqrt{10010}i}{12936} & 0 & 0 & 0 & 0\\
0 & -\frac{\sqrt{30030}i}{539} & 0 & -\frac{2\sqrt{15015}i}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}i}{588} & 0 & \frac{\sqrt{5005}i}{6468} & 0 & 0 & 0\\
0 & 0 & \frac{2\sqrt{15015}i}{539} & 0 & \frac{\sqrt{30030}i}{539} & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}i}{6468} & 0 & \frac{\sqrt{3003}i}{588} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{30030}i}{539} & 0 & -\frac{\sqrt{3003}i}{539} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{10010}i}{12936} & 0 & -\frac{23\sqrt{2002}i}{12936} & 0\\
0 & 0 & 0 & 0 & \frac{\sqrt{3003}i}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{30030}i}{12936} & 0 & \frac{\sqrt{1430}i}{1848}\\
-\frac{\sqrt{1430}i}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0\\
0 & \frac{23\sqrt{2002}i}{12936} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & -\frac{4\sqrt{5005}i}{7007} & 0 & 0 & 0 & 0 & 0\\
-\frac{5\sqrt{30030}i}{12936} & 0 & -\frac{\sqrt{3003}i}{588} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5005}i}{7007} & 0 & \frac{\sqrt{1001}i}{14014} & 0 & 0 & 0 & 0\\
0 & \frac{13\sqrt{10010}i}{12936} & 0 & -\frac{\sqrt{5005}i}{6468} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}i}{14014} & 0 & \frac{2\sqrt{15015}i}{7007} & 0 & 0 & 0\\
0 & 0 & -\frac{\sqrt{5005}i}{6468} & 0 & \frac{13\sqrt{10010}i}{12936} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15015}i}{7007} & 0 & \frac{\sqrt{1001}i}{14014} & 0 & 0\\
0 & 0 & 0 & -\frac{\sqrt{3003}i}{588} & 0 & -\frac{5\sqrt{30030}i}{12936} & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}i}{14014} & 0 & -\frac{4\sqrt{5005}i}{7007} & 0\\
0 & 0 & 0 & 0 & \frac{23\sqrt{2002}i}{12936} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{5005}i}{7007} & 0 & \frac{\sqrt{2145}i}{2002}\\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1022 | symmetry | $-\frac{3\sqrt{35}x^3yz}{2}+\frac{3\sqrt{35}xy^3z}{2}$ |

$\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued …*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1023 | symmetry | $\frac{\sqrt{105}x^4z}{4} - \frac{\sqrt{105}x^2z^3}{2} - \frac{\sqrt{105}y^4z}{4} + \frac{\sqrt{105}y^2z^3}{2}$ |

$$\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}i}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 1024 | symmetry | $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,4)$ = $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{286}}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & -\frac{\sqrt{2002}}{924} & 0 & 0 & 0 & -\frac{\sqrt{1430}}{462} & 0 & 0 & 0 \\ -\frac{\sqrt{858}}{154} & 0 & 0 & 0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & \frac{2\sqrt{143}}{231} & 0 & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 \\ 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & 0 & \frac{\sqrt{858}}{154} & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & 0 & \frac{2\sqrt{143}}{231} & 0 \\ 0 & 0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{462} & 0 & 0 & 0 & -\frac{\sqrt{2002}}{924} \\ 0 & 0 & 0 & \frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{286}}{924} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & \frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 & 0 \\ \frac{5\sqrt{286}}{924} & 0 & 0 & 0 & -\frac{\sqrt{1430}}{462} & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 \\ 0 & -\frac{\sqrt{1430}}{462} & 0 & 0 & 0 & \frac{5\sqrt{286}}{924} & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 \\ 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & -\frac{\sqrt{5005}}{2002} \\ 0 & 0 & 0 & \frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{2002} & 0 & 0 \end{bmatrix}$ $-\frac{\sqrt{105}x^3yz}{2}-\frac{\sqrt{105}xy^3z}{2}+\sqrt{105}xyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|

$$\mathbb{M}_{5,2}^{(1,1;a)}(E_g,4)$$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & \frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & 0 \\
\frac{\sqrt{858}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 \\
0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & \frac{\sqrt{858}i}{154} & 0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 & 0 & \frac{2\sqrt{143}i}{231} & 0 \\
0 & 0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{2002}i}{924} \\
0 & 0 & 0 & -\frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}i}{924} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 & -\frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 \\
-\frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 \\
0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & \frac{5\sqrt{286}i}{924} & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 \\
0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{2002} \\
0 & 0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}i}{2002} & 0 & 0
\end{bmatrix}$$